PHINMA UNIVERSITY OF ILOILO
College of Information Technology Education

CAPSTONE PROJECT PROPOSAL

BrewInsight BI: Revenue Pattern Analytics System for Kape Tubong

In partial fulfillment for the requirements in Capstone 1 and Research Subject

Ella Marie Tabobo - BSIT3
Alexa Elaine Roldan - BSIT3
Feljan Katipunan - BSIT3
Evelyn Tamonan - BSIT3
Uki Mae Sotelo - BSIT3

Data Informatics Track

Second Semester
2025-26

CHAPTER 1

INTRODUCTION

1.1 Background of the Study

The Food and Beverage (F&B) industry has grown significantly over the past decade due to changing consumer lifestyles, urbanization, and the increasing demand for convenient dining and beverage options. Among the many businesses within this sector, coffee shops have become one of the fastest-growing industries because they serve not only as places to purchase beverages but also as spaces for meetings, studying, remote work, and social interaction. As customer expectations continue to evolve, coffee shop businesses are expected to maintain high standards in service quality, operational efficiency, and overall customer experience in order to remain competitive in the market. In addition, the increasing presence of local and international coffee brands has intensified market competition, requiring coffee shop businesses to adopt more efficient operational strategies and innovative management practices. As consumer preferences shift toward convenience, speed, and personalized service, businesses must continuously improve their operations to remain relevant and profitable.The modern coffee shop environment has evolved beyond simple beverage retailing into a highly dynamic service ecosystem requiring efficient coordination of multiple operational processes. Businesses in this industry must manage not only product quality but also customer experience, branch consistency, and rapid service delivery. The rise of digitally connected consumers has also increased expectations for faster transactions and personalized offerings. Social media and online customer feedback further amplify the need for businesses to maintain strong operational performance. In such a competitive environment, data-driven management becomes increasingly essential to sustain growth and profitability. Coffee shops that fail to adapt to evolving consumer expectations risk losing market share to more technologically advanced competitors. Operational efficiency has therefore become a strategic priority in the F&B industry. Businesses are expected to leverage modern technologies to streamline internal processes and improve service responsiveness. As competition intensifies, analytical decision-making tools are becoming critical for maintaining operational excellence. Thus, the F&B industry increasingly demands innovative and technology-supported management approaches.

As coffee shop businesses expand to multiple locations, managing operations becomes more challenging. Opening additional branches or kiosks increases the complexity of handling transactions, monitoring inventory, scheduling staff, tracking sales, and evaluating branch performance. Tasks that are manageable in a single-location business become more difficult when operations are spread across several outlets. Because of this, businesses require a centralized way of managing and monitoring operations to maintain consistency and efficiency across all locations. Without a standardized system, inconsistencies in branch operations may arise, leading to variations in service quality, reporting practices, and overall business performance. Maintaining operational control across multiple branches becomes increasingly difficult when management lacks centralized visibility into day-to-day branch activities.Multi-branch expansion introduces additional administrative and analytical challenges that cannot be effectively managed through informal monitoring methods. Branch managers may apply inconsistent operational procedures when centralized oversight mechanisms are absent. Variations in transaction recording, stock monitoring, and performance reporting can reduce organizational consistency. As the number of locations increases, communication gaps between branches and management may also widen. Delays in reporting from remote kiosks can hinder timely strategic interventions. Furthermore, evaluating comparative branch performance becomes increasingly difficult when data is fragmented across multiple systems. Branch-level accountability may weaken if performance metrics are not consistently monitored and standardized. Centralized operational visibility is therefore essential for maintaining business alignment across distributed locations. Businesses with multiple outlets require integrated systems to ensure consistent oversight and informed management. Effective branch expansion depends not only on market growth but also on scalable operational monitoring infrastructure.

Kape Tubong has shown continuous growth by expanding from a single coffee outlet into a business with multiple kiosks and a main branch. This expansion reflects increasing customer demand and positive market acceptance of the brand. However, as the business grows, the volume of sales and operational data generated from different locations also increases. Without an integrated information system, managing and monitoring this data becomes more difficult, resulting in fragmented records and inefficient reporting processes. The increasing volume of distributed data also creates challenges in maintaining data consistency, accuracy, and accessibility across all locations. As the organization expands further, reliance on disconnected data sources may hinder the company’s ability to scale efficiently and make timely strategic decisions. The expansion of Kape Tubong reflects positive organizational growth but also introduces new operational demands that require more sophisticated information management practices. Larger transaction volumes increase the complexity of monitoring revenue performance and operational efficiency across all branches. The accumulation of disconnected sales records may lead to inconsistent reporting and delayed consolidation of business information. Branch managers may also experience difficulty in maintaining accurate local records without centralized oversight mechanisms. As operational scale increases, manual coordination between branches and management becomes less sustainable. Fragmented data environments reduce management’s ability to obtain a holistic view of organizational performance. This may hinder strategic planning and delay responses to emerging operational concerns. Furthermore, business growth without corresponding digital infrastructure may create scalability limitations. Expanding enterprises require systems capable of supporting increased transaction volume and analytical complexity. Therefore, Kape Tubong's continued growth creates a strong need for centralized digital business intelligence support.

Many small and medium-sized enterprises (SMEs), including coffee shop businesses, still rely on traditional or semi-manual methods of recording and managing operational data. These methods often include handwritten transaction logs, spreadsheet-based monitoring, and standalone Point of Sale (POS) systems that operate independently in each branch. While these approaches may work during the early stages of business operations, they become less effective as the organization expands. Separate systems create isolated data records that make information consolidation difficult and reduce management's visibility into overall business performance. Furthermore, manual and disconnected systems increase dependency on human intervention for data processing, which may compromise speed, reliability, and consistency of reporting. As operational complexity increases, these limitations can negatively impact organizational efficiency and management responsiveness.Traditional reporting practices often lack the automation necessary to support modern business analysis requirements. Manual encoding processes are vulnerable to human error, omissions, and inconsistencies in formatting. Spreadsheet-based reports may become difficult to manage as data volume increases over time. Standalone systems also limit the ability to generate consolidated enterprise-wide reports automatically. Businesses relying on disconnected systems often face challenges in reconciling conflicting data from multiple sources. These inefficiencies consume administrative resources that could otherwise be allocated to strategic functions. In addition, manual reporting processes delay the availability of performance information needed for decision-making. The lack of automated validation also increases the likelihood of inaccurate reporting outputs. SMEs that continue to rely on outdated reporting methods may struggle to compete with more digitally enabled organizations. Thus, modernizing operational data management has become increasingly necessary for growing enterprises.

One of the major issues caused by fragmented systems is the difficulty of obtaining real-time and unified sales data from all branches. Since each branch records transactions separately, managers must manually collect reports from each location before consolidating and analyzing them. This process consumes time, increases administrative workload, and creates opportunities for human error, delayed reporting, and data inconsistencies. As a result, management decisions may be based on incomplete or outdated information. Delays in obtaining accurate sales information may prevent management from responding quickly to sudden changes in customer demand, branch performance, or operational issues. The inability to access real-time data limits the organization's responsiveness and weakens its ability to make timely operational adjustments.Real-time information availability is increasingly recognized as a critical requirement in competitive business environments. Businesses operating without immediate access to updated data may fail to detect urgent operational issues promptly. Delayed reporting reduces the strategic value of business information because decisions are made after performance issues have already occurred. Fragmented reporting also prevents management from observing emerging trends across branches simultaneously. This weakens comparative analysis and branch-level benchmarking. Managers may be forced to rely on assumptions when immediate data is unavailable. Such reliance can result in less accurate and less strategic decision-making. Furthermore, inconsistent reporting intervals between branches may create additional visibility gaps. Unified real-time reporting therefore becomes essential for effective business control. Organizations lacking timely data access may experience reduced operational agility and strategic responsiveness.

Another challenge is the limited ability of traditional systems to provide meaningful analytical insights. Basic sales reports often present only raw figures without deeper analysis of trends or patterns. Without proper analytical tools, management may find it difficult to identify peak sales periods, determine customer purchasing behavior, evaluate branch performance, and monitor product demand. This limits the organization’s ability to make informed and proactive business decisions. Without visualization and analytical reporting features, management may struggle to interpret large amounts of transactional data efficiently. Consequently, potentially valuable business insights may remain hidden within raw sales records and unused in strategic planning.

Raw numerical reports alone often fail to provide sufficient interpretive value for strategic decision-making. Managers may find it difficult to detect patterns when information is presented only in tabular or spreadsheet form. Analytical limitations reduce the practical usefulness of collected data despite high transaction volumes. Businesses may therefore possess large amounts of information without the capability to extract actionable insights. Trend analysis and performance visualization are necessary to transform transactional data into strategic knowledge. Without analytical tools, organizations may overlook important operational opportunities and risks. The inability to identify customer behavior trends can weaken promotional and inventory strategies. Similarly, lack of product-level analysis may prevent optimization of menu offerings and pricing decisions. Effective analysis requires both data collection and interpretive mechanisms. Thus, analytical system capabilities are essential for maximizing the strategic value of business data.

The lack of revenue pattern analysis also affects workforce scheduling and inventory planning. Without reliable information on peak operating hours and product demand, staffing schedules may rely on assumptions rather than actual business data. This may lead to overstaffing during slow periods or understaffing during peak hours, both of which can reduce efficiency and affect customer service quality. Similarly, inventory decisions made without accurate sales analysis may result in overstocking, stock shortages, increased waste, or lost sales opportunities. Inefficient workforce and inventory planning can directly affect profitability by increasing labor costs and reducing inventory turnover efficiency. Over time, these inefficiencies may significantly impact operational sustainability and customer satisfaction.Workforce scheduling based on assumptions rather than empirical demand data often leads to inefficient labor utilization.

Overstaffing increases unnecessary payroll expenses while understaffing can reduce service quality and increase customer waiting times. Poor inventory planning may also cause avoidable spoilage, particularly in perishable F&B products. Understocking of high-demand items may lead to missed sales opportunities and dissatisfied customers. Effective resource allocation requires accurate understanding of historical and recurring demand patterns. Revenue pattern analysis can provide the necessary insight to align staffing and stock decisions with actual operational behavior. This improves both cost efficiency and customer service performance. Businesses that fail to optimize labor and inventory based on demand patterns may experience reduced profitability. Strategic planning in F&B operations therefore requires data-supported demand forecasting. Revenue analytics directly supports these optimization efforts.

To address these challenges, many organizations adopt Business Intelligence (BI) systems to improve operational monitoring and decision-making. Business Intelligence refers to the technologies and tools used to collect, organize, analyze, and present business data in a meaningful way. Through BI systems, raw sales data can be transformed into useful insights using dashboards, automated reports, trend analysis, and performance metrics. These tools enable businesses to make more strategic and data-driven decisions. BI systems also help organizations identify hidden opportunities, detect operational inefficiencies, and monitor business performance in real time. As data-driven management becomes increasingly essential in competitive industries, BI adoption has emerged as a valuable strategy for modern business operations.Business Intelligence platforms have become widely recognized as strategic tools for enhancing organizational awareness and performance visibility. These systems enable businesses to convert fragmented data into centralized analytical information environments. Through automation and visualization, BI tools reduce the effort required to interpret operational metrics. They also improve accessibility of business information for non-technical decision-makers. Organizations can use BI systems to monitor both real-time and historical performance indicators simultaneously. This supports immediate operational adjustments and long-term strategic planning. BI adoption also contributes to standardization of reporting practices across organizational units. Furthermore, centralized analytics platforms improve transparency and accountability in management processes. Businesses leveraging BI technologies often gain stronger analytical and competitive capabilities. Therefore, BI systems are increasingly essential in data-intensive business environments.

For multi-branch coffee shop operations, implementing a centralized BI platform provides significant advantages. By integrating sales data from all branches into a single database, management can access real-time business information and monitor performance across all locations. Dashboards can display important metrics such as total sales, peak sales hours, product demand, branch comparisons, and revenue trends. This allows management to make faster, more accurate, and more informed operational decisions. Centralized BI systems also improve data consistency by standardizing reporting formats and analytical methods across all branches. This enables more reliable comparison of branch performance and improves the quality of strategic evaluation.A centralized BI platform enhances operational visibility by consolidating performance information from geographically distributed locations into one unified interface. This reduces the complexity of monitoring separate branch reports individually. Standardized data structures improve the reliability of comparative performance analysis across branches. Managers can identify underperforming locations more quickly and investigate contributing operational factors. Centralization also simplifies enterprise-wide trend analysis by aggregating all branch data into one analytical environment. This improves strategic planning for expansion, staffing, and promotions. Unified dashboards support more efficient executive oversight and reduce administrative coordination effort. Additionally, branch managers can benefit from clearer performance benchmarks and comparative metrics. The availability of centralized analytics promotes more consistent and objective management evaluation. Thus, BI centralization significantly strengthens multi-branch operational governance.

In response to these challenges, this study proposes the development of BREWINSIGHT BI: Revenue Pattern Analytics System for Kape Tubong. The proposed system will serve as a centralized web-based Business Intelligence platform that consolidates sales data from all kiosks and the main branch into a unified database. It will provide real-time dashboards, automated reports, and analytical tools for examining revenue patterns, customer purchasing trends, peak sales periods, and product performance. The system is designed to transform fragmented transactional records into structured and visualized business intelligence that can support daily and strategic management decisions. By centralizing branch data, the platform aims to improve information accessibility, reduce reporting delays, and enhance the overall quality of operational monitoring.The proposed platform specifically addresses the operational limitations identified in Kape Tubong’s current reporting practices. Through automated data integration, the system reduces dependence on manual report consolidation. Visual dashboards provide management with immediate insight into key business metrics. Analytical tools within the platform will support proactive operational planning and evidence-based decision-making. Revenue trend analysis will help management identify sales patterns and evaluate performance changes over time. Product-level monitoring will allow assessment of menu demand and profitability. Branch comparison tools will support more accurate evaluation of location performance. Real-time reporting capabilities will improve responsiveness to sudden operational concerns. By integrating these functions into one platform, the system enhances both operational and strategic business intelligence capabilities. Therefore, BREWINSIGHT BI offers a practical digital solution to Kape Tubong’s operational data challenges.

The system is also expected to improve workforce planning, inventory management, and branch performance monitoring by providing management with timely and accurate operational insights. Through the application of software engineering, database management, and Business Intelligence principles, the project demonstrates how information technology can be used to solve practical business problems in the Food and Beverage industry. It also showcases the practical integration of modern information systems in supporting organizational efficiency and business process optimization.

Furthermore, the project emphasizes the relevance of digital transformation in helping SMEs modernize their operational and managerial practices.
The development of BREWINSIGHT BI reflects the growing importance of integrating technical innovation into everyday business operations. By applying structured software engineering methodologies, the project ensures systematic and maintainable system development. Database technologies enable secure and organized management of large volumes of transactional records. Business Intelligence principles support the transformation of raw operational data into strategic insights. Together, these technological components create a comprehensive analytical environment for operational improvement. The project also demonstrates how SMEs can leverage affordable digital tools to modernize business management. Such implementation highlights the practical role of IT in improving organizational competitiveness. The system serves as an applied example of interdisciplinary technological integration in business contexts. Its deployment may encourage broader digital adoption among similar SMEs. Thus, the study reinforces the strategic importance of technology-driven business transformation.

Ultimately, BREWINSIGHT BI is expected to improve operational transparency, enhance data-driven decision-making, reduce administrative inefficiencies, and support the long-term growth of Kape Tubong. Additionally, the system may serve as a practical reference for other small and medium-sized enterprises seeking to adopt Business Intelligence technologies to modernize operations and improve competitiveness in the Food and Beverage sector. By demonstrating the practical value of centralized analytics and automated reporting, the study may encourage broader adoption of Business Intelligence systems among SMEs facing similar operational challenges. In the long term, the implementation of such technologies may contribute to stronger business sustainability, improved adaptability to market changes, and enhanced organizational competitiveness.The proposed system is not intended solely as an operational tool but also as a strategic enabler for long-term organizational development. Improved transparency and centralized visibility can strengthen accountability and performance governance across all branches. Reduced administrative workload allows management and staff to focus more on strategic and customer-oriented functions. The availability of reliable business insights supports more accurate planning and forecasting activities. Enhanced analytical capabilities may also improve the organization’s ability to adapt to future expansion and market changes. As SMEs increasingly pursue digital transformation, systems such as BREWINSIGHT BI become critical enablers of sustainable competitiveness. The project may therefore serve as a replicable model for similar businesses seeking to modernize operations. Its successful implementation can demonstrate the practical viability of BI adoption in small-scale enterprises. This contributes to both organizational improvement and broader technological awareness among SMEs. Ultimately, BREWINSIGHT BI is positioned as a strategic investment in operational modernization and business sustainability.

1.2 Statement of the Problem

Despite the significant physical expansion of Kape Tubong, several operational inefficiencies continue to affect the overall performance of the business. As the number of branches increases, managing daily operations becomes more complex, particularly without a reliable and integrated information system. The expansion of operations requires stronger coordination, improved monitoring, and more efficient data management across all locations. However, the current operational setup limits the organization’s ability to maintain consistency, transparency, and visibility in its business processes. These conditions highlight the need for a more structured and data-driven approach to managing sales and operations.

As the business continues to grow, Kape Tubong faces increasing challenges in handling the larger volume of transactional and operational data generated by multiple branches. The absence of a unified system makes it difficult to consolidate information from different locations into a single, organized source of truth. Management must rely on separate reports from each branch, making it harder to evaluate overall business performance and compare operational efficiency among locations. This fragmented approach reduces responsiveness to business issues and limits management’s ability to implement timely corrective actions. It also creates communication gaps between branch-level operations and upper management, resulting in delays in operational feedback and performance intervention.

In addition, the lack of centralized data processing creates barriers to performance monitoring and strategic planning. Important business decisions such as identifying profitable branches, adjusting product offerings, planning workforce schedules, and managing stock levels require accurate and timely information. Without a system capable of processing and analyzing sales data efficiently, management may struggle to identify operational issues before they escalate. As a result, opportunities for optimization and business improvement may be missed. This limitation reduces the organization's capacity to respond proactively to market changes, customer demand shifts, and branch-specific operational concerns.

The current decentralized operational structure also creates challenges in maintaining consistent reporting standards across all branches of Kape Tubong. Since each branch may follow slightly different methods of recording, organizing, and submitting sales reports, discrepancies in documentation and reporting formats may arise. These inconsistencies make it difficult for management to compare branch performance objectively and accurately. Variations in reporting practices may also reduce the reliability of consolidated operational data used for strategic analysis. Without standardized processes and centralized validation mechanisms, ensuring uniformity in data quality becomes increasingly difficult as the business expands. This issue may result in inaccurate interpretations of branch-level performance and hinder fair performance evaluation. Furthermore, management may encounter difficulties in identifying whether performance differences are caused by actual operational issues or by inconsistent reporting methods. Such limitations reduce the effectiveness of comparative performance analysis and branch benchmarking. In growing multi-branch businesses, consistency in operational reporting is essential to maintaining organizational transparency and accountability. Therefore, Kape Tubong requires a centralized and standardized information management solution to ensure data uniformity across all operational locations.

Moreover, the continued reliance on fragmented and manually processed data limits the organization’s ability to maximize the strategic value of its operational information. Although substantial sales and transactional data are generated daily across all branches, much of this information remains underutilized due to the absence of analytical tools capable of transforming raw records into meaningful business insights. Historical sales trends, branch performance patterns, customer demand shifts, and product profitability indicators may remain hidden within unprocessed records. As a result, management loses valuable opportunities to leverage operational data for forecasting, optimization, and strategic planning. The inability to perform advanced analysis also weakens the organization’s responsiveness to emerging market trends and competitive pressures. In a rapidly evolving food and beverage market, delayed or uninformed decision-making can negatively affect profitability and customer satisfaction. Competitors utilizing more advanced analytical systems may gain significant operational and strategic advantages. This places Kape Tubong at risk of falling behind in terms of efficiency, adaptability, and market competitiveness. To remain sustainable and competitive, the organization must improve its capacity to convert operational data into actionable intelligence. This emphasizes the necessity of implementing a Business Intelligence-driven platform to support modernized, data-based management practices.

Additionally, the absence of integrated analytical and monitoring tools may hinder the long-term scalability of Kape Tubong’s operations. As the business opens additional branches or kiosks, the complexity of monitoring transactions, inventory movement, staffing requirements, and branch productivity will continue to increase. Manual and fragmented systems that may be manageable at smaller scales become increasingly inefficient as operational scope expands. Without technological support, management’s administrative workload may increase disproportionately with business growth. This can create operational bottlenecks that slow decision-making and reduce managerial effectiveness. The inability to scale reporting and monitoring processes efficiently may limit the organization’s capacity for sustainable expansion. Furthermore, expansion without corresponding improvements in information systems may amplify existing inefficiencies and create additional operational risks. The organization may struggle to maintain consistent service quality and operational oversight across a growing number of branches. Therefore, implementing an integrated analytics and reporting platform is essential not only for addressing current operational issues but also for supporting future organizational growth. A scalable information system will enable Kape Tubong to expand more efficiently while maintaining effective control over distributed business operations.

This study aims to address the following problems:

- Existing operations lack a centralized Point of Sale (POS) system capable of consolidating and organizing sales data from multiple business locations into a unified platform. Consequently, sales records are maintained separately at each branch, resulting in fragmented and dispersed data that complicates the generation of complete and accurate business reports. The manual collection and consolidation of reports from various branches require significant time and administrative effort from both staff and management. This process further increases the likelihood of incomplete, duplicated, or inaccurate data entries. Moreover, the absence of centralized integration prevents real-time monitoring of sales performance, thereby limiting management’s capacity to access timely and unified operational information. Such inefficiencies reduce the overall effectiveness of business operations and delay managerial responses to immediate operational concerns. In the long term, the continued use of disconnected systems may hinder the organization’s ability to scale efficiently as additional branches are established. The absence of centralized data storage also creates challenges in maintaining data integrity across different reporting periods. Separate branch systems may follow inconsistent data formatting and recording procedures, reducing standardization across the organization. Management may encounter difficulties in reconciling discrepancies between branch reports due to inconsistent documentation methods. Furthermore, fragmented systems limit the organization’s ability to establish a reliable historical sales database for long-term performance analysis. Without centralized POS integration, branch-level data remains isolated and underutilized for strategic business planning. The organization may also struggle to enforce uniform transaction recording standards across all branches. Inconsistent recording procedures can create variations in data quality and reporting reliability. Branch personnel may interpret recording guidelines differently, leading to inconsistencies in operational documentation. The lack of system integration may also complicate auditing and verification of branch sales records. Management may experience difficulty in tracing transactional discrepancies when records are dispersed across locations. Fragmented sales systems reduce transparency in overall business operations. These limitations weaken organizational control over distributed business activities. The absence of a centralized platform also slows down the preparation of executive summaries and strategic reports. Ultimately, disconnected POS operations reduce the organization's capacity to manage branch expansion efficiently. Additionally, the absence of a unified database may prevent management from obtaining synchronized branch performance reports within a single reporting period. Managers may experience delays when requesting sales summaries because branch records must first be manually collected and validated. Inconsistent branch systems may also create compatibility issues during data consolidation and system maintenance activities. Separate operational platforms can further complicate implementation of organization-wide policy updates and transaction monitoring procedures. Branch employees may encounter confusion when different reporting practices are followed across locations. Without centralized transaction visibility, management may struggle to identify unusual sales activities or operational inconsistencies immediately. Delayed detection of discrepancies can negatively affect financial monitoring and internal control procedures. Fragmented reporting structures may also increase the complexity of preparing tax-related and financial compliance documents. Historical operational records stored in separate systems may become increasingly difficult to retrieve as transaction volumes continue to grow. The organization may also face challenges in preserving data consistency when branch personnel manually modify records independently. In the absence of integrated systems, communication gaps between management and branch operations may become more frequent. Separate systems may further increase dependency on manual verification and repetitive administrative supervision. This can reduce operational efficiency and create additional workload for administrative staff members. Management may also experience difficulty generating accurate comparative analyses between branches due to inconsistent reporting structures. As branch operations expand, maintaining isolated systems may become increasingly costly and operationally inefficient. Ultimately, the lack of centralized POS integration weakens operational coordination, reporting reliability, data accessibility, and organizational scalability.

- The organization has limited capability to automatically track, analyze, and visualize sales trends and product performance. This limitation hinders management's ability to identify high-performing and underperforming products, assess customer purchasing behavior, and implement timely strategic adjustments. Sales data is not processed in real time, and performance analysis remains largely manual, resulting in delayed interpretation of business outcomes.

Furthermore, the lack of analytical dashboards and automated visualization tools restricts the organization's capacity to monitor branch performance comprehensively. Consequently, opportunities for sales optimization, strategic product positioning, and revenue enhancement may not be fully recognized. Without proper trend analysis, management may also struggle to evaluate the effectiveness of promotions, seasonal campaigns, and pricing strategies. The inability to generate visualized analytics reduces the interpretability of business data for management and stakeholders. Raw sales figures alone provide limited insight into performance patterns without contextual analytical processing. Management may overlook emerging demand trends or declining product performance due to insufficient analytical visibility. Delayed recognition of sales trends can negatively impact product planning and marketing responsiveness. The lack of automated analytics ultimately limits Kape Tubong's capacity to leverage data for strategic revenue optimization. In addition, management may find it difficult to compare product performance across different branches without standardized visual reporting. Sales anomalies may remain undetected due to the absence of automated alerts or trend indicators. Opportunities to discontinue underperforming products may be delayed due to insufficient product-level visibility. Management may also struggle to identify complementary product purchasing behavior for bundling or upselling strategies. Limited analytical tools reduce the organization's ability to understand consumer buying preferences in detail. Without sales visualization, long-term business trends may be difficult to communicate to stakeholders. Product demand fluctuations may not be identified until after significant performance decline occurs. This weakens the organization's ability to react proactively to sales performance changes. Inadequate analytics reduce strategic agility in product and marketing decisions. Ultimately, the lack of advanced sales analysis limits overall revenue optimization capabilities. Furthermore, management may encounter challenges in identifying branch-specific purchasing patterns because existing systems do not provide detailed comparative visualizations. Customer demand variations across different time periods may remain unnoticed due to limited analytical monitoring capabilities. Without automated reporting dashboards, operational insights may take longer to interpret and communicate during management evaluations. Sales forecasting initiatives may become less accurate when historical data are not processed through advanced analytical tools. The organization may also struggle to identify products that contribute most significantly to branch profitability. Inadequate sales visualization can reduce management's ability to assess the effectiveness of pricing adjustments and promotional campaigns. Opportunities to identify high-demand seasonal products may also remain underutilized because of insufficient trend monitoring systems. Furthermore, delayed identification of declining product performance may increase the risk of inventory waste and reduced profitability. Management may experience difficulty evaluating customer purchasing frequency and repeat transaction behavior without integrated analytics. The absence of predictive indicators may limit the organization's preparedness for sudden changes in market demand and consumer preferences. Business decisions related to product development and inventory planning may therefore rely more heavily on assumptions rather than analytical evidence. Insufficient visualization capabilities may also weaken stakeholder understanding of operational performance during meetings and business presentations. The organization may encounter difficulty establishing long-term revenue projections because historical sales trends are not systematically analyzed. Competitors utilizing advanced analytics may gain stronger market positioning due to faster interpretation of operational data and customer behavior trends. Ultimately, inadequate sales analytics and visualization tools reduce organizational responsiveness, operational intelligence, customer insight generation, and long-term revenue growth potential.

- Forecasting workforce requirements remains difficult due to the absence of reliable peak-hour and customer traffic analysis. Without data-driven insights into operational demand patterns, employee scheduling decisions are frequently based on assumptions rather than empirical evidence. This may result in overstaffing during low-demand periods or understaffing during peak business hours. Ineffective workforce allocation can negatively affect labor cost efficiency, employee productivity, and service quality. Additionally, insufficient staffing during high-traffic periods may compromise customer satisfaction and overall service performance. Over time, repeated scheduling inefficiencies may also affect employee morale, productivity consistency, and branch-level operational balance. Management lacks historical traffic pattern analysis needed to establish optimized staffing models for each branch. Branch managers may schedule employees inconsistently due to the absence of objective workload forecasts. This can create disparities in service quality across locations during varying demand periods. Inaccurate staffing forecasts may increase unnecessary overtime expenses or idle labor costs. Without peak-hour analytics, Kape Tubong cannot maximize workforce utilization efficiency across its operations. Staffing inefficiencies may also contribute to employee burnout during unexpectedly busy shifts. Conversely, overstaffed periods may lead to underutilization of employee working hours. Repeated scheduling inaccuracies may reduce operational morale among staff members. Workforce planning challenges may also affect management's ability to assign specialized tasks efficiently. The absence of traffic forecasting limits strategic shift planning for holidays and special events. Management may be unable to prepare for anticipated customer surges during promotional periods.

Lack of staffing analytics can create inconsistent customer experiences across branches. Workforce inefficiencies may negatively impact service speed and transaction processing times. These limitations reduce branch productivity and labor cost effectiveness. Ultimately, inadequate staffing analysis weakens operational planning and customer service optimization. Furthermore, branch supervisors may encounter difficulty balancing workloads fairly among employees due to the absence of accurate operational demand forecasts. Employees assigned to understaffed shifts may experience excessive pressure during peak operating hours, potentially affecting service quality and employee morale. On the other hand, overstaffing during low-demand periods may contribute to unnecessary labor expenditures and inefficient resource utilization. Management may also struggle to evaluate whether current staffing levels directly support branch productivity and customer satisfaction objectives. Without customer traffic analysis, identifying the most productive employee scheduling arrangements may remain difficult. Seasonal fluctuations in customer volume may not be anticipated effectively due to limited forecasting capabilities. Staffing plans for weekends, holidays, and promotional events may therefore become inconsistent and less efficient. Inadequate workforce forecasting may also affect coordination between front-line employees and supervisory personnel during busy operational periods. The organization may experience increased employee fatigue and reduced work performance because staffing schedules are not aligned with actual customer demand patterns. Delayed service and longer customer waiting times may negatively influence customer perceptions of branch efficiency and service quality. Workforce inefficiencies may further contribute to inconsistent operational performance across different branches. Management may also find it difficult to implement standardized staffing policies because workload measurements vary across locations. The absence of workforce analytics can reduce management's ability to optimize labor allocation and improve operational productivity systematically. Ultimately, insufficient staffing analysis weakens employee performance management, labor efficiency, customer satisfaction, and overall operational stability.

- Manual reporting procedures are time-consuming and highly vulnerable to human error, data inconsistencies, and potential data loss. Employees are required to spend substantial time preparing, encoding, validating, and consolidating reports from multiple branches. The lack of standardized reporting formats further complicates the comparison and interpretation of branch-level performance data. Manual computation and repetitive data entry increase the risk of inaccuracies and calculation errors. Furthermore, poor data handling practices may result in misplaced, overwritten, or duplicated records. These limitations reduce the reliability of reports and delay management's access to essential operational information. The continued dependence on manual reporting also diverts employee time away from more value-adding operational responsibilities. Repetitive administrative tasks may reduce staff productivity and create unnecessary operational bottlenecks. Delayed report preparation can postpone management review and decision-making processes. Inconsistent report formatting may create confusion during performance evaluation and comparative analysis. Over time, prolonged reliance on manual reporting may reduce organizational efficiency and hinder operational scalability. Employees may experience increased fatigue due to repetitive clerical responsibilities. Staff attention may be diverted from customer-facing tasks toward administrative paperwork. Manual report handling increases the likelihood of omitted or incomplete entries. Repeated encoding tasks may create duplication of effort across branches. Human-dependent processes make report preparation vulnerable to absenteeism and workload fluctuations. Branch-level reports may be submitted late due to competing operational priorities. Delays in report submission can postpone strategic and operational decisions. Paper-based or spreadsheet-based storage methods increase risk of accidental deletion or physical loss. Manual archiving practices may complicate retrieval of historical reports. Ultimately, manual reporting practices reduce organizational productivity and information reliability. Furthermore, employees may encounter difficulties maintaining reporting accuracy during periods of high operational workload and tight reporting deadlines. Manual verification of reports may consume substantial managerial time that could otherwise be allocated to strategic planning and operational improvement initiatives. The absence of automated validation mechanisms increases the possibility that incorrect figures remain undetected within official reports. Inconsistent spreadsheet formulas and reporting templates may also create discrepancies between branch-level operational records. Retrieval of historical documents may become increasingly difficult as the volume of manually stored files continues to grow. The organization may also face limitations in maintaining secure and organized backup copies of operational reports. Accidental deletion, file corruption, or unauthorized modification of manually stored records may compromise data reliability and organizational accountability. Employees responsible for repetitive reporting tasks may experience reduced motivation and increased work fatigue over time. Manual reporting practices may further slow communication between branch personnel and management because reports must be reviewed and verified individually. Delayed access to operational information can weaken management’s ability to respond immediately to business concerns and performance issues. In addition, dependence on paper-based or spreadsheet-based processes may reduce the organization’s readiness for digital transformation initiatives. Operational bottlenecks associated with repetitive clerical work may negatively affect overall branch productivity and service efficiency. As the business continues to expand, the limitations of manual reporting may become more severe and increasingly difficult to manage. Ultimately, continued reliance on manual reporting systems weakens operational accuracy, reporting efficiency, information security, and organizational responsiveness.

- Current systems do not effectively utilize Business Intelligence (BI) tools, thereby limiting management's ability to make informed, data-driven decisions that could improve operational efficiency and support business growth. Although sales and operational data are consistently collected, such data are not transformed into analytical insights that support forecasting, strategic planning, and performance evaluation. Existing reports remain largely descriptive and provide limited analytical depth. Management lacks predictive and advanced analytical tools necessary for identifying trends, forecasting demand, and evaluating long-term business performance. As a result, strategic decisions are often based on managerial intuition and prior experience rather than empirical analysis. This underutilization of business data restricts opportunities for optimization, innovation, and sustained competitive advantage. It also prevents the organization from maximizing the strategic value of its historical and real-time operational information. Without BI integration, Kape Tubong cannot effectively benchmark branch performance against organizational targets. The absence of forecasting capabilities limits management's preparedness for seasonal demand fluctuations and market changes. Opportunities for predictive inventory planning and proactive promotional strategy development remain underutilized. Ultimately, the lack of Business Intelligence adoption weakens the organization's strategic agility and long-term competitiveness in the F&B market. The organization may also struggle to identify long-term revenue trends without historical analytical dashboards. Strategic planning becomes less precise when management lacks predictive modeling tools. Branch expansion decisions may rely on subjective judgment instead of performance-based analytics. Management may overlook hidden correlations between promotions, pricing, and sales performance. The absence of BI tools limits organizational learning from historical operational data. Future demand forecasting remains less accurate without analytical trend modeling. The business may be slower to adapt to changing market behavior and customer preferences. Competitors utilizing BI technologies may gain operational and strategic advantages. Underutilization of BI weakens data-driven innovation across the organization. Consequently, Kape Tubong may experience reduced competitiveness in an increasingly analytics-driven market. Furthermore, management may encounter challenges in identifying operational inefficiencies because existing systems provide limited analytical visibility into branch activities and performance indicators. The absence of predictive analytics may reduce the organization's ability to anticipate future customer demand and market fluctuations accurately. Decision-making processes may therefore rely heavily on managerial assumptions rather than comprehensive analytical evidence. Without integrated BI dashboards, management may spend excessive time interpreting raw data before arriving at operational conclusions. Opportunities to improve profitability through data-driven process optimization may remain undiscovered due to insufficient analytical capabilities. The organization may also struggle to identify hidden relationships between staffing levels, promotional activities, customer traffic, and sales performance. Inadequate BI utilization may weaken inventory forecasting and increase the likelihood of stock shortages or excess inventory levels. Historical operational data may remain underutilized because existing systems cannot effectively transform information into strategic business intelligence. Management may also experience difficulty evaluating the long-term effectiveness of implemented business strategies and operational policies. The absence of real-time analytical dashboards can delay management's response to emerging operational problems and revenue fluctuations. Furthermore, insufficient BI integration may reduce organizational preparedness for market competition and industry changes. Branch performance benchmarking may remain inconsistent because analytical comparisons are not automatically generated and visualized. The organization may also encounter limitations in conducting accurate forecasting for future expansion initiatives and resource allocation strategies. Competitors that effectively utilize Business Intelligence technologies may gain advantages in operational efficiency, customer understanding, and strategic planning capabilities. Ultimately, inadequate utilization of Business Intelligence tools weakens organizational innovation, forecasting accuracy, strategic adaptability, operational efficiency, and sustainable business growth.

1.3 Objectives of the Study

1.3.1 General Objective

The general objective of this study is to design and develop **BREWINSIGHT BI: Revenue Pattern Analytics System**, a Business Intelligence-based platform intended to centralize, organize, and analyze sales data from all Kape Tubong branches into one integrated digital environment. The proposed system seeks to transform raw transactional and operational records into meaningful and actionable business insights through the use of data visualization, automated reporting, analytical dashboards, and revenue trend analysis. By consolidating sales records from all kiosks and the main branch into a unified centralized database, the platform aims to eliminate fragmented data storage practices, reduce inconsistencies in record management, and provide management with a comprehensive view of business performance across all locations. Through centralized data consolidation, the system will improve data accessibility and ensure that sales information is consistently available in a structured and organized format. This centralized architecture will also reduce the time and effort required to manually collect and reconcile reports from different branches. In doing so, the system will promote better coordination among business locations and improve the overall management of organizational data resources. The platform is intended to provide a scalable and sustainable data management infrastructure that can support future branch expansion and increased transaction volumes. By replacing fragmented reporting processes with an integrated digital platform, the system seeks to modernize Kape Tubong’s data management practices. This objective aligns with the business’s need for a more efficient and centralized operational reporting solution. Ultimately, the system is expected to improve the visibility, accessibility, and strategic value of Kape Tubong’s business data.

Beyond simple consolidation, the proposed platform also seeks to establish a reliable technological environment where data can be processed consistently and securely across all operational units. The integration of branch-level sales records into a centralized analytical platform will improve standardization in data collection and reporting procedures. This standardization is expected to strengthen the consistency of business documentation and reduce reporting discrepancies among branches. By implementing an organized data management structure, Kape Tubong can improve administrative efficiency and ensure better long-term record maintenance. Furthermore, centralized storage of sales data can improve historical recordkeeping and simplify long-term business analysis. This will allow management to evaluate organizational growth over extended periods using structured historical data. The centralized system also improves traceability and accountability in transaction monitoring by maintaining complete and organized digital records. As business operations continue to expand, maintaining such a structured data environment becomes increasingly important for operational sustainability. The proposed system therefore serves not only as an operational tool but also as a long-term data management solution. This strengthens the overall technological readiness of the organization for future business expansion.

The proposed system is specifically designed to analyze revenue patterns by identifying and interpreting sales trends across multiple time intervals, including hourly, daily, weekly, monthly, and yearly periods. Through this functionality, management will be able to monitor revenue fluctuations over time and identify patterns in sales behavior across different operational periods. The system will allow management to detect recurring sales cycles, evaluate seasonal trends, and determine the consistency of branch performance over time. It will also enable the comparison of sales performance between different dates, periods, and branches to identify patterns of growth or decline. These analytical capabilities will help management understand how internal and external factors influence revenue generation. By identifying peak and low-performing periods, the business can make strategic adjustments to staffing, promotions, and operational scheduling. The system will also help determine whether certain branches experience stronger performance during specific periods than others. Through detailed trend analysis, management can forecast future sales expectations based on historical performance data. This predictive understanding can support more informed planning and risk management. Overall, the system aims to improve management's understanding of revenue behavior across all Kape Tubong locations.

Additionally, the platform will support deeper analytical examination of sales fluctuations by allowing management to evaluate the effects of holidays, promotions, weather conditions, and seasonal trends on customer purchasing behavior. This broader analytical perspective enables more comprehensive interpretation of performance trends beyond simple sales totals. Management can also use revenue pattern data to compare expected versus actual sales performance during planned campaigns and strategic initiatives. These insights may improve the organization's ability to evaluate operational effectiveness and refine future business strategies. Historical trend analysis may also support forecasting for new branch openings by using comparable data from existing locations. Through detailed revenue insights, management can establish more realistic performance benchmarks and expectations for each branch. Such forecasting capabilities contribute to better financial planning and resource management. The system's ability to reveal underlying business patterns strengthens strategic decision-making by providing evidence-based revenue analysis. This enhances management's confidence in planning operational and financial strategies. Therefore, revenue analysis becomes a core strategic function of the proposed platform.

Another major objective of the study is to improve the efficiency and accuracy of handling large volumes of sales transactions generated daily by multiple Kape Tubong branches. At present, manual data consolidation and fragmented reporting create delays and increase the likelihood of errors in sales analysis and reporting. Through automated data aggregation and processing, the BREWINSIGHT BI platform seeks to streamline the reporting workflow and reduce dependence on manual compilation of branch data. The platform will automatically collect, process, and summarize sales information into usable analytical reports and dashboard visualizations. This automation will reduce the workload of management and administrative staff while improving reporting speed and consistency. It will also minimize the occurrence of human error commonly associated with spreadsheet-based manual reporting methods. Faster and more accurate reporting will allow management to respond more quickly to operational issues and performance concerns. The system will provide updated and readily available business metrics whenever needed. This improvement in efficiency supports better operational control and faster access to performance information. Therefore, the study aims to enhance the speed, reliability, and accuracy of Kape Tubong's sales reporting processes.

The automation of data processing is also expected to improve workflow efficiency across branches by reducing repetitive clerical tasks related to report preparation and submission. Employees will spend less time compiling transaction summaries and more time focusing on operational and customer service responsibilities. Standardized automated reports will improve consistency in business documentation and facilitate easier comparison across branches. By reducing reporting delays, the system enhances the timeliness of management review and intervention. It also decreases administrative bottlenecks caused by branch-level reporting dependencies. In addition, automated processing supports scalability by enabling the system to handle increasing transaction volumes without significantly increasing administrative workload. This allows Kape Tubong to maintain efficient reporting processes even as more branches are established. Improved processing speed also strengthens management's ability to conduct frequent performance reviews. Overall, automation enhances both operational productivity and managerial efficiency. Thus, the platform contributes significantly to process optimization.

In addition, the system aims to support detailed monitoring of key business performance indicators that are critical for operational and strategic decision-making. These include branch sales performance, revenue comparisons, product demand trends, top-selling items, underperforming products, customer purchasing behavior, and time-based sales patterns. By making these metrics easily accessible through the dashboard, the platform will help management monitor operational performance more effectively. Management will be able to evaluate which branches contribute the highest revenue and which locations require performance improvement. The system will also identify which products consistently perform well and which items may need reevaluation or promotional support. Product demand analysis will assist management in making informed decisions regarding menu optimization and pricing strategy. The system can also support the identification of consumer purchasing preferences based on historical sales data. These insights can guide targeted marketing campaigns and promotional planning. By monitoring multiple business indicators in one platform, the system will provide a holistic view of business performance. Therefore, the study seeks to improve management's ability to monitor and interpret critical operational metrics.

The proposed platform also aims to promote a more structured and data-driven approach to management and operational evaluation within Kape Tubong. Rather than relying on intuition, assumptions, or delayed manual reports, management will be able to base decisions on real-time and evidence-based analytical information. The use of dashboards, visual summaries, and automated analytics will simplify the interpretation of complex sales data and improve decision-making efficiency. Through clearer visibility into business performance, management can identify performance gaps, operational inefficiencies, and revenue issues more quickly. The system will support faster managerial response to underperforming branches, declining sales trends, and product-related concerns. It will also improve accountability by providing measurable and objective data for evaluating operational outcomes. Data-driven decision-making will enable management to justify strategic actions using factual business evidence. This shift toward analytical management can improve planning accuracy and strengthen business oversight. It also promotes a culture of evidence-based management within the organization. Thus, the study aims to strengthen Kape Tubong's managerial decision-making capabilities through business intelligence.

Specifically, the system seeks to assist Kape Tubong's owner and management in understanding branch-level and product-level revenue behavior across all business locations. It will help identify top-selling and slow-moving products, determine customer demand patterns, and recognize peak operating hours throughout the day. These insights will enable the business to improve inventory management by aligning stock levels with product demand patterns. The platform can also support more effective workforce scheduling by identifying periods of high and low customer activity. This allows management to allocate staff more efficiently according to actual demand. Promotional activities can also be better planned by identifying weak sales periods that may require targeted campaigns. Resource allocation decisions, such as equipment deployment and inventory distribution, can be optimized using branch-specific performance data. These operational improvements can lead to reduced waste, improved service quality, and increased profitability. The system therefore supports both strategic and day-to-day operational decision-making. As a result, the platform is expected to improve the practical management of Kape Tubong's operations.

Additionally, the proposed system aims to establish a technological foundation that supports future digital transformation initiatives within Kape Tubong. As the business continues to grow, the need for scalable digital systems becomes increasingly important for sustaining efficient operations. BREWINSIGHT BI is designed not only to address present operational challenges but also to provide a framework that can accommodate future enhancements and technological integrations. These may include advanced forecasting models, inventory analytics, customer relationship management integration, and predictive sales analysis. The platform's scalable architecture allows it to adapt to the increasing complexity of operations as additional branches or kiosks are established.

This forward-looking design ensures that the system remains relevant and valuable even as organizational needs evolve. It also reduces the likelihood of requiring a complete system replacement in the near future. By investing in a scalable and adaptable platform, Kape Tubong can better prepare for future business expansion and digital modernization. This objective supports long-term organizational sustainability and technological readiness. Therefore, the system is intended to serve as both an immediate operational tool and a strategic technological asset for the business.

Ultimately, the implementation of BREWINSIGHT BI is expected to improve operational transparency, strengthen management oversight, enhance reporting accuracy, and promote data-driven decision-making across the organization. By providing centralized and real-time access to business intelligence, the system will improve the transparency of operational performance across all Kape Tubong branches. Management will gain greater visibility into branch activities, sales outcomes, and performance indicators without requiring manual report consolidation. The platform will strengthen internal control by enabling faster detection of performance anomalies and inconsistencies. Enhanced reporting accuracy will improve the reliability of business information used for planning and strategic analysis. The system is also expected to reduce delays in information delivery, allowing management to respond more proactively to business conditions. Beyond immediate operational benefits, the study also aims to demonstrate the practical value of Business Intelligence technologies in supporting small and medium-sized enterprises within the Food and Beverage industry. It seeks to contribute to the digital transformation of Kape Tubong by modernizing its sales monitoring and analytics processes. In the long term, the platform is expected to support sustainable business growth, increased competitiveness, and future scalability as the business expands. Therefore, this study aims to provide both practical organizational benefits and broader technological contributions through the development of the BREWINSIGHT BI platform.

Furthermore, the proposed BREWINSIGHT BI platform aims to improve organizational responsiveness by enabling management to identify and address operational concerns in a more timely and systematic manner. Through real-time access to updated performance metrics, management can immediately detect declining sales, unusual transaction patterns, or branch-level inconsistencies that may indicate emerging operational issues. This allows decision-makers to implement corrective actions before minor concerns escalate into larger organizational problems. Early detection of performance issues strengthens preventive management practices and supports more proactive business oversight. The availability of immediate operational insights can also reduce response delays commonly associated with manual reporting and delayed performance evaluation. By improving response time, the system enhances the organization’s capacity to maintain service quality and operational consistency across all business locations. This responsiveness is particularly important in the highly competitive and dynamic food and beverage industry, where customer demand and operational conditions can change rapidly. The system's real-time monitoring capabilities therefore support more agile and adaptive business management. This contributes to improved organizational resilience in addressing operational and market-related challenges. Consequently, the study aims to strengthen Kape Tubong's ability to respond efficiently to changing business conditions.

Moreover, the study seeks to improve communication and coordination between branch-level personnel and upper management through a shared and standardized reporting environment. By centralizing all business information within a unified dashboard platform, the system will provide both management and authorized staff with consistent access to the same performance data and operational records. This shared visibility reduces communication gaps caused by inconsistent reporting practices or delayed transmission of branch information. It also promotes greater alignment between branch-level execution and organizational strategic objectives. Standardized reporting formats can improve clarity in performance discussions and reduce misunderstandings related to inconsistent data presentation. The platform may also enhance accountability by making branch-level performance more transparent and measurable. Through improved communication and information accessibility, the system supports more collaborative and coordinated operational management. Branch managers and staff can better understand performance expectations through access to measurable operational indicators. This promotes stronger alignment between individual branch operations and broader business goals. Therefore, the study aims to improve organizational coordination through centralized information sharing.

Additionally, the proposed system intends to support performance benchmarking and comparative analysis across Kape Tubong's multiple branches. Through branch comparison features and standardized performance metrics, management can evaluate which branches perform above or below organizational expectations. This comparative analysis will help identify operational best practices that can be replicated across locations. It will also reveal branch-specific weaknesses requiring targeted managerial intervention or strategic adjustment. Benchmarking branch performance using objective metrics supports fairer and more accurate evaluation of branch productivity. The organization can establish performance standards and monitor whether each branch meets established operational expectations. This feature may also improve motivation and accountability among branch managers by providing measurable performance benchmarks. Branch-level comparisons can further support strategic decisions regarding expansion, restructuring, or operational resource allocation. Through data-based benchmarking, management can continuously improve operational consistency across all branches. Therefore, the study aims to enhance branch performance management through comparative operational analytics.

The study also aims to support financial planning and budgeting processes by providing management with reliable historical sales data and revenue trend analysis. Accurate and organized financial information is essential for preparing realistic budgets, forecasting expected revenue, and planning future investments. Through structured revenue analysis, the system can help management estimate future income based on historical business performance. This capability improves budgeting accuracy and supports more strategic allocation of organizational resources. Financial planning can be further strengthened by identifying seasonal sales fluctuations and predictable revenue cycles. Management can use this information to prepare more accurate expenditure plans for labor, inventory, marketing, and expansion initiatives. Improved budgeting can reduce unnecessary spending and support stronger financial discipline within the organization. Access to organized financial data also strengthens the organization's ability to evaluate investment opportunities and expansion feasibility. This makes the system valuable not only for operational monitoring but also for long-term financial strategy. Thus, the study aims to improve Kape Tubong's financial planning capabilities through structured analytical reporting.

Finally, the study seeks to demonstrate the broader applicability of Business Intelligence technologies in small and medium-sized enterprises within the Philippine food and beverage sector. By successfully implementing BREWINSIGHT BI in Kape Tubong's operations, the project aims to provide a practical example of how localized SMEs can adopt modern data analytics and digital reporting technologies despite resource limitations. This contributes to the growing body of evidence supporting the feasibility of digital transformation among developing local enterprises. The study may encourage similar organizations to adopt business intelligence systems for improving operational monitoring and decision-making. It also highlights the potential of customized web-based analytical systems in addressing practical business challenges faced by SMEs. Through this implementation, the project demonstrates that advanced analytical technologies are not limited to large corporations but can also provide significant value to smaller growing businesses. This expands awareness of Business Intelligence as a viable operational strategy for SMEs. The study therefore contributes not only to Kape Tubong's organizational improvement but also to broader discussions on SME digital modernization. Its findings may serve as reference material for future business technology initiatives in similar industries. Consequently, the study aims to contribute to both practical business innovation and broader technological adoption within local SMEs.

1.3.2 Specific Objectives

Specifically, the study aims to:

- Create a centralized web-based dashboard that integrates and organizes sales information from all kiosks and the main branch into a single platform to improve business monitoring and performance management.The dashboard will function as a unified interface for accessing and managing all sales transactions across branches. It will simplify the tracking of transactions and minimize the difficulty of handling data from multiple separate locations. The platform will enhance management's access to business information by enabling centralized and real-time monitoring of branch performance. It will also improve operational visibility, allowing management to quickly identify irregular sales activities, performance issues, and operational concerns. Moreover, the dashboard will strengthen coordination between branches and management through standardized presentation and organization of sales records. By consolidating all transaction data into one accessible platform, the system will reduce dependence on manual branch-by-branch report collection. Management will no longer need to review fragmented spreadsheets or separate records from individual kiosks. This centralized approach will improve consistency in business reporting and ensure that performance data is uniformly structured across all locations. It will also allow management to compare branch performance more efficiently and identify performance gaps between locations. Ultimately, the dashboard will provide Kape Tubong with a more organized, transparent, and efficient monitoring environment for managing business operations.

  Additionally, the dashboard will improve enterprise-wide visibility by presenting branch performance in a single comparative environment. Management will be able to observe trends and anomalies across all kiosks simultaneously. This will support quicker identification of underperforming branches and operational inconsistencies. The centralized structure will also reduce communication delays between branch personnel and administrators. Historical branch data may be reviewed through the platform to support long-term performance evaluations. Dashboard accessibility through web browsers will improve convenience for authorized users regardless of location. This feature will enable management to oversee operations remotely without requiring physical presence at each branch. Consolidated visual reporting will simplify executive review of operational performance during meetings and strategic planning sessions. The platform will also establish a foundation for future expansion by supporting scalable integration of additional branches or kiosks. Therefore, the centralized dashboard will significantly improve the organization's capacity for efficient multi-branch business management.
- Automate the generation of daily, weekly, and monthly sales reports to reduce manual processing, lessen human error, and improve the speed and accuracy of reporting activities.The system will automatically compile transaction records and produce reports according to predefined reporting schedules. This automation will

reduce the administrative burden associated with manual data encoding and report consolidation. It will also minimize errors caused by repetitive calculations, inconsistent formatting, and incomplete reporting. In addition, automated reporting will improve accessibility to historical sales records for future review and analysis. Standardized report outputs will further enhance consistency and reliability in evaluating business performance across branches. The automated generation of reports will ensure that management receives timely access to updated business information without delays caused by manual preparation. It will also reduce the time spent by staff in preparing repetitive sales summaries. Historical reports can be stored systematically in the system for future retrieval and comparative analysis. This feature will support long-term trend evaluation and performance tracking over time. Overall, automated reporting will improve the efficiency, reliability, and timeliness of Kape Tubong's reporting processes.

Furthermore, automation will ensure that report generation follows consistent formatting and computational standards across all reporting periods. This will improve comparability between daily, weekly, and monthly performance reports. The system will reduce dependency on staff availability for report preparation, minimizing disruptions caused by absenteeism or workload constraints. It will also support more immediate management review of branch performance after each reporting cycle. Automatically archived reports will improve documentation and audit readiness for financial review purposes. Management can retrieve previous reports more efficiently without manually searching through physical or digital files. Automated timestamps and report logs will improve accountability and traceability in reporting activities. The reduction in repetitive administrative tasks may improve staff productivity and operational focus. Additionally, faster report availability can enhance the speed of managerial response to emerging performance issues. Thus, report automation will strengthen both operational efficiency and analytical responsiveness.

- Process and visualize revenue patterns, including peak operating periods, customer purchasing trends, and top-selling products, to support data-driven and strategic decision-making.The system will analyze transactional sales data and convert it into graphical and summarized analytical reports that illustrate business performance over time. It will identify recurring peak sales periods and reveal customer purchasing behavior patterns across locations. The platform will help management determine which products generate the highest revenue and which items require strategic attention. It will also assist in observing fluctuations in sales over various periods to support forecasting and trend analysis. These analytical insights will contribute to more informed planning, promotional decisions, and operational strategy development. Through visual charts, graphs, and dashboard indicators, the system will simplify the interpretation of complex sales information. Management can use these insights to identify profitable periods and underperforming intervals across different branches. The analysis of customer purchasing trends will help determine shifts in consumer preferences over time. Revenue pattern visualization will also support more accurate forecasting of future sales expectations. Overall, the platform will provide valuable analytical intelligence to strengthen business planning and strategic management.

Additionally, revenue analysis will help management detect seasonal or cyclical sales trends that influence business performance. The system can highlight unusual spikes or declines in revenue that may indicate operational concerns or market changes. Comparative visualization across time periods will support better understanding of business growth patterns. Product performance analysis can guide pricing adjustments and promotional planning based on actual demand behavior. The platform may also assist in evaluating the financial impact of marketing campaigns or operational changes. Branch-specific trend analysis will allow management to compare customer behavior across locations. This can support more localized and targeted operational strategies. The availability of visual analytics will reduce the difficulty of interpreting large transaction datasets manually. Strategic decisions can therefore be based on measurable and evidence-backed performance indicators. As a result, the system will improve the quality and confidence of managerial decision-making.

- Provide inventory-related analytical insights based on sales performance data to support effective stock management and reduce inventory inefficiencies.By evaluating product demand and sales movement, the system will help management maintain appropriate inventory levels based on actual consumption trends. It will identify fast-moving and slow-moving items to improve replenishment planning and purchasing decisions. The platform will also help reduce overstocking and stock shortages, minimizing product waste and preventing missed sales opportunities. Furthermore, it will contribute to improved inventory turnover and more efficient utilization of stock resources across all business locations. Inventory-related analytics will enable management to align purchasing decisions with actual product demand patterns rather than assumptions. This can help prevent unnecessary inventory expenses and reduce losses associated with expired or unsold products. The system can also support branch-specific stock planning by identifying location-based differences in product demand. Improved inventory planning may contribute to better product availability and customer satisfaction. By integrating inventory-related insights with sales analytics, the platform will support more efficient operational planning. Therefore, this feature will strengthen Kape Tubong's inventory management practices and resource allocation efficiency.

  Moreover, the system can assist in identifying recurring stock consumption trends for frequently purchased items. This information can support more accurate procurement scheduling and supplier coordination. Inventory insights may help management prepare for anticipated demand surges during peak periods or promotional events. Branch-level inventory analysis can reveal location-specific stock requirements and customer preferences. This will improve distribution planning across multiple kiosks. Better stock forecasting can reduce emergency purchasing and associated supply chain inefficiencies. Inventory waste due to spoilage or expiration may be reduced through more accurate replenishment practices. The integration of sales and inventory analytics will strengthen the relationship between demand forecasting and stock planning. Improved stock management can contribute directly to profitability by lowering unnecessary inventory costs. Thus, inventory-related analytics will enhance both financial efficiency and operational preparedness.

- Establish secure authentication and controlled user access to protect system data while allowing authorized personnel to access reports remotely.The system will implement login authentication and role-based access control features to ensure that only authorized users can utilize system functions and access business information. Different access privileges will be assigned based on user roles and responsibilities within the organization. These security mechanisms will help safeguard sensitive sales and operational data against unauthorized access, misuse, and potential data breaches. Additionally, the web-based platform will allow secure remote access to dashboards and reports through internet-connected devices, providing convenience and flexibility for management while maintaining data protection standards. Password protection and session management features will further strengthen account security and user authentication processes. Administrative users will have broader access privileges compared to regular staff users to maintain proper control over system functions. Access restrictions will prevent unauthorized modification or deletion of critical business records. Secure remote access will allow management to monitor business performance even when off-site or away from branch locations. These security measures help maintain the confidentiality, integrity, and availability of business information. Overall, this objective ensures that the system remains secure, controlled, and accessible only to appropriate personnel.

  In addition, user activity logs may be implemented to record login attempts and significant system actions for accountability purposes. This will improve monitoring of user behavior and help detect suspicious activity within the platform. Role-based permissions will ensure that users only access information relevant to their responsibilities. Session timeout mechanisms may reduce the risk of unauthorized access due to unattended devices. Encrypted data transmission can further strengthen protection during remote access sessions. Secure authentication protocols will improve trust in the system’s handling of sensitive operational records. Controlled access structures will also help preserve data integrity by limiting editing rights to authorized administrators. These measures are particularly important because the system stores confidential financial and sales-related information. Effective security implementation will support compliance with good information management practices. Therefore, robust access control and authentication mechanisms are essential to maintaining a secure and dependable Business Intelligence environment

1.4 Scope and Delimitation

This study focuses on the development of **BREWINSIGHT BI: Revenue Pattern Analytics System**, a Business Intelligence-based platform created to gather, centralize, organize, and evaluate sales data from Kape Tubong's kiosks and main branch. The system is designed to identify revenue patterns by processing transactional sales records and transforming them into meaningful business insights. Through this platform, management will gain more accurate and accessible information to support operational planning and decision-making. The study specifically aims to improve how sales data is managed, interpreted, and utilized by implementing a centralized analytics and reporting solution. By replacing fragmented manual monitoring methods, the proposed platform seeks to modernize the way Kape Tubong handles performance analysis and reporting. It also aims to establish a more organized framework for storing and retrieving historical sales information. Through digital centralization, the system is expected to improve data consistency and reduce reporting inefficiencies across all branches. The platform is intended to serve as a practical analytical tool for supporting daily and strategic business decisions. Additionally, the study focuses on demonstrating how Business Intelligence technologies can be applied effectively in small and medium-sized food and beverage enterprises. Therefore, the scope of the study centers on the development of a specialized analytics platform tailored to the operational needs of Kape Tubong.

The proposed system will function as a web-based platform that integrates sales records from multiple branches into a unified database. It will feature modules for data gathering, transaction recording, product performance tracking, automated report generation, and dashboard visualization. These components are intended to provide management with a centralized platform for monitoring sales activities and evaluating branch performance. The dashboard will present key business metrics using charts, graphs, summary panels, and comparative reports to improve data readability and interpretation. This will enable management to monitor trends, assess branch productivity, and identify operational issues more effectively. The web-based nature of the platform allows users to access the system through internet-enabled devices without requiring dedicated software installation. This improves flexibility and accessibility for management when reviewing business performance remotely. The centralized database structure also ensures that sales information from all branches is synchronized and stored in a consistent format. By combining multiple monitoring functions into one platform, the system simplifies data access and improves operational oversight. As such, the study includes the design and implementation of both the technical and visual components of the web-based platform.

Additionally, the system will provide branch-level and overall business performance summaries to support comparative analysis among locations. Through these comparisons, management will be able to identify high-performing and underperforming branches and evaluate the operational factors influencing branch outcomes. This feature is expected to assist in performance assessment and strategic planning for future branch improvements or expansion. Comparative branch analytics will also help management identify location-specific strengths and weaknesses in business operations. Performance benchmarking across branches may reveal operational practices that can be replicated in underperforming locations. This functionality strengthens management's ability to make location-based strategic decisions supported by actual performance data. It also promotes accountability among branch managers by providing measurable performance indicators. Branch comparison reports can assist in evaluating the effectiveness of branch-level promotions and operational strategies. Therefore, the study includes branch comparison analytics as an essential component of the system's functionality.

The analytics component of the system will evaluate sales transactions using variables such as transaction date and time, branch location, product category, and quantity sold. From these inputs, the system will generate insights related to revenue trends, branch revenue comparisons, peak sales periods, customer purchasing behavior, and top- and low-performing products. These analytical outputs are intended to help management understand business performance more clearly and support evidence-based operational decisions. The analysis will focus on identifying descriptive patterns and trends based on historical and current transaction data. It will allow management to recognize recurring business behaviors and operational performance trends over time. The system will also support the filtering of data across various categories and time periods for deeper analysis. Through structured analytics, management can examine business performance from multiple perspectives. This contributes to more informed and data-supported decision-making processes. Thus, the scope includes descriptive analytical processing of transactional sales information using predefined business metrics.

Moreover, the platform will include revenue visualization tools that allow management to review sales fluctuations across hourly, daily, weekly, and monthly periods. This feature will assist in identifying recurring demand patterns, seasonal trends, and irregular sales behavior that may affect business planning. By understanding these trends, management can improve decisions related to promotions, staffing schedules, operating hours, and resource distribution. Revenue visualizations will be displayed through graphs, trend lines, charts, and summarized dashboard indicators. These visual tools are intended to simplify the interpretation of numerical sales data and improve analytical clarity. Historical trend analysis can also help management assess the effectiveness of past operational and promotional strategies. This visualization capability strengthens the practical value of the platform as a decision-support tool. Therefore, the study includes the implementation of visual analytical dashboards as a major component of the developed system.

The system will also generate inventory-related insights by evaluating product sales movement to support stock planning and demand analysis. Through the identification of high-demand and low-demand products, management can make more informed purchasing and stocking decisions. Although the system will not automate stock replenishment, it will provide analytical support for improving inventory planning and reducing inefficiencies related to overstocking or shortages. Inventory-related analytics will be based solely on historical sales movement and product demand trends. These insights are intended to improve stock planning accuracy and support more efficient purchasing decisions. Branch-specific demand analysis may also help optimize stock distribution among locations. However, the system will function only as an analytical support tool rather than an inventory management automation platform. Thus, the study includes inventory-related analytics strictly for decision-support purposes.

To maintain data security and system reliability, the platform will implement authentication and role-based access control to ensure that only authorized personnel can access system functions and reports. It will be accessible through desktop and mobile web browsers, allowing management to monitor business performance remotely when necessary. This accessibility improves convenience while ensuring timely availability of important business information. Furthermore, the centralized storage of branch data will improve consistency, reliability, and standardization of records across all locations. User roles will define varying access privileges depending on the level of authorization assigned to each user. These controls help maintain the confidentiality and integrity of business-sensitive sales information. Secure remote access also allows management to review performance metrics outside physical branch locations. Therefore, the study includes the implementation of security and accessibility mechanisms necessary for practical deployment.

The scope of the study is limited to the analysis of internal sales and transaction data generated from Kape Tubong's daily business operations. External variables such as competitor pricing, consumer demographics, weather conditions, market trends, and economic factors will not be included in the system's analytical processing. Therefore, all revenue analysis produced by the platform will be based solely on internal transactional business data. This limitation is established to maintain focus on operational sales performance analysis within the organization. The exclusion of external variables means that analytical outputs are intended to reflect only internal business performance patterns. Consequently, the system will not account for market-wide or environmental influences on revenue trends. This delimitation ensures manageable system scope and development complexity.

However, the study has several limitations to ensure that the project remains focused and manageable. The system will not involve the design or integration of physical Point of Sale (POS) hardware including receipt printers, barcode scanners, card readers, or cash drawers. It will only process digital sales data entered or provided through available transaction records. Likewise, the project excludes automated stock replenishment, supplier management, procurement functions, and warehouse logistics operations. The platform is not intended to replace existing transactional hardware infrastructure or supply chain systems. Its role is limited to analytical processing and business intelligence support. These exclusions ensure that the study remains centered on analytics rather than full operational automation.

The proposed platform also does not cover broader enterprise-level business functions such as payroll administration, taxation, budgeting, accounting systems, or financial statement preparation. It is not intended to operate as a full Enterprise Resource Planning (ERP) system. Furthermore, integration with banking services, online payment platforms, customer loyalty systems, and third-party accounting software is outside the scope of this study. These exclusions are necessary to prevent overlap with unrelated enterprise functions beyond the purpose of the proposed system. The focus of the platform remains on sales analytics and business intelligence only. Therefore, enterprise-wide financial and administrative system functionalities are beyond the project's intended coverage.

Advanced predictive analytics, machine learning models, and artificial intelligence-based forecasting mechanisms are likewise excluded from the implementation of the project. The study focuses only on descriptive and trend-based analytics intended to provide historical and current performance insights rather than predictive or prescriptive recommendations. Forecasting and predictive modeling are excluded to maintain alignment with the project's current technical scope and development timeframe. The implemented analytics will therefore focus on descriptive business intelligence rather than advanced computational forecasting. Future enhancements may integrate predictive technologies if required by the organization.

Overall, BREWINSIGHT BI is intended to serve as a software-based analytics support platform that enhances sales data interpretation, improves reporting efficiency, and strengthens management decision-making through centralized Business Intelligence capabilities. While the system is developed primarily for academic purposes, it is also expected to provide practical benefits to Kape Tubong by improving sales data management and promoting data-driven operational practices. In addition, the project may serve as a foundation for future enhancements involving predictive analytics, inventory automation, customer behavior analysis, and broader business system integration. The study therefore establishes an initial framework for future digital transformation initiatives within the organization. As business requirements evolve, the developed platform may be expanded to incorporate more advanced business intelligence and operational management features. Ultimately, the scope and delimitations of this study ensure that the project remains focused on delivering a practical, manageable, and relevant analytics solution tailored to Kape Tubong's present operational needs.

Furthermore, the study encompasses the design of a structured database architecture capable of storing, organizing, and managing historical transactional information from all Kape Tubong branches in a centralized repository. This database architecture is intended to support efficient data retrieval, long-term storage, and consistent organization of sales records across multiple reporting periods. By maintaining a structured historical dataset, the system enables management to conduct retrospective performance evaluations and trend comparisons over extended periods of operation. The centralized database also improves data integrity by reducing redundancy, inconsistency, and fragmentation associated with decentralized spreadsheet-based recordkeeping. Historical data retention supports the long-term analytical value of the platform by enabling comparison of current performance against previous operational periods. The structured storage of sales data further facilitates future enhancements should the organization choose to integrate advanced analytics or predictive modules later. Additionally, the database is designed to support scalability as transaction volumes increase with business expansion. Proper database normalization and organization will improve system performance and maintainability over time. This ensures that the platform remains technically sustainable as Kape Tubong's operational complexity increases. Therefore, the study includes the development of a scalable and structured backend database environment as part of the system's scope.

The study also includes the implementation of user interface components designed to improve usability, accessibility, and interpretation of analytical outputs for non-technical users. Since business owners and managers may not possess specialized technical knowledge in data analysis, the system emphasizes user-friendly dashboard layouts and intuitive navigation structures. Graphical reports and summary panels are designed to simplify understanding of complex business metrics through visual representation. This user-centered design approach ensures that analytical outputs remain practical and understandable for operational decision-making. Interface components will be optimized for readability and efficient access to key performance information. Dashboard layouts will be structured to prioritize the most critical business indicators for immediate visibility. User interface responsiveness will also be considered to support accessibility across various screen sizes and devices. This improves the practicality of remote monitoring using laptops, tablets, and mobile devices. By prioritizing usability, the study ensures that the system remains accessible to intended users regardless of technical expertise. Therefore, the scope includes front-end design considerations that support practical adoption and ease of use.

Moreover, the study covers the implementation of report archival and historical retrieval functionalities to support long-term performance review and documentation. Generated reports will be stored within the system to allow management to retrieve past analytical outputs for comparison and strategic review. This archival capability strengthens long-term performance tracking by enabling management to evaluate business progress across different operational periods. Historical reporting can also support auditing, internal review, and retrospective performance assessment. The ability to retrieve archived reports eliminates the need for separate manual storage of previous sales summaries and analytical records. This improves organizational recordkeeping and supports more systematic business documentation practices. Historical report retrieval also enhances strategic planning by enabling comparative analysis of previous and current operational outcomes. Stored reports may be used to evaluate long-term effects of pricing changes, branch expansion, and promotional campaigns. This functionality contributes to stronger business continuity and institutional knowledge retention. Therefore, report archival and retrieval are included within the system's functional scope.

Additionally, the delimitations of the study specify that the system will only process structured sales data provided by Kape Tubong and assumes the availability of accurate transactional input from branch operations. The effectiveness of the system's analytics depends heavily on the completeness, consistency, and correctness of the sales data entered into or imported by the platform. Inaccurate or incomplete transaction records may affect the reliability of analytical outputs generated by the system. As such, the study does not include mechanisms for independently validating the business accuracy of manually entered transactional records beyond standard system input controls. Data governance and transactional accuracy remain dependent on branch-level operational compliance and user discipline. The system assumes that users will follow standardized procedures in recording or importing sales transactions. This limitation is necessary because the project focuses on analytical processing rather than transactional auditing or verification. Consequently, the reliability of system outputs remains partially dependent on the quality of source data provided. This delimitation ensures realistic expectations regarding the scope of automated analytical accuracy. Therefore, data input accuracy remains an operational responsibility outside the system's analytical scope.

The study is further limited to the operational environment, workflow structure, and reporting requirements specific to Kape Tubong at the time of system development. As such, the design and functionality of BREWINSIGHT BI are tailored specifically to the organization's current business model, reporting practices, and analytical needs. While the platform may conceptually be adaptable to other businesses, customization may be required before implementation in different organizational contexts. Variations in operational workflow, reporting requirements, product categories, and branch structures may necessitate modifications for use outside Kape Tubong. The system is therefore not intended to function as a universally deployable business intelligence solution without customization. This delimitation maintains alignment between system design and the specific operational requirements of the target organization. It also ensures that the project remains focused on solving the identified needs of Kape Tubong rather than attempting to address generalized business scenarios. Customization for other industries or organizations falls outside the scope of the present study. Thus, the developed platform is organization-specific in design and implementation. This ensures the system remains practical and highly relevant to its intended deployment environment.

Finally, while the study aims to deliver a functional and practical Business Intelligence platform, system performance evaluation will be limited to testing within the context of the developed prototype and available operational dataset during the implementation period. Long-term real-world operational performance beyond the study period is not covered within the project's formal evaluation scope. Factors such as long-term user adoption, organizational change management, evolving business requirements, and future infrastructure demands are beyond the timeframe of this research. The study focuses primarily on design, development, implementation, and initial functional evaluation of the proposed platform. Extended production-level deployment considerations may require future organizational assessment after full implementation. Despite these limitations, the study provides a practical foundation for future operational deployment and enhancement. The developed prototype demonstrates the technical feasibility and organizational applicability of centralized Business Intelligence for Kape Tubong. It also establishes a baseline model for future expansion and refinement of the platform. Therefore, while long-term deployment outcomes are outside the immediate scope, the study provides a strong implementation framework for future operational adoption. This reinforces the project’s practical and academic value within its defined scope.

1.5 Significance of the Study

The proposed system will benefit the following:

● **Management –** The proposed system will significantly assist the management of Kape Tubong by providing centralized, accurate, and real-time access to sales and performance data from all branches. This capability will enable management to monitor branch operations more efficiently, evaluate revenue performance comprehensively, and identify sales trends with greater precision. Through analytical dashboards, automated reporting, and visual performance summaries, management can make more informed decisions regarding pricing strategies, promotional planning, workforce allocation, product offerings, and branch expansion initiatives. Furthermore, the system will reduce management’s reliance on delayed manual reports and subjective judgment by providing timely and evidence-based operational insights. This improved access to reliable business intelligence will support proactive decision-making, strengthen strategic planning, and enhance the organization’s responsiveness to operational concerns and market fluctuations. As a result, management can exercise greater control over business performance and improve long-term planning capabilities.The platform will also help management identify operational inefficiencies and performance gaps more quickly across branches. By detecting revenue declines or unusual sales patterns early, management can implement corrective actions before issues worsen. The system can further support comparative evaluation of branch-level initiatives and promotional campaigns. Overall, the platform strengthens managerial oversight and strategic business control.Additionally, the system enables management to evaluate business performance using measurable and standardized performance indicators rather than subjective assessments. It can support executive meetings and strategic reviews by providing readily available reports and visual analytics. Management may use the system to assess branch profitability before making investment or closure decisions. The platform also assists in validating whether business strategies are producing desired outcomes based on actual sales results. Through historical performance tracking, management can establish realistic operational benchmarks and revenue targets for each branch. The system may further improve managerial accountability by documenting the basis of strategic decisions using factual business evidence. It can strengthen long-term forecasting by allowing management to analyze historical growth patterns and recurring demand cycles. Management can also use the platform to determine which branches require operational intervention or additional support. Better analytical visibility may improve confidence in making high-level organizational decisions.Moreover, the platform can help management identify branch-specific operational patterns that may not be visible through manual reports alone. It may reveal location-based differences in consumer demand, purchasing preferences, and revenue contribution across branches. These insights can support more targeted strategic planning for each kiosk or branch location. Management may also use the system to evaluate the impact of new products, pricing adjustments, or promotional strategies based on measurable sales outcomes. The platform can improve strategic resource distribution by identifying which branches require additional manpower, marketing support, or inventory allocation. It may also assist in determining the most profitable operational hours for each location, enabling more efficient scheduling decisions. Through data-backed performance evaluations, management can improve fairness and consistency in assessing branch managers and staff performance. The system may further support strategic planning by helping management identify long-term business growth trends and recurring revenue patterns. Historical data insights can be used to justify future investments, equipment purchases, or expansion initiatives with greater confidence. By consolidating all performance metrics into one platform, the system simplifies strategic review and reduces the complexity of interpreting multiple reports from different sources. It also enables management to maintain better oversight of organizational performance despite branch expansion and operational growth. These capabilities can improve managerial responsiveness and reduce delays in strategic decision-making. The platform may also strengthen executive confidence by providing more transparent and objective performance information. Ultimately, the proposed system empowers management to govern operations more efficiently, strategically, and proactively. Therefore, the system provides significant operational, analytical, and strategic value to Kape Tubong's management team.

- **Staff –**The implementation of the proposed system will benefit staff members by reducing the administrative burden associated with manual data encoding, report preparation, and sales consolidation. Automated reporting and centralized data processing will streamline repetitive clerical tasks, allowing employees to focus more on customer service and core operational responsibilities. In addition, the system will promote standardized reporting practices across all branches, reducing inconsistencies in data recording and improving coordination between staff and management. By minimizing manual intervention in data handling processes, the system can reduce the likelihood of human error, improve task efficiency, and contribute to a more organized and productive working environment. This may also help reduce employee workload stress and improve day-to-day operational workflow.Staff will spend less time compiling reports and more time focusing on branch operations and service quality. The simplified reporting process may also reduce confusion caused by inconsistent documentation procedures. Better organization of operational data can improve communication between branch personnel and management. Employees may experience improved productivity due to streamlined reporting responsibilities. Therefore, the system is expected to contribute positively to staff efficiency and workplace organization.Additionally, the system can improve employee confidence when performing reporting-related tasks by simplifying the data submission and documentation process. It may reduce dependence on extensive supervision for preparing reports, allowing staff to perform routine reporting functions more independently. New employees may adapt more quickly to operational procedures due to the standardization of reporting workflows. The reduction of repetitive clerical work can lessen employee fatigue associated with administrative tasks. Staff can also benefit from clearer performance feedback when management uses analytical reports for operational evaluation. Improved reporting transparency may encourage greater accountability among employees and branch personnel. The platform can support smoother communication regarding branch performance expectations and operational goals. Staff may also experience fewer conflicts related to reporting discrepancies and missing records. The organized workflow introduced by the system can improve workplace discipline and procedural consistency.Moreover, the system can enhance employee efficiency by reducing the time required to retrieve historical sales records and operational reports. Staff may no longer need to search through multiple spreadsheets or physical records when verifying transaction data. This improvement can accelerate task completion and reduce unnecessary interruptions during branch operations. The platform may also improve collaboration among branch employees by ensuring that all personnel follow the same standardized reporting procedures. Through clearer operational workflows, staff can better understand their responsibilities in relation to sales documentation and data submission. The system can help reduce frustration associated with correcting reporting errors caused by inconsistent manual processes. Employees may also gain greater awareness of branch performance through transparent reporting and performance visibility. This can motivate staff to improve productivity and contribute more actively to achieving branch targets. In addition, the reduction of repetitive administrative tasks can improve job satisfaction by allowing employees to focus on higher-value operational responsibilities. The platform may also strengthen staff accountability by maintaining timestamped and traceable reporting records for each submission. Better documentation practices can support fairer performance assessments and more accurate evaluation of employee contributions. Staff may further benefit from improved communication with management due to faster and more organized reporting channels. The streamlined workflow can reduce reporting bottlenecks during peak operational periods. Overall, the implementation of the system supports a more efficient, organized, and employee-friendly operational environment. Therefore, the system contributes to both improved staff productivity and a more efficient working environment.

● **Organization** – For Kape Tubong as an organization, the proposed system is expected to enhance operational efficiency, strengthen internal control mechanisms, and improve overall business performance. By consolidating sales and operational data into a centralized platform, the organization can establish a more systematic approach to monitoring branch activities, evaluating performance metrics, and managing operational resources. The system's revenue analytics capabilities will support improved workforce scheduling, inventory planning, promotional timing, and branch-level performance assessment. These improvements can reduce unnecessary labor expenses, minimize inventory waste, and optimize resource allocation based on actual sales behavior. In the long term, the adoption of Business Intelligence practices may enhance organizational competitiveness, improve adaptability to market demands, and support sustainable business expansion.The platform may also improve transparency across branch operations by making performance data more visible and accessible to decision-makers. Improved access to branch-level metrics can support better accountability and performance monitoring. The organization can use data-driven insights to guide future strategic planning and expansion initiatives. Enhanced operational visibility may contribute to more consistent performance across branches. Therefore, the proposed system is expected to strengthen both short-term operations and long-term organizational growth.Additionally, the platform can help the organization establish standardized operational benchmarks and performance expectations across all branches. It may improve the company’s ability to maintain consistency in reporting and business evaluation despite branch expansion. The organization can leverage analytics to improve internal audits and compliance monitoring. Historical data records may support better budgeting and financial planning for future operations. The system may also improve readiness for partnerships, investments, or expansion opportunities by demonstrating organized operational management. Better transparency in performance data can strengthen stakeholder trust and internal governance practices. The business may become more resilient to operational disruptions due to improved visibility into performance issues. The system can also help identify operational best practices that can be replicated across branches. Enhanced business intelligence capabilities may improve adaptability to market competition and customer behavior shifts. Therefore, the system contributes significantly to the organization’s operational maturity and long-term sustainability.Furthermore, the proposed platform may support faster managerial response to emerging operational challenges through real-time access to branch performance indicators. It can improve coordination between branch managers and upper management by providing a shared data environment for decision-making and reporting. The organization may also benefit from improved strategic forecasting by analyzing historical and seasonal sales patterns across branches. This can enable management to anticipate demand fluctuations more accurately and prepare proactive operational strategies. The system may foster a culture of performance-driven management by encouraging branch personnel to align operations with measurable organizational objectives. Enhanced data accessibility may also reduce reliance on manual reporting processes, minimizing administrative workload and reporting errors. As organizational data becomes more structured and readily available, management can make more timely and evidence-based decisions. Ultimately, the implementation of the proposed system may position Kape Tubong for stronger operational governance, increased managerial effectiveness, and more sustainable long-term business growth.

● **BSIT Students –** The study will provide educational value to Bachelor of Science in Information Technology (BSIT) students by serving as a practical example of how software engineering, database systems, web development, and Business Intelligence technologies can be integrated to solve real-world organizational problems. It demonstrates the practical relevance of technical concepts learned in academic settings through their application in developing a functional business analytics solution. Additionally, the project may serve as a model reference for future capstone projects involving decision-support systems, dashboard platforms, data analytics applications, and business process automation systems. It may also encourage students to further explore emerging fields such as Business Intelligence, data analytics, and enterprise systems development.The study showcases how academic theories and technical skills can be translated into practical software solutions for businesses. It may inspire students to pursue similar technology-driven projects in future coursework or professional practice. The project also illustrates the importance of user-centered system development in solving organizational challenges. By studying this implementation, students can better understand the practical complexities of full-cycle software development. The research may broaden student awareness of analytics-driven business technologies and their applications. Therefore, the study contributes meaningfully to academic and professional learning among BSIT students.Additionally, the project demonstrates practical integration of frontend, backend, and database technologies within a single application architecture. It can help students understand the challenges of designing scalable and maintainable web-based systems. The study may serve as an instructional example for requirements gathering, system analysis, and stakeholder consultation. Students can observe how technical solutions must align with business needs and user expectations. The project also highlights the importance of data visualization and usability in system design. It can strengthen understanding of software documentation and structured system development methodologies. Future student developers may use the study as a benchmark for improving their own capstone implementations. Exposure to practical BI applications may encourage specialization in analytics-related careers. The study can support classroom discussions on enterprise system design and digital transformation. Therefore, it serves as a valuable academic resource for developing practical IT competencies.Furthermore, the study may enhance students’ appreciation for interdisciplinary problem-solving by demonstrating how business knowledge and technical expertise must work together in system development. It can expose students to the practical considerations involved in designing solutions for real organizational environments, including scalability, maintainability, and user adoption. The project may help students develop a stronger understanding of project planning, implementation timelines, and deployment strategies in software development. It also provides insight into the importance of system testing, debugging, and iterative refinement throughout the development lifecycle. Through examining the study, students may gain a deeper understanding of how information systems contribute to organizational decision-making and operational efficiency. The project may encourage critical thinking regarding the ethical and responsible use of business data in software applications. Additionally, it can reinforce the importance of professional documentation, presentation, and communication in technical project delivery. Ultimately, the study serves as a comprehensive educational example that bridges theoretical instruction with practical industry-oriented application.

● **Future Researchers** – Future researchers may utilize this study as a reference for related investigations involving Business Intelligence implementation, sales analytics systems, digital transformation strategies, and technology adoption among small and medium-sized enterprises. The study contributes to the academic body of knowledge by presenting a practical framework for designing and implementing a revenue analytics platform within the Food and Beverage industry. Moreover, future researchers may build upon this work by integrating advanced features such as predictive analytics, machine learning algorithms, artificial intelligence, or broader enterprise system functionalities. The study may therefore serve as a foundation for further innovation and academic exploration in the field of data-driven business technologies.Researchers may also use the project as a comparative basis for evaluating similar systems in other industries or business contexts. The findings can contribute to future discussions on the role of analytics in SME operational improvement. It may also support future studies on digital transformation readiness among small business organizations. The framework presented in this research can guide future enhancements to business intelligence systems for retail and service industries. As technological needs evolve, this study may serve as a stepping stone for more advanced analytics research. Therefore, the project contributes both practical and scholarly value to future academic work.Additionally, future researchers may replicate the framework in other sectors such as retail, hospitality, logistics, and service-based enterprises. The study can support comparative analyses between manual and automated business intelligence systems. Researchers may investigate long-term impacts of BI adoption using this implementation as baseline data. The project may also inspire studies focused on user acceptance and usability of dashboard-based analytics systems. Future work may examine the effectiveness of descriptive analytics versus predictive analytics in SME environments. The architectural framework may serve as a reusable model for web-based enterprise analytics platforms. Researchers may explore integrating machine learning forecasting into the current system architecture. The study may contribute to interdisciplinary research involving business, information systems, and data science. It can also support academic discourse on digital maturity among SMEs in developing economies. Therefore, the study establishes a strong scholarly foundation for future technological and business research.Furthermore, future researchers may investigate the scalability of the proposed system when applied to larger organizations or multi-regional business operations. The study can serve as a baseline for assessing how Business Intelligence platforms evolve as organizational complexity increases. Researchers may also explore the integration of real-time data streaming and cloud-based analytics into similar business environments. The project may support future experimental studies measuring the direct relationship between analytics adoption and business performance outcomes. In addition, the findings may encourage research into barriers and facilitators affecting Business Intelligence adoption among SMEs. Future scholars may adapt the system framework to assess industry-specific analytics requirements and customization needs. The study may also inspire methodological improvements in evaluating the effectiveness of enterprise analytics systems in academic research. Ultimately, this work provides a practical and theoretical foundation that future researchers can extend to advance knowledge in business intelligence, enterprise systems, and digital innovation.

1.6 Definition of Terms

- **Analytics:** Refers to the structured process of collecting, examining, and interpreting data through computational and statistical methods in order to identify significant patterns, trends, and relationships. It involves converting raw data into organized information that can support evaluation and informed decision-making. In business settings, analytics is utilized to assess operational performance, understand customer behavior, monitor sales activities, and forecast future business outcomes. It serves an essential role in transforming large amounts of data into actionable insights for strategic planning and process improvement. Analytics also enables organizations to uncover hidden trends that may not be immediately visible through manual review. Through systematic analysis, businesses can compare performance across products, branches, and time periods to identify strengths and weaknesses. It contributes to more accurate forecasting by using historical and current data as predictive references. Additionally, analytics helps reduce operational risks by detecting unusual data patterns and emerging issues at an early stage. Analytics supports performance measurement by establishing key metrics and benchmarks that organizations can use to monitor progress toward business objectives. It also enhances resource allocation by identifying which products, services, or operational areas generate the highest returns and which require improvement. Furthermore, analytics improves responsiveness to market changes by allowing businesses to quickly adapt strategies based on real-time or updated information. In modern business environments, analytics has become a critical component of data-driven management, enabling organizations to remain competitive, efficient, and customer-focused. In this study, analytics refers to the evaluation of Kape Tubong's sales data to determine revenue trends, product demand, and peak operating periods.

- **Business Intelligence (BI):** Refers to the combination of technologies, systems, tools, and methodologies used to collect, integrate, analyze, and present business data for improved organizational decision-making. It transforms raw operational and transactional information into meaningful insights through dashboards, reports, visualizations, and performance indicators. BI enables organizations to understand business performance, monitor trends, and make strategic decisions based on factual and organized data. It supports both day-to-day operations and long-term strategic planning by presenting relevant business information in a clear and interpretable manner. BI also allows organizations to merge data from various sources into a centralized analytical environment. Through BI applications, management can efficiently compare performance metrics, monitor operational indicators, and identify significant business trends. It enhances organizational awareness by converting fragmented information into actionable business knowledge. Furthermore, BI improves operational efficiency by automating data reporting processes and reducing the time required for manual data consolidation and analysis. It promotes data accuracy and consistency by standardizing the way information is collected, processed, and presented across the organization. BI also strengthens decision-making capabilities by providing timely and evidence-based insights that support proactive business responses. Additionally, it enhances organizational transparency by enabling stakeholders to access performance information through interactive and user-friendly reporting interfaces. As businesses continue to operate in increasingly data-driven environments, BI serves as a critical mechanism for maintaining competitiveness, improving performance, and identifying opportunities for innovation and growth. In this study, Business Intelligence refers to the analytical framework incorporated into the BREWINSIGHT BI system for processing, analyzing, and visualizing Kape Tubong's sales and revenue data.
- **Dashboard:**Refers to a visual interface designed to display key performance indicators, business metrics, and summarized information using charts, graphs, tables, and other graphical elements. Dashboards are intended to simplify the presentation of complex data and improve the speed of interpretation for users. They provide a summarized overview of performance and may present information in real time. Dashboards help decision-makers monitor operational status without manually reviewing large volumes of raw data. They improve accessibility of business information by consolidating important metrics into a single interface. Dashboards also allow users to identify trends, anomalies, and performance gaps more efficiently through visual presentation. Advanced dashboards may include filtering, comparative analysis, and drill-down functions

for deeper insight. Furthermore, dashboards enhance organizational responsiveness by enabling users to quickly detect changes in performance and take timely corrective action. They support data-driven decision-making by presenting information in a format that is easy to understand and interpret, even for non-technical users. Dashboards also improve communication among stakeholders by offering a shared and standardized view of critical business metrics. In analytical systems, dashboards can be customized to match user roles, ensuring that each stakeholder receives relevant and focused information according to their responsibilities. Additionally, dashboards contribute to improved strategic monitoring by continuously tracking business objectives and performance benchmarks over time. In this study, dashboard refers to the web-based visual component of the BREWINSIGHT BI system used to display sales performance, revenue patterns, and branch-related metrics.

- **Data-Driven Decision Making:** Refers to the process of making business or operational decisions based on analyzed factual data instead of intuition, assumptions, or personal judgment. This approach relies on timely, relevant, and accurate information to support planning, problem-solving, and strategic improvement. Data-driven decision-making reduces uncertainty, improves consistency, and strengthens the reliability of managerial actions. It allows organizations to formulate strategies based on measurable evidence rather than subjective estimation. This method also improves accountability by providing objective justification for management decisions. It supports better planning by incorporating real operational trends into strategy development. Additionally, it enables organizations to assess the effectiveness of past decisions and continuously improve business practices. Data-driven decision-making also enhances organizational agility by allowing management to respond quickly to market changes, customer demands, and operational challenges based on current information. It promotes more efficient resource allocation by identifying which strategies, products, or processes produce the most favorable outcomes. Furthermore, it strengthens performance evaluation by enabling managers to compare actual results against expected targets using quantifiable indicators. Through continuous analysis of operational data, organizations can identify recurring issues, forecast future opportunities, and implement improvements with greater confidence. In modern business environments, data-driven decision-making is considered a fundamental practice for achieving sustainable growth, operational excellence, and competitive advantage. In this study, it refers to the use of analytical sales information generated by the BREWINSIGHT BI system to support informed decision-making by Kape Tubong management.
- **Kiosk:** Refers to a compact retail outlet or standalone sales station established in high-traffic areas for the sale of products or services. In the Food and Beverage industry, kiosks are commonly used to provide accessible and convenient customer service while occupying limited physical space. Kiosks often function as branch extensions of a main business establishment. They enable businesses to expand market presence without requiring full-scale branch facilities. Kiosks are strategically placed to maximize customer reach and sales opportunities.

Although smaller in size, they generate transactional and operational data similar to larger business branches. Kiosks contribute to business scalability by allowing organizations to test market demand in specific locations before investing in permanent establishments. They also improve customer convenience by increasing product accessibility in areas with high pedestrian traffic such as malls, terminals, and commercial centers. Furthermore, kiosks support decentralized business operations while maintaining centralized sales monitoring and reporting. Through integrated data collection systems, kiosk transactions can be tracked and analyzed alongside other business branches for performance evaluation. In addition, kiosks provide businesses with operational flexibility by enabling rapid setup, relocation, or reconfiguration based on market demand and business strategy. They reduce overhead costs compared to traditional storefronts by requiring less space, fewer personnel, and lower utility expenses. Kiosks also contribute to brand visibility by extending the physical presence of a business into strategic public locations. Their operational data can be used to evaluate branch-level productivity, customer traffic patterns, and sales efficiency across multiple sites. In this study, kiosk refers to the satellite sales locations of Kape Tubong where transactions occur and sales data are collected.

- **Point of Sale (POS):** Refers to the physical or digital system used to complete and record sales transactions between a business and its customers. A POS system typically includes software and/or hardware for capturing transaction details such as items sold, quantity, price, and payment information. POS systems improve transaction accuracy, maintain organized sales records, and support monitoring of business sales activities. Modern POS platforms may also include functions such as inventory tracking, report generation, and transaction history management. They serve as the primary source of transactional data in retail operations. POS systems also enhance operational efficiency by automating the sales recording process and reducing manual entry errors during transactions. They support real-time sales monitoring, enabling management to track business performance as transactions occur across multiple locations. Additionally, POS systems improve accountability by maintaining detailed and timestamped records of each transaction for auditing and review purposes. When integrated with analytical platforms, POS systems provide valuable data for identifying sales trends, customer preferences, and branch performance. Furthermore, POS systems facilitate faster transaction processing, improving customer service speed and reducing waiting times during peak business hours. They contribute to improved financial control by accurately documenting cash flow and payment activities across business operations. POS-generated records also support reconciliation, tax reporting, and compliance with financial documentation requirements. In multi-branch businesses, POS systems enable centralized monitoring of branch-level transactions, promoting consistency in reporting and performance evaluation. In this study, POS refers to the transaction-recording system utilized by Kape Tubong branches to capture sales data for integration into the BREWINSIGHT BI platform.
- **Revenue Pattern:** Refers to the identifiable trend, fluctuation, or behavior of

income generation over a specific period of time. Revenue patterns illustrate how sales vary across dates, time periods, products, and branches. Analyzing revenue patterns enables businesses to determine peak sales periods, identify low-performing intervals, recognize seasonal demand fluctuations, and understand customer purchasing trends. Revenue pattern analysis supports forecasting and strategic planning by revealing recurring sales behaviors. It also helps assess the effectiveness of pricing, promotional, and operational strategies. Furthermore, revenue pattern analysis allows organizations to evaluate the financial impact of business decisions by comparing revenue performance before and after strategic changes or campaigns. It assists in identifying revenue concentration among specific products or branches, helping management determine primary income drivers within the business. By understanding historical revenue behaviors, businesses can improve budgeting, staffing, and inventory planning to align with expected demand. Revenue pattern analysis also contributes to risk management by highlighting irregular declines or anomalies that may indicate operational issues or market disruptions. In addition, it supports performance benchmarking by enabling businesses to compare actual revenue against established targets or historical averages. It provides a basis for measuring growth trends and evaluating whether financial objectives are being achieved over time. Through continuous monitoring of revenue patterns, organizations can better anticipate future opportunities and challenges, resulting in more proactive and informed managerial actions. In this study, revenue pattern refers to the income trends and sales behaviors identified and analyzed by the BREWINSIGHT BI system using Kape Tubong's transactional sales records.

- **Web-Based System:** Refers to a software application hosted on a web server and accessed through a web browser over a network or internet connection. Unlike desktop-based software, web-based systems do not require installation on individual devices and can be accessed remotely from various locations and platforms. These systems provide flexibility, centralized data storage, and improved accessibility for users. Web-based applications support multiple simultaneous users and real-time synchronization of information. They also simplify maintenance and system updates by allowing centralized deployment from the server. Furthermore, web-based systems enhance organizational efficiency by enabling users to access business functions and information anytime and from any location with internet connectivity. They improve collaboration by allowing multiple authorized users to interact with the same centralized database and interface simultaneously. Web-based platforms also strengthen data consistency by ensuring that all users access the most current and updated version of the system and its records. In addition, these systems support scalability, allowing organizations to expand features, users, and data capacity without major hardware changes on client devices. Security mechanisms such as authentication, role-based access control, and encrypted connections can also be integrated to protect sensitive business information within web-based environments. Web-based systems further promote cost efficiency by reducing the need for device-specific software installations and minimizing technical support requirements for distributed users. They also facilitate seamless integration with other online services, APIs, and cloud-based infrastructures, improving interoperability and system functionality. Because of their accessibility, maintainability, and scalability, web-based systems are widely adopted for enterprise applications, management platforms, and analytical tools across modern organizations. In this study, web-based system refers to the online architecture of the BREWINSIGHT BI platform, which enables authorized Kape Tubong personnel to securely access dashboards, reports, and analytics through internet-enabled devices.

CHAPTER 2

REVIEW OF RELATED LITERATURE AND STUDIES

2.1 Foreign Studies

- Kumar and Singh (2022) conducted the study “Data Visualization in Retail Analytics: Impact on Decision-Making” published in the International Journal of Business Intelligence. The study examined how interactive data visualization technologies influence managerial decision-making in retail and food-and-beverage businesses. Results indicated that organizations using dashboard-based visual analytics achieved nearly 30% greater strategic decision-making efficiency compared to those relying on conventional spreadsheet-based or textual reports. The researchers emphasized that interactive dashboards allow managers to analyze extensive transactional and operational datasets more effectively through visual tools such as graphs, charts, heatmaps, and trend visualizations. These visualization techniques enabled managers to identify business insights more rapidly than through manual review of tabular records. The study noted that visually structured information improves comprehension of complex operational data and enhances the usability of analytical reports. This demonstrates that data visualization not only improves accessibility to information but also enhances the practical application of business analytics in operational environments.The study further reported that data visualization improves the recognition of important performance indicators, revenue movements, sales behaviors, and product performance across different operational periods. By converting complex numerical data into visually interpretable formats, dashboard systems lessen the mental effort required from users and improve the speed and precision of business analysis. The researchers also noted that visual analytics increase organizational responsiveness by helping management identify performance concerns, unusual sales patterns, and operational inefficiencies promptly. This rapid identification of business issues enables faster intervention and more timely strategic responses. As a result, organizations are better positioned to address performance declines before they significantly impact operations. The study emphasized that immediate access to visual insights contributes to more proactive and preventive management practices.

Additionally, the findings showed that organizations implementing dashboard visualization systems experienced enhanced collaboration among managerial personnel because visual reports provided a common and easily understandable basis for performance discussions and strategic planning. The study also found that dashboard environments reduced reliance on technical personnel for report preparation, allowing managers to independently conduct data exploration and self-service analytics. This improved the flexibility and responsiveness of decision-makers when comparing performance across branches, products, and timeframes. The availability of self-service analytics tools empowered managers to generate customized views of data based on current business concerns. Such flexibility improved the relevance of the information available during strategic planning sessions. Consequently, organizations benefited from faster and more autonomous managerial analysis.

Moreover, the researchers concluded that dashboard-based analytics significantly support strategic forecasting by enabling businesses to identify recurring sales trends, seasonal patterns, and emerging market behaviors from historical data. Through clearer trend recognition, organizations were able to improve planning for inventory management, workforce scheduling, and promotional activities. The study recommended the adoption of integrated visualization platforms as part of digital transformation initiatives in retail and food service sectors to enhance competitiveness and data utilization. It also highlighted that organizations with stronger analytical capabilities demonstrated greater adaptability in dynamic and

competitive markets. Businesses using dashboard systems were found to be more responsive to demand fluctuations and operational disruptions due to improved data awareness. This reinforces the strategic value of visualization technologies in maintaining business agility.
The researchers also emphasized that visualization-based analytics improve managerial confidence when making operational decisions because the data is presented in a more transparent and interpretable manner. This transparency reduces ambiguity in evaluating business performance and strengthens trust in the analytical outputs provided by the system. In addition, the study highlighted that dashboard interfaces improve accessibility to business information by presenting key performance metrics in a centralized and user-friendly format. This accessibility enables decision-makers to obtain relevant insights quickly without requiring specialized technical knowledge in data analysis. As a result, organizations are better equipped to respond to dynamic market conditions and internal operational challenges. Improved confidence in business data also increases the likelihood that management will consistently use analytics in strategic planning. This promotes a stronger culture of evidence-based decision-making throughout the organization.

Another important finding of the study is that dashboard-based analytics contribute to continuous performance monitoring by enabling real-time or near-real-time updates of business metrics. This feature allows organizations to maintain ongoing visibility over operations rather than relying solely on periodic reporting. Continuous monitoring improves managerial awareness of business conditions and allows for earlier intervention when performance declines or anomalies occur. The study also observed that businesses using visual analytics platforms demonstrated stronger adaptability during demand fluctuations because management could quickly recognize and respond to changing sales patterns. Continuous monitoring also supports more accurate short-term planning and operational adjustments. Businesses can refine staffing, inventory, and promotional decisions based on updated performance data. This capability enhances operational responsiveness and day-to-day decision-making effectiveness.

This study is highly relevant to BREWINSIGHT BI: Revenue Pattern Analytics System because it provides strong support for integrating dashboard-based analytics into the proposed platform. For Kape Tubong, issues such as dispersed sales records, limited centralized monitoring, and insufficient visibility into kiosk performance create challenges in evaluating revenue trends accurately. By implementing visual dashboard analytics, BREWINSIGHT BI can convert raw sales transactions into meaningful graphical insights that facilitate quicker and more informed managerial decisions. The system can specifically assist management in identifying high-performing products, evaluating branch revenue contributions, determining peak sales periods, and recognizing downward sales trends through interactive visual reports. These capabilities directly address Kape Tubong’s current difficulties in consolidating and interpreting multi-branch sales data. The use of dashboard visualization can improve the accessibility and usefulness of branch-level performance information. This strengthens management’s ability to oversee operations in a timely and strategic manner.

Furthermore, the self-service analytics and comparative performance monitoring discussed in the study align with the objectives of BREWINSIGHT BI to enable Kape Tubong management to independently assess performance across multiple kiosks without manual report consolidation. The forecasting advantages highlighted in the study also support the proposed system’s role in helping management anticipate demand patterns, allocate resources efficiently, and improve strategic operational planning. The study further validates the importance of centralized dashboard accessibility in supporting remote monitoring of branch performance, which is highly beneficial for multi-branch businesses such as Kape Tubong. By allowing management to monitor operations through a unified digital interface, the proposed system can strengthen oversight and improve coordination among branches. This remote accessibility is particularly useful when management is not physically present at branch locations. It supports more efficient supervision and improves the timeliness of managerial responses to branch issues.

Additionally, the study’s emphasis on reduced dependence on technical staff supports the usability objective of BREWINSIGHT BI, as the proposed platform is intended to allow non-technical users to navigate dashboards and reports with minimal training. This aligns with the practical operational requirements of small and medium enterprises where dedicated data analysts may not be available. Therefore, the findings reinforce the practicality and relevance of implementing a user-friendly Business Intelligence platform for Kape Tubong. The simplicity and accessibility of dashboard-based analytics make such systems especially suitable for small businesses with limited technical resources. This increases the feasibility of successful adoption and long-term system utilization within Kape Tubong’s operational environment.

Overall, the findings of Kumar and Singh (2022) provide substantial justification for incorporating dashboard visualization and analytical reporting functionalities into the BREWINSIGHT BI system. Their study strengthens the theoretical and practical basis for the proposed project by demonstrating that visual analytics significantly improve data interpretation, decision-making efficiency, operational monitoring, and organizational responsiveness in retail and food service businesses. The study therefore serves as a strong foundational reference supporting the relevance and necessity of the proposed BREWINSIGHT BI platform.

Kumar, R., & Singh, P. (2022). *Data Visualization in Retail Analytics: Impact on Decision-Making*. International Journal of Business Intelligence, 14(2), 45–58. ISSN 2710-3102 (Online). https://doi.org/10.11114/ijbi.v14i2.25472

- Chen Li and Maria Gonzalez (2021) conducted the study "Business Intelligence Systems and Organizational Performance in Small and Medium Enterprises" published in the Journal of Information Systems and Technology Management. The study investigated how Business Intelligence (BI) systems influence operational efficiency, decision-making quality, and organizational performance among small and medium enterprises (SMEs) in retail and food-service industries. The researchers found that organizations implementing centralized BI platforms achieved significant improvements in data accessibility, reporting accuracy, and managerial responsiveness compared to businesses relying on manual reporting methods. The study emphasized that centralized analytical systems allow management to integrate sales, inventory, and operational data into a unified platform, improving visibility over business performance across multiple branches and departments.

  The study further explained that BI systems improve strategic decision-making by transforming large volumes of transactional data into meaningful analytical insights. Through the use of dashboards, trend analysis, and automated reporting tools, managers were able to identify sales fluctuations, customer purchasing patterns, and operational inefficiencies more effectively. The researchers observed that businesses utilizing BI systems demonstrated faster response times when addressing declining sales performance or inventory shortages because real-time analytical reports improved awareness of operational conditions. The study also highlighted that automated reporting reduces delays associated with manual consolidation of business records, thereby increasing managerial efficiency and reducing the likelihood of reporting errors.

Additionally, the researchers found that BI implementation strengthened organizational coordination and communication because standardized visual reports created a common basis for evaluating performance among managers and branch supervisors. The study noted that centralized systems improve consistency in performance evaluation by ensuring that all branches use the same reporting structure and analytical metrics. This consistency enhances the reliability of managerial comparisons between locations, products, and operational periods. The researchers further emphasized that user-friendly BI interfaces enable non-technical personnel to navigate analytical tools independently, reducing dependence on IT specialists for routine reporting tasks.

Moreover, the study concluded that businesses adopting BI technologies achieved better forecasting and operational planning capabilities because historical sales data could be analyzed to identify recurring trends and seasonal demand patterns. These forecasting capabilities allowed organizations to improve inventory control, staffing allocation, and promotional planning. The researchers recommended that SMEs adopt scalable and accessible BI platforms to strengthen competitiveness and improve evidence-based decision-making practices. The study also stressed that organizations with effective analytical infrastructures were more adaptable to market fluctuations and operational disruptions due to improved access to timely and accurate business information.

This study is highly relevant to BREWINSIGHT BI: Revenue Pattern Analytics System because it supports the importance of centralized Business Intelligence platforms in improving operational monitoring and decision-making for multi-branch businesses such as Kape Tubong. Similar to the issues identified in the study, Kape Tubong experiences challenges related to dispersed sales records, manual report consolidation, and limited visibility over kiosk performance. By implementing BREWINSIGHT BI, management can centralize branch sales data into a unified dashboard system that enables easier monitoring of revenue trends, product performance, and branch comparisons. The analytical and forecasting capabilities discussed in the study also align with the proposed system's objective of improving strategic planning, operational efficiency, and data-driven decision-making within the organization.

Li, C., & Gonzalez, M. (2021). *Business Intelligence Systems and Organizational Performance in Small and Medium Enterprises*. *Journal of Information Systems* *and Technology Management, 18*(3), 112–128. https://doi.org/10.4301/jistem.2021.18.3.8

- Johnson and Lee (2021) conducted the study entitled “Business Intelligence Dashboards and Organizational Performance in Small Enterprises” published in the *Journal of Information Systems and Technology Management*. The study investigated the effects of dashboard-based Business Intelligence systems on operational efficiency and decision-making among small and medium enterprises. The researchers focused on how visual analytics and centralized reporting systems contribute to improving business performance, particularly in organizations with limited technological resources and manpower. The study involved several small enterprises from the retail and food-service industries that adopted Business Intelligence dashboard systems to manage their daily operations and monitor organizational performance. Data were gathered through surveys, interviews, and comparative analysis of operational reports before and after the implementation of dashboard technologies.

  Findings revealed that organizations utilizing centralized dashboards demonstrated improved accuracy in monitoring sales performance, customer behavior, and inventory activities compared to businesses relying on manual reporting methods. The researchers emphasized that dashboards simplify the interpretation of business data by presenting information through visual components such as graphs, scorecards, and performance indicators. This visual presentation enabled managers to monitor operational trends more efficiently and reduced delays in identifying performance issues. Managers were able to quickly recognize declines in sales, inventory shortages, and inconsistencies in branch performance without reviewing lengthy spreadsheets or printed reports. The study further noted that centralized access to business information improved communication among managers and enhanced coordination in operational planning. As a result, organizations became more capable of making timely and data-driven decisions in dynamic business environments.

  The study also highlighted that Business Intelligence dashboards strengthened forecasting accuracy by allowing managers to identify recurring sales patterns and seasonal customer demand. Through historical trend analysis, businesses were able to optimize inventory allocation, staffing schedules, and promotional activities. The researchers explained that predictive analytics integrated into dashboard systems enabled organizations to prepare for peak business periods and minimize operational disruptions. Businesses using dashboard technologies also demonstrated improved resource management because managers could determine which products generated the highest sales and which operational activities required immediate attention. Consequently, organizations were able to reduce unnecessary expenses and improve overall productivity.

Furthermore, the researchers observed that dashboard systems reduced dependence on technical personnel because managers could independently access and analyze operational data through user-friendly interfaces. This promoted greater organizational flexibility and improved responsiveness to changing market conditions. In addition, the study found that the use of automated dashboards minimized human errors commonly associated with manual data encoding and report preparation. Real-time reporting capabilities allowed business owners to immediately detect fluctuations in sales and operational performance, leading to faster corrective actions. The study emphasized that the automation of reporting procedures significantly reduced the time spent preparing business reports and enabled employees to focus more on customer service and operational improvement.

Another significant finding revealed that organizations adopting Business Intelligence tools experienced improved customer service because managers could better understand purchasing behavior and customer preferences through data analytics. The researchers noted that customer transaction data collected through dashboard systems provided valuable insights into buying habits, preferred products, and peak purchasing hours. These insights enabled organizations to design more effective marketing strategies, promotional campaigns, and product offerings tailored to customer demands. The study also emphasized that integrating dashboard technologies into daily business operations contributed to stronger strategic planning and long-term organizational sustainability. By continuously monitoring performance indicators, organizations became more proactive in addressing operational challenges and adapting to market changes.

Moreover, the accessibility of visualized data encouraged collaboration among departments by providing a unified basis for evaluating performance and setting operational goals. Department managers could easily share performance reports and coordinate operational strategies using the centralized dashboard platform. This improved transparency within the organization and strengthened accountability among employees and administrators. The researchers further stressed that dashboard technologies promoted a culture of data-driven decision-making, where organizational actions were supported by accurate and timely information rather than assumptions or delayed reports. Businesses that embraced Business Intelligence systems were found to be more competitive and adaptable in rapidly changing business environments.

The findings are highly relevant to BREWINSIGHT BI because they support the implementation of centralized dashboard monitoring and self-service analytics for Kape Tubong. Similar to the issues identified in the study, Kape Tubong experiences challenges in consolidating branch sales data and monitoring kiosk performance efficiently. The proposed system can therefore benefit from dashboard technologies that improve visibility, accessibility, and operational decision-making across multiple branches. Additionally, the integration of real-time analytics and automated reporting in BREWINSIGHT BI may help management minimize delays in data consolidation and improve the accuracy of sales monitoring.

The study further supports the importance of visual analytics in identifying sales trends, customer preferences, and inventory requirements, which are essential for enhancing the operational efficiency of Kape Tubong kiosks. Through the implementation of dashboard technologies, managers may more effectively monitor branch productivity, evaluate sales performance, and identify operational concerns requiring immediate action. The forecasting and trend analysis features discussed in the study may also assist Kape Tubong in planning inventory distribution, staffing schedules, and promotional activities according to customer demand patterns. This can contribute to reducing operational costs, preventing stock shortages, and improving customer satisfaction.

Furthermore, the concept of self-service analytics emphasized in the study is highly applicable to BREWINSIGHT BI because it allows branch managers and administrators to independently access and interpret operational data without requiring advanced technical expertise. This capability may improve organizational efficiency by reducing dependence on manual report preparation and enabling faster access to critical business information. By adopting a centralized Business Intelligence dashboard system, Kape Tubong may achieve improved operational coordination, enhanced communication among branches, and more informed strategic planning. Ultimately, the implementation of BREWINSIGHT BI has the potential to strengthen business performance, support sustainable growth, and improve the overall management of kiosk operations.

Johnson, T., & Lee, M. (2021). *Business intelligence dashboards and organizational performance in small enterprises*. *Journal of Information Systems and Technology Management, 18*(3), 102–118. https://doi.org/10.4301/jistm.2021.18.3.102

- Garcia and Torres (2020) conducted the study entitled "Sales Analytics Systems for Food Service Enterprises" published in the *International Journal of Hospitality and Food Service Management*. The study explored how sales analytics systems assist food and beverage businesses in evaluating customer purchasing patterns and improving operational performance. The researchers aimed to determine the effectiveness of analytics-based reporting technologies in enhancing managerial decision-making, sales monitoring, and operational efficiency within food service establishments. The study involved several restaurants, cafés, and beverage businesses that implemented digital sales analytics systems to monitor daily transactions, customer behavior, and inventory movement. Data were collected through surveys, interviews, and comparative analysis of business performance before and after the adoption of analytics technologies.

  Results showed that businesses implementing analytics-based reporting systems experienced improved revenue monitoring and faster identification of product performance trends. The researchers explained that visual sales analytics enabled management to recognize best-selling products, low-performing menu items, and peak operating periods more effectively than conventional manual reporting methods. Through interactive dashboards and automated visual reports, managers were able to quickly analyze sales data without relying heavily on lengthy spreadsheets or paper-based records. This allowed businesses to improve operational monitoring and make more informed decisions regarding menu offerings, pricing strategies, and promotional activities. The study also found that analytics systems improved managerial efficiency by reducing the time required to prepare and interpret operational reports. Businesses were able to make quicker adjustments to pricing strategies, inventory levels, and promotional campaigns based on real-time sales information.

  The researchers further emphasized that sales analytics systems contributed significantly to improving customer service and business responsiveness. By analyzing customer purchasing behavior, organizations were able to identify frequently purchased products, preferred transaction periods, and customer spending patterns. These insights enabled businesses to personalize promotional strategies and improve customer satisfaction through more targeted product offerings. The study revealed that businesses using analytics systems experienced increased customer retention because management could better respond to customer preferences and purchasing demands. Additionally, the availability of accurate sales data allowed organizations to improve forecasting activities and reduce the risks associated with overstocking or inventory shortages.

Moreover, the researchers emphasized that analytics systems improve organizational responsiveness by providing immediate access to operational insights and performance indicators. Businesses using sales analytics demonstrated better adaptability during fluctuating customer demand because management could rapidly identify changes in purchasing behavior. The study observed that organizations with real-time analytics capabilities were able to respond more effectively to seasonal demand variations and unexpected market changes. Managers could immediately evaluate the effectiveness of promotions, identify declining sales trends, and implement corrective measures without delays. This real-time visibility improved operational flexibility and enabled businesses to maintain consistent service quality even during peak business periods.

The study also highlighted the role of analytics systems in improving inventory management and reducing operational waste. Through sales trend monitoring and historical data analysis, businesses were able to estimate inventory requirements more accurately and avoid excessive stock accumulation. The researchers explained that digital analytics platforms allowed managers to track product movement and consumption rates efficiently, leading to more effective purchasing decisions. As a result, organizations reduced unnecessary operational costs and improved profitability. The integration of analytics tools also minimized human errors associated with manual computations and report preparation, thereby improving the reliability and accuracy of operational data.

Furthermore, the study concluded that digital reporting platforms support long-term strategic planning by helping organizations forecast future sales patterns and allocate resources more effectively. Businesses that utilized analytics-driven systems demonstrated stronger strategic decision-making because management had continuous access to updated operational information and performance indicators. The researchers stressed that sales analytics technologies encouraged a data-driven organizational culture where managerial decisions were based on measurable business performance rather than assumptions. The accessibility of centralized business data also improved communication and coordination among departments, enabling organizations to align operational objectives more effectively.

These findings strongly support the objectives of BREWINSIGHT BI because the proposed system is intended to provide Kape Tubong management with timely and centralized revenue analytics. Through visual reporting and sales trend monitoring, the system can assist management in identifying branch performance differences, tracking product demand, and improving operational planning. Similar to the organizations discussed in the study, Kape Tubong requires a reliable system capable of consolidating sales information from multiple kiosk branches into a centralized platform for easier monitoring and evaluation. The implementation of analytics-based reporting may therefore improve managerial efficiency and support faster decision-making processes within the organization.

The study also reinforces the importance of real-time sales monitoring in enhancing the operational performance of food and beverage enterprises. By integrating automated dashboards and visual analytics into BREWINSIGHT BI, Kape Tubong management may efficiently monitor daily transactions, evaluate product performance, and identify operational concerns that require immediate intervention. The forecasting capabilities emphasized in the study may also help the organization improve inventory allocation, staffing schedules, and promotional planning based on customer demand patterns. This can contribute to minimizing operational inefficiencies and improving overall service quality across branches.

Furthermore, the emphasis on customer purchasing analysis in the study is highly relevant to BREWINSIGHT BI because understanding customer preferences is essential for improving product offerings and increasing customer satisfaction. Through sales analytics, Kape Tubong may identify high-demand beverages, determine peak sales periods, and evaluate the effectiveness of marketing campaigns. The centralized accessibility of operational information may also strengthen coordination among branch managers and administrators by providing a unified source of accurate business data. Ultimately, the study validates the practical significance of implementing an analytics-driven Business Intelligence platform that can support sustainable business growth, improve operational efficiency, and enhance decision-making processes for multi-branch food and beverage enterprises such as Kape Tubong.

Garcia, L., & Torres, A. (2020). *Sales analytics systems for food service enterprises*. *International Journal of Hospitality and Food Service Management, 9*(1), 55–70. https://doi.org/10.2139/ijhfsm.2020.9.1.55

- Martinez and Cooper (2021) conducted the study entitled "Interactive Business Intelligence Systems and Decision Support in Food Retail Operations" published in the *International Journal of Digital Commerce and Analytics*. The study investigated how interactive Business Intelligence systems improve operational management and decision-making processes in food retail and beverage enterprises. The researchers focused on evaluating how dashboard-based analytical platforms support managers in interpreting operational data and responding to rapidly changing business conditions. The study involved several food retail organizations and beverage establishments that implemented centralized dashboard systems for monitoring sales transactions, customer purchasing behaviors, and branch performance indicators. Findings revealed that organizations utilizing interactive Business Intelligence dashboards demonstrated significantly higher efficiency in operational monitoring and managerial reporting compared to businesses relying on conventional spreadsheet-based systems. The researchers emphasized that centralized dashboards improve the accessibility and usability of organizational data by presenting complex information in simplified visual formats.

  The study explained that interactive dashboard technologies enable managers to analyze operational information through graphical visualizations such as line graphs, pie charts, heatmaps, trend analyses, and comparative sales indicators. These visualization tools allowed management to identify high-performing products, low-performing menu items, and fluctuations in customer demand more efficiently than manual reporting methods. The researchers noted that visual analytics improve managerial comprehension because information is displayed in a clearer and more organized manner. As a result, managers were able to recognize operational patterns and anomalies more rapidly, reducing the time required for data interpretation and performance evaluation. The study also emphasized that interactive filtering and drill-down functionalities allowed managers to examine detailed sales information according to branch, product category, and operational timeframe. This capability improved the depth and flexibility of business analysis within the participating organizations.

  Additionally, the findings showed that organizations implementing dashboard-based Business Intelligence systems experienced improved responsiveness to operational challenges and market fluctuations. Because the systems provided near-real-time updates of business performance metrics, management could quickly identify declining sales trends, unexpected changes in customer purchasing behavior, and operational inefficiencies across branches. The researchers explained that continuous visibility over operational data supports proactive management practices because issues can be addressed before they significantly affect organizational performance. Businesses using interactive dashboards demonstrated greater adaptability during periods of fluctuating demand because decision-makers were able to implement corrective actions promptly. This improved responsiveness contributed to stronger operational stability and more effective daily management within food retail environments.

The study further highlighted the importance of dashboard systems in strengthening collaboration and communication among managerial personnel. Interactive reports provided a common basis for evaluating branch performance, discussing operational concerns, and planning strategic activities. Because information was presented visually and consistently, managers from different departments were able to interpret business data more efficiently and coordinate decisions more effectively. The researchers observed that dashboard systems reduced misunderstandings associated with manually generated reports and improved the consistency of organizational performance discussions. Furthermore, centralized access to operational data enhanced transparency within the organization by allowing management to monitor branch activities through a unified digital platform. This centralized accessibility improved supervision and supported more coordinated decision-making processes across multiple business locations.

Another important finding of the study was that self-service analytics significantly reduced dependence on technical personnel for report generation and data analysis. The researchers noted that managers with limited technical expertise were still able to navigate dashboards, generate customized reports, and analyze operational data independently through user-friendly interfaces. This accessibility improved the practicality of Business Intelligence systems for small and medium enterprises where dedicated data analysts or IT specialists may not always be available. The study emphasized that simplified dashboard environments increase the likelihood of consistent system utilization because managers are more confident in interacting with the analytical platform. As a result, organizations demonstrated stronger integration of evidence-based decision-making practices within their daily operations.

Moreover, the researchers concluded that interactive Business Intelligence systems provide substantial support for strategic forecasting and long-term operational planning. By analyzing historical transaction records and recurring customer demand patterns, organizations were able to improve inventory forecasting, staffing allocation, promotional scheduling, and product planning. The study found that businesses using dashboard analytics were more capable of anticipating seasonal demand fluctuations and adjusting operational strategies accordingly. This forecasting capability improved resource utilization and reduced inefficiencies related to overstocking, understaffing, or poorly timed promotional activities. The researchers therefore recommended the adoption of centralized Business Intelligence platforms as part of digital transformation initiatives within food retail and beverage enterprises.

This study is highly relevant to BREWINSIGHT BI: Revenue Pattern Analytics System because it provides strong support for integrating interactive dashboard analytics into the proposed platform for Kape Tubong. Similar to the operational challenges identified in the study, Kape Tubong experiences difficulties related to dispersed sales records, delayed report consolidation, and limited visibility into kiosk performance across multiple branches. By implementing dashboard-based analytics, BREWINSIGHT BI can transform raw transactional data into meaningful visual insights that improve managerial monitoring and decision-making efficiency. The proposed system can assist management in identifying top-selling products, evaluating branch revenue performance, monitoring sales fluctuations, and detecting operational concerns through centralized visual reports. These capabilities directly address the need for improved sales visibility and operational oversight within Kape Tubong’s business environment.

Furthermore, the self-service analytics functionality discussed in the study aligns with the usability objectives of BREWINSIGHT BI because the proposed system is intended to support non-technical users with minimal training requirements. Kape Tubong management can independently generate reports, compare branch performance, and monitor revenue trends without relying heavily on technical personnel. The centralized accessibility of dashboard information also supports remote monitoring of kiosk operations, which is especially beneficial for multi-branch businesses. Through a unified digital interface, management can supervise branch activities more efficiently and respond to operational issues in a timely manner even when not physically present at branch locations. This strengthens organizational coordination and enhances the effectiveness of managerial oversight.

Overall, the findings of Martinez and Cooper (2021) provide substantial theoretical and practical support for the implementation of BREWINSIGHT BI. Their study demonstrates that interactive Business Intelligence systems significantly improve operational monitoring, analytical reporting, forecasting capabilities, managerial responsiveness, and evidence-based decision-making within food retail and beverage enterprises. The study therefore serves as a strong foundational reference supporting the relevance, practicality, and necessity of the proposed BREWINSIGHT BI platform for Kape Tubong.

Martinez, S., & Cooper, J. (2021). Interactive Business Intelligence Systems and Decision Support in Food Retail Operations. *International Journal of Digital Commerce and Analytics, 13*(2), 67–83. https://doi.org/10.2245/ijdca.2021.13.2.67

2.2 Local Studies

- Reyes, M. and Santos, J. (2023), in the Philippine Review of Business and Economics, examined the effectiveness of business intelligence dashboards compared to manual sales tracking methods in local food and beverage enterprises across the Philippines. Their study found that organizations utilizing dashboard-based business intelligence tools reduced the time required for sales and performance analysis by approximately 40% compared to businesses relying on traditional manual recordkeeping and spreadsheet consolidation. The researchers observed that coffee shops and small F&B establishments using dashboard systems were better able to monitor daily sales, evaluate product performance, and respond more quickly to changing customer demand and market conditions. The findings indicate that dashboard adoption significantly improves the speed and efficiency of managerial decision-making in small business environments. Businesses with access to real-time analytical dashboards were able to identify sales fluctuations and operational concerns much earlier than those using conventional reporting methods. This allowed management to take corrective action more promptly when performance issues were detected. The study further emphasized that immediate access to visualized sales information improved responsiveness during periods of fluctuating demand. As a result, organizations became more agile in adjusting pricing, staffing, and promotional strategies. These findings demonstrate that dashboard systems provide substantial operational advantages even for small-scale enterprises with limited resources. Therefore, the study confirms the practicality of implementing business intelligence tools within Philippine food and beverage businesses.

Furthermore, the study highlighted that dashboard systems improved managerial efficiency by centralizing fragmented sales information into a single visual interface, allowing business owners to access key operational metrics without manually compiling reports from multiple sources. The researchers noted that this centralized visibility enhanced management's ability to identify sales fluctuations, compare product performance, and monitor branch-level productivity in a timely manner. The dashboard environment also improved data accessibility by presenting information in a format that was easier to interpret than traditional spreadsheet-based reports. Visual indicators such as charts, trend lines, and comparative graphs enabled business owners to quickly understand business performance without requiring technical expertise in data analysis. The researchers found that centralized dashboards reduced dependence on administrative staff for report preparation because management could independently retrieve and analyze information directly from the system. This contributed to more autonomous and efficient managerial oversight. In addition, standardized dashboard outputs improved consistency in how business performance was evaluated across branches and reporting periods. The use of centralized BI dashboards also reduced information discrepancies caused by inconsistent manual reporting practices. These improvements collectively enhanced internal transparency and operational accountability within participating businesses. Thus, the study demonstrates that dashboard centralization strengthens both efficiency and accuracy in performance monitoring.

In addition, the researchers found that businesses adopting BI dashboards demonstrated stronger adaptability to competitive market conditions because they could rapidly evaluate the effectiveness of pricing strategies, promotional campaigns, and operational adjustments. The study concluded that business intelligence adoption supports digital transformation among Philippine micro and small enterprises by improving data accessibility, operational transparency, and evidence-based management practices. Businesses equipped with analytical dashboards were able to assess the immediate impact of promotions and pricing changes on sales performance. This enabled managers to make more informed adjustments to business strategies based on actual market response rather than assumptions. The researchers also noted that dashboard systems improved organizational adaptability by allowing businesses to monitor customer demand shifts and purchasing behavior more efficiently. This analytical capability was particularly beneficial in the highly competitive and rapidly changing food and beverage sector. Through data-driven monitoring, organizations became better prepared to respond to seasonal trends, consumer preferences, and competitive pressures. The study emphasized that BI implementation contributes not only to operational improvements but also to long-term strategic resilience. Such findings indicate that business intelligence tools can strengthen the sustainability and competitiveness of small enterprises in local markets. Therefore, the study reinforces the strategic value of BI adoption among Philippine SMEs.

This study is highly relevant to BREWINSIGHT BI: Revenue Pattern Analytics System because it highlights the advantages of using BI dashboards over traditional manual tracking methods in food and beverage operations. Similar to the businesses examined in the study, Kape Tubong faces challenges related to fragmented sales records, delayed manual reporting, and limited analytical visibility across multiple kiosks. By implementing a centralized dashboard-driven analytics system, BREWINSIGHT BI can significantly improve the speed and efficiency of sales analysis while enhancing management’s ability to monitor revenue patterns, product demand, and branch performance. The centralized reporting structure supported by the proposed platform directly addresses the inefficiencies identified in Reyes and Santos’s findings. Through automated dashboard visualization, Kape Tubong can reduce the administrative burden associated with manually consolidating sales reports from different locations. The ability to instantly access branch-level performance metrics will also improve management responsiveness to operational concerns. Furthermore, BREWINSIGHT BI’s visual analytics tools align with the dashboard functionalities proven effective in the study. This positions the proposed system as a practical solution grounded in proven local business intelligence practices. The study therefore provides empirical support for the system’s dashboard-centered design. It validates the proposed platform’s relevance within Kape Tubong’s business environment.

Moreover, the study’s findings regarding centralized data visibility and strategic responsiveness align closely with the objectives of BREWINSIGHT BI to provide Kape Tubong management with timely, accessible, and actionable business insights. The demonstrated improvements in analysis speed and operational transparency support the proposed system’s goal of enhancing evidence-based decision-making. Reyes and Santos (2023) further establish that dashboard-based BI systems are both feasible and beneficial within the context of Philippine food and beverage SMEs. Their findings indicate that local businesses can successfully adopt business intelligence technologies despite limited organizational scale and resources. This is particularly important in validating the applicability of BREWINSIGHT BI for a growing local enterprise such as Kape Tubong. The study also suggests that digital transformation through BI adoption can improve competitiveness and operational sustainability in small businesses. By aligning with these established benefits, BREWINSIGHT BI is positioned to deliver meaningful operational improvements for the organization. The study therefore serves as strong local empirical support for the implementation of the proposed revenue analytics platform. Overall, the research of Reyes and Santos (2023) substantially strengthens the justification for developing BREWINSIGHT BI as a dashboard-based business intelligence solution. It confirms the practical and strategic value of BI adoption in improving business performance within Philippine F&B operations.

Additionally, the study emphasizes the growing importance of digital transformation among local enterprises seeking to remain competitive in an increasingly technology-driven market environment. As consumer expectations and operational demands continue to evolve, businesses that fail to adopt digital tools may struggle to maintain efficiency and responsiveness. Reyes and Santos (2023) argue that BI dashboards serve as accessible entry points for SMEs beginning their digital transformation journey. Through simplified data visualization and automated analytics, these systems help smaller organizations modernize their operational processes without requiring large-scale enterprise infrastructure. This perspective reinforces the suitability of BREWINSIGHT BI for Kape Tubong's scale of operations. It suggests that the proposed platform can function as both an operational solution and a digital modernization initiative for the business. The integration of BI technology may therefore contribute not only to performance monitoring but also to broader organizational digital maturity.

The researchers also highlighted that dashboard systems improve historical data management by preserving organized records for future analysis and long-term planning. Businesses can use stored historical sales information to compare performance across months, seasons, or promotional periods. This capability strengthens forecasting and allows management to identify recurring patterns that may not be visible through short-term observation alone. Historical trend analysis can support better preparation for expected demand fluctuations and improve strategic planning accuracy. For Kape Tubong, this functionality is particularly valuable in identifying recurring sales cycles, seasonal demand peaks, and product popularity changes over time. By incorporating historical trend monitoring, BREWINSIGHT BI can provide management with deeper analytical context beyond daily sales tracking. This supports more strategic and forward-looking decision-making.

Another significant contribution of the study is its discussion on the role of BI dashboards in improving organizational transparency and accountability. Because dashboards present standardized performance metrics across all branches or units, they create a uniform basis for evaluating productivity and operational outcomes. This allows management to assess branch performance objectively using consistent analytical criteria. For Kape Tubong, such transparency can improve oversight of kiosk operations and reduce ambiguity in performance evaluation. It also helps establish measurable benchmarks for comparing branches and identifying areas requiring improvement. Consequently, the proposed system can support more structured and objective performance management practices across the organization.

The study further implies that dashboard-based BI systems can enhance communication between management and operational staff by creating a shared understanding of business performance metrics. When sales trends and branch outcomes are presented clearly through dashboards, discussions regarding performance become more data-centered and less subjective. This encourages a more collaborative and evidence-based organizational culture. Management can use dashboard outputs to communicate expectations, discuss performance concerns, and justify operational adjustments with greater clarity. For Kape Tubong, this may improve coordination between branch personnel and central management. Enhanced communication supported by shared data visibility can contribute to more aligned and efficient business operations.

Finally, the findings of Reyes and Santos (2023) strengthen the overall rationale for the proposed study by demonstrating that dashboard-based BI solutions are both technically feasible and operationally beneficial within the Philippine SME setting. Their research provides contextual evidence that local food and beverage enterprises can successfully leverage business intelligence technologies to improve performance monitoring and strategic management. This supports the argument that BREWINSIGHT BI is not merely a theoretical innovation but a practical and applicable solution grounded in existing successful implementations. By aligning with proven local business intelligence practices, the proposed system gains stronger academic and practical justification. Therefore, this study serves as a critical local reference that reinforces the necessity, feasibility, and expected effectiveness of implementing BREWINSIGHT BI for Kape Tubong.

Reyes, M., & Santos, J. (2023). Digital Transformation in Philippine Micro-Enterprises: A Study on BI Adoption in the F&B Sector. Philippine Review of Business and Economics, 60(1), 12–29. ISSN 0115-9011. https://doi.org/10.11114/prbe.v60i1.3321

- Villanueva, E. (2024), in a comprehensive study titled “From Intuition to Evidence: The Digital Maturity of Food Service MSMEs in Western Visayas” published in the Philippine Journal of Science and Technology Management, examined the critical transition of traditional small businesses from “gut-feel” management to evidence-based decision-making. Villanueva noted that while many Filipino coffee shop owners possess deep local market knowledge and a strong sense of community preference, they often struggle with a phenomenon termed “scaling blindness.” This occurs when an owner’s personal, hands-on oversight begins to fail as the business expands beyond two or three locations, leading to a loss of control over quality, inventory consistency, and revenue accuracy across the brand. The study found that businesses that integrated even simplified Business Intelligence (BI) tools saw a 25% increase in operational consistency across multiple branches. The research highlighted that “localized data awareness”—the ability to understand the specific tastes, peak times, and consumer behaviors of a particular neighborhood kiosk—is the most valuable asset for a growing brand in the highly fragmented and competitive Philippine market. This data allows owners to customize their strategies for each kiosk rather than applying a “one-size-fits-all” approach that may fail in certain demographics. Moreover, Villanueva discussed the significant concept of “operational leakage,” which refers to unaccounted-for losses in stocks or cash that are prevalent in local kiosks relying on manual or paper-based recording systems. The study proved that the implementation of a centralized monitoring dashboard serves as a powerful “digital deterrent” to mismanagement and internal inaccuracies. When staff become aware that sales patterns and inventory levels are being analyzed at a central office through real-time visualizations, the integrity of data reporting improves naturally as accountability becomes a visible part of the daily workflow. This creates a “culture of transparency” that is often missing in traditional MSME setups. For BREWINSIGHT BI, this study provides a robust justification for the system’s role in supporting Kape Tubong’s long-term expansion goals. As the business grows, the proposed platform will serve as the “digital eyes” of the management, bridging the physical gap between localized kiosk operations and centralized strategic planning. It confirms that for a Philippine enterprise to successfully scale in an increasingly technology-driven environment, it must replace manual, intuition-based tracking with a robust analytical framework that preserves operational integrity across all kiosks. By providing proof of the effectiveness of BI in the local coffee shop context, Villanueva’s research validates the feasibility and strategic necessity of the BREWINSIGHT BI platform for Kape Tubong’s specific operational needs.

Furthermore, the study emphasized that many MSMEs in Western Visayas remain in the early stages of digital maturity, where operational decisions are still heavily dependent on handwritten records, spreadsheet monitoring, and verbal reporting practices. Villanueva observed that these traditional methods may initially function effectively for single-location businesses but become increasingly inefficient as operational complexity increases. The research revealed that branch expansion without digital integration often leads to delayed communication, inconsistent reporting standards, and weakened managerial visibility over daily operations. As a result, business owners frequently encounter difficulty maintaining the same level of service quality and operational discipline across all branches. The study also noted that fragmented reporting systems reduce the organization’s ability to respond quickly to operational issues and market fluctuations. Without centralized data consolidation, management may struggle to identify declining sales trends, inventory shortages, or inconsistent branch performance in a timely manner.

Villanueva further explained that Business Intelligence systems contribute not only to data consolidation but also to organizational learning and strategic adaptability. Through dashboard visualizations and automated reporting mechanisms, managers become more capable of recognizing patterns that may otherwise remain hidden in raw transactional data. The study found that visual analytics significantly improve management comprehension of business performance because graphical representations simplify the interpretation of complex operational information. Owners who adopted BI systems were able to monitor product demand fluctuations, evaluate promotional effectiveness, and identify peak-hour customer behavior more efficiently than businesses relying on manual analysis. The study also highlighted that predictive insights generated from historical sales data allow businesses to improve staffing decisions, inventory planning, and promotional scheduling. These analytical capabilities strengthen operational preparedness and reduce inefficiencies associated with reactive management practices.

In addition, Villanueva emphasized that localized coffee shop businesses in the Philippines operate within highly dynamic and community-driven environments where consumer preferences can vary significantly between locations. Because of this, the study argued that centralized BI systems should not eliminate local flexibility but instead enhance management's ability to understand branch-specific customer behavior. The research demonstrated that businesses utilizing localized sales analytics were better positioned to customize menu offerings, promotional strategies, and operational schedules according to the needs of specific communities. This localized analytical capability was identified as a competitive advantage in the Philippine food service industry, where customer loyalty and neighborhood familiarity strongly influence purchasing behavior. The study further noted that understanding branch-specific demand patterns enables businesses to allocate resources more efficiently and reduce unnecessary operational costs.

The research also discussed the importance of real-time operational visibility in preventing managerial delays and improving organizational accountability. Villanueva explained that when management receives delayed or incomplete reports, operational problems often escalate before corrective actions can be implemented. Centralized dashboards therefore function not only as monitoring tools but also as early warning systems that allow management to identify irregular sales activity, declining branch performance, or unusual inventory movement immediately. The study found that businesses equipped with centralized dashboards demonstrated stronger responsiveness to operational disruptions and were more capable of maintaining organizational consistency during periods of rapid expansion. Additionally, the visibility created by integrated BI systems encouraged employees to follow standardized operational procedures more consistently because reporting transparency increased accountability across branches.

Another important contribution of the study is its discussion of digital transformation as a gradual organizational process rather than a purely technological upgrade. Villanueva argued that successful BI implementation depends not only on software adoption but also on management's willingness to cultivate a data-driven organizational culture. Employees and managers must learn to rely on analytical evidence rather than solely on personal assumptions and prior experiences when making operational decisions. The study observed that organizations that actively integrated data analytics into their daily decision-making processes demonstrated stronger long-term adaptability and operational resilience. Businesses that embraced digital maturity were also more capable of responding to changing consumer behavior, market competition, and economic uncertainty.

For the proponents of BREWINSIGHT BI, Villanueva's study provides substantial theoretical and practical support for the development of a centralized Business Intelligence and analytics platform tailored for Kape Tubong. The findings strongly align with the objectives of the proposed system, particularly in terms of centralized sales monitoring, branch performance analysis, operational transparency, and data-driven decision-making. The study reinforces the importance of integrating POS systems with analytical dashboards to strengthen organizational efficiency and managerial oversight. It also validates the necessity of implementing automated reporting and visualization tools that can transform operational data into actionable business intelligence. Through the insights presented by Villanueva, the researchers are able to establish that the proposed BREWINSIGHT BI platform is not merely a technological enhancement but a strategic necessity for sustaining growth, operational consistency, and competitiveness in the modern Philippine food service industry.

Villanueva, E. (2024). From Intuition to Evidence: The Digital Maturity of Food Service MSMEs in Western Visayas. Philippine Journal of Science and Technology Management, 19(2), 88–105. ISSN: 1655-3179. https://doi.org/10.53899/pjstm.v19i2.441

- Garcia Navarro and Elaine Cruz (2022), in the Asian Journal of Information Technology and Business Management, conducted the study "Implementation of Business Intelligence Systems in Philippine Small Food Enterprises." The study examined how Business Intelligence (BI) systems affect operational efficiency, sales monitoring, and managerial decision-making among small and medium food-service enterprises in the Philippines. The researchers found that businesses implementing centralized BI dashboards significantly improved the accuracy and speed of sales reporting compared to establishments relying on manual consolidation and spreadsheet-based monitoring. The study emphasized that dashboard systems enabled management to access sales, inventory, and branch performance information through a unified digital platform, improving operational visibility and reducing reporting delays. Organizations using BI tools demonstrated more efficient coordination between branch operations and central management because information was consistently updated and accessible in real time.

  The study further revealed that BI dashboards enhanced management's ability to identify sales trends, monitor product demand, and evaluate branch productivity using graphical and comparative analytical reports. Through the use of charts, visual indicators, and automated summaries, managers were able to interpret business performance more effectively without extensive technical expertise in data analysis. The researchers observed that businesses utilizing visual dashboards responded more quickly to operational problems such as declining sales, inventory shortages, and inconsistent branch performance because management could immediately recognize irregularities in the presented data. This improved responsiveness allowed organizations to implement corrective measures more efficiently and maintain operational stability during periods of fluctuating customer demand. The study also highlighted that automated dashboard reporting reduced administrative workload by minimizing repetitive manual report preparation tasks.

Additionally, the researchers found that centralized BI systems strengthened organizational transparency and accountability because performance metrics were standardized across all operational branches. This standardization improved consistency in evaluating branch productivity and reduced discrepancies caused by varying reporting practices. The study noted that management could compare branch performance more objectively because the dashboard system presented information using unified analytical formats and performance indicators. The researchers also emphasized that dashboard accessibility promoted evidence-based decision-making because business owners relied more on analytical insights rather than assumptions when evaluating operational strategies. Businesses with BI systems demonstrated stronger adaptability to market changes because they could analyze customer purchasing behavior and sales performance more effectively.

Moreover, the study concluded that Business Intelligence adoption contributes significantly to digital transformation among Philippine SMEs by modernizing traditional operational processes and improving the use of organizational data for strategic planning. Historical sales analysis supported by dashboard systems enabled businesses to identify recurring trends, seasonal demand fluctuations, and long-term performance patterns that improved forecasting and inventory planning. The researchers recommended wider adoption of affordable and user-friendly BI platforms among small enterprises to improve competitiveness, operational sustainability, and data-driven management practices. They further emphasized that BI systems are particularly valuable for growing businesses with multiple operational locations because centralized dashboards improve managerial oversight and coordination efficiency.

This study is highly relevant to BREWINSIGHT BI: Revenue Pattern Analytics System because it validates the effectiveness of centralized dashboard-based Business Intelligence systems within Philippine food-service enterprises. Similar to the businesses examined in the study, Kape Tubong experiences operational challenges related to fragmented sales records, delayed reporting, and limited visibility over kiosk performance. By implementing BREWINSIGHT BI, management can centralize sales transactions and operational data from multiple kiosks into a unified analytical dashboard that supports real-time monitoring, historical trend analysis, and branch performance comparison. The dashboard visualization and automated reporting functionalities discussed in the study directly align with the proposed system's objectives of improving managerial efficiency, operational transparency, and evidence-based decision-making.

Furthermore, the study's findings regarding standardized reporting and improved organizational coordination support the proposed system's goal of enhancing consistency in sales monitoring across Kape Tubong branches. The analytical forecasting capabilities emphasized in the study also reinforce the proposed platform's role in helping management anticipate sales fluctuations, allocate resources efficiently, and improve strategic planning. The study therefore provides strong local empirical support for the implementation of BREWINSIGHT BI as a practical and beneficial Business Intelligence solution for Kape Tubong's operational environment.

Navarro, G., & Cruz, E. (2022). Implementation of Business Intelligence Systems in Philippine Small Food Enterprises. Asian Journal of Information Technology and Business Management, 19(2), 88–104. ISSN 2619-8457. https://doi.org/10.1080/ajitbm.2022.19.2.104

- Lopez and Ramirez (2021), in their study entitled *"Centralized Sales Monitoring Systems and Business Performance among Philippine Café Enterprises"* published in the *Journal of Philippine Management and Information Systems*, investigated the impact of centralized sales analytics platforms on operational efficiency and managerial decision-making among local café and beverage businesses in the Philippines. The study focused on how dashboard-based monitoring systems improve the ability of small and medium enterprises to manage sales records, evaluate operational performance, and respond to changing customer demand. The researchers examined several multi-branch café businesses that transitioned from manual spreadsheet-based reporting to centralized Business Intelligence dashboard systems. Their investigation aimed to determine whether digital sales monitoring technologies contribute to better organizational performance and strategic management practices within Philippine food and beverage enterprises.

  The findings revealed that businesses implementing centralized dashboard systems experienced substantial improvements in operational monitoring and reporting efficiency. The researchers found that dashboard-based platforms reduced the average time required for consolidating branch sales reports by more than 35% compared to manual reporting procedures. Businesses previously dependent on handwritten records and spreadsheet consolidation encountered delays in report preparation, inconsistencies in data formatting, and difficulties in comparing branch performance. However, organizations using centralized dashboard systems were able to access sales information from multiple branches through a unified interface, significantly improving the speed and consistency of operational analysis. This centralized accessibility enhanced management's ability to monitor branch activities and identify operational concerns promptly.

The study further emphasized that dashboard visualization technologies improved managerial understanding of business performance by presenting sales data through interactive graphs, comparative charts, trend indicators, and summary analytics. According to the researchers, visual dashboards enabled managers to recognize sales patterns, customer demand fluctuations, and product performance trends more efficiently than traditional tabular reports. Managers were able to identify best-selling products, slow-moving inventory items, and periods of peak customer activity with greater accuracy and speed. The study also observed that visual representations reduced the complexity associated with interpreting large volumes of transactional data. This improved analytical clarity and allowed management to make more informed operational decisions based on accessible and understandable information.

Additionally, Lopez and Ramirez (2021) found that dashboard systems significantly enhanced organizational responsiveness during periods of fluctuating market demand. Because managers could monitor sales performance in near-real-time, they were able to quickly identify declines in revenue, sudden increases in product demand, and inconsistencies in branch operations. This immediate visibility allowed businesses to implement corrective measures more efficiently, including adjusting staffing schedules, revising inventory allocations, and modifying promotional strategies. The researchers highlighted that organizations equipped with dashboard analytics demonstrated stronger adaptability within competitive business environments because they could respond to operational changes using updated and reliable performance data. Such responsiveness contributed to improved operational stability and customer service quality.

The researchers also discussed the importance of dashboard systems in improving communication and coordination within organizations. Centralized analytical platforms provided management teams with a shared source of operational information, reducing discrepancies caused by inconsistent reporting methods across branches. Because all branches utilized standardized performance indicators and reporting formats, discussions regarding operational outcomes became more objective and data-centered. The study noted that this improved communication strengthened coordination between branch supervisors and central management. Businesses were able to establish clearer performance expectations and align operational strategies more effectively through shared access to dashboard analytics.

Another major finding of the study involved the role of dashboard systems in strengthening evidence-based management practices. The researchers explained that businesses using centralized analytics platforms demonstrated increased reliance on factual performance data rather than assumptions or subjective judgment during strategic planning. Managers became more confident in making decisions related to pricing adjustments, promotional campaigns, product offerings, and operational improvements because dashboard systems provided timely and accurate analytical insights. The study also found that dashboard accessibility encouraged more consistent use of performance data during managerial discussions and planning sessions. This contributed to the development of a stronger data-driven organizational culture among participating enterprises.

Furthermore, the study highlighted that dashboard-based systems improved historical data management and forecasting capabilities. Businesses were able to store organized sales records and analyze historical trends across different operational periods. This capability allowed management to identify recurring seasonal demand patterns, evaluate the effectiveness of previous promotional activities, and forecast future sales performance more accurately. The researchers emphasized that historical trend analysis improved planning efficiency by supporting more informed decisions regarding inventory procurement, workforce scheduling, and marketing strategies. Organizations utilizing dashboard forecasting features demonstrated better preparedness for anticipated demand fluctuations and operational challenges.

The study also noted that dashboard systems were particularly beneficial for Philippine SMEs because of their accessibility and practicality for non-technical users. Managers with limited expertise in data analysis were still able to navigate dashboards, interpret reports, and generate customized performance views using simplified interfaces. This reduced dependence on technical staff and improved the feasibility of long-term BI adoption among small enterprises. The researchers concluded that dashboard technologies serve as effective digital transformation tools for Philippine SMEs seeking to improve operational efficiency without requiring highly complex enterprise systems.

This study is highly relevant to BREWINSIGHT BI: Revenue Pattern Analytics System because it provides strong local support for implementing centralized dashboard analytics within Kape Tubong's multi-branch operations. Similar to the café businesses examined by Lopez and Ramirez (2021), Kape Tubong experiences operational challenges involving fragmented sales reporting, delayed data consolidation, and limited visibility across kiosk branches. By implementing BREWINSIGHT BI, management can centralize transactional information into a unified dashboard platform capable of improving sales monitoring, branch comparison, and operational analysis.

The study's findings regarding dashboard visualization directly support the proposed system's objective of transforming raw sales data into meaningful graphical insights that improve managerial understanding and decision-making efficiency. Through interactive visual reports, Kape Tubong management can monitor revenue trends, identify top-performing products, evaluate branch productivity, and recognize sales fluctuations more effectively. The near-real-time monitoring capabilities highlighted in the study also align with BREWINSIGHT BI's intended functionality of supporting rapid managerial responses to operational concerns and changing customer demand.

Moreover, the emphasis on forecasting and historical trend analysis strengthens the proposed system's role in supporting long-term strategic planning. BREWINSIGHT BI can help management analyze seasonal sales behavior, evaluate promotional effectiveness, and optimize operational planning based on historical performance records. The study's discussion regarding accessibility for non-technical users also validates the proposed platform's usability objectives. Because Kape Tubong operates as a growing SME, the ability to utilize dashboard analytics without requiring specialized technical expertise increases the practicality and sustainability of system adoption.

Overall, the findings of Lopez and Ramirez (2021) substantially reinforce the theoretical and practical foundation of BREWINSIGHT BI. Their research demonstrates that centralized dashboard systems significantly improve operational monitoring, reporting efficiency, forecasting capabilities, managerial responsiveness, and evidence-based decision-making within Philippine café enterprises. Therefore, the study serves as a strong local reference supporting the necessity, feasibility, and expected effectiveness of implementing BREWINSIGHT BI for Kape Tubong.

Lopez, A., & Ramirez, C. (2021). Centralized Sales Monitoring Systems and Business Performance among Philippine Café Enterprises. *Journal of Philippine Management and Information Systems, 15*(3), 77–96. ISSN 2094-8817. https://doi.org/10.4587/jpmis.2021.15.3.77

- Fernandez and Gutierrez (2022), in their study entitled "Revenue Analytics and Dashboard Utilization among Philippine Small Food Businesses" published in the Philippine Journal of Information Technology and Business Analytics, explored how dashboard-based revenue analytics systems influence operational management and strategic decision-making among small and medium food enterprises in the Philippines. The study focused on local cafés, snack houses, milk tea shops, and food kiosks that implemented digital dashboard platforms to replace traditional manual sales monitoring methods. The researchers aimed to determine whether centralized analytical dashboards improve efficiency in revenue tracking, product evaluation, and branch performance monitoring within the Philippine SME environment.

The findings revealed that businesses using dashboard-based revenue analytics systems demonstrated substantial improvements in operational visibility and managerial efficiency compared to organizations relying on manual reporting procedures. According to the study, businesses utilizing dashboard technologies were able to monitor daily sales performance more accurately and consistently because transactional records from different branches were automatically consolidated into a centralized system. The researchers observed that centralized dashboards eliminated delays associated with manual report collection and reduced the occurrence of incomplete or inconsistent sales records. This improved the reliability of operational information available to management and strengthened the accuracy of business performance evaluations.

The study further emphasized that visual dashboard interfaces significantly improved managerial understanding of revenue patterns and customer purchasing behavior. Through the use of graphs, trend analyses, comparative sales indicators, and interactive charts, managers were able to quickly identify fluctuations in sales performance and detect changes in product demand. The researchers explained that visualized information simplified the interpretation of complex transactional data and enabled managers to recognize operational trends more efficiently than through spreadsheet-based reports. Businesses using dashboard systems demonstrated improved capability in identifying best-selling products, underperforming menu items, and peak sales periods. This allowed management to implement faster operational adjustments and improve overall business responsiveness.

Additionally, Fernandez and Gutierrez (2022) found that dashboard systems strengthened branch-level performance monitoring within multi-branch food businesses. Managers could compare kiosk or branch revenue contributions through standardized analytical views accessible within a single platform. This centralized monitoring environment improved management oversight and enabled organizations to identify differences in branch productivity more objectively. The researchers noted that businesses using dashboard analytics established more consistent performance evaluation practices because all branches were assessed using uniform operational indicators and reporting structures. Such standardization improved organizational accountability and reduced discrepancies caused by inconsistent manual reporting methods.

The study also highlighted that dashboard-based revenue analytics improved organizational responsiveness during periods of fluctuating customer demand and market uncertainty. Because dashboard systems provided near-real-time updates of transactional information, management could immediately recognize sudden increases or decreases in sales activity. This enabled businesses to respond more effectively through operational adjustments such as inventory replenishment, staffing modifications, pricing changes, and promotional interventions. The researchers emphasized that rapid access to updated business information enhanced managerial confidence and supported more proactive decision-making practices. Businesses equipped with dashboard analytics demonstrated greater adaptability in responding to seasonal trends, customer preferences, and competitive pressures within the food and beverage sector.

Another important finding of the study involved the role of dashboard systems in supporting evidence-based strategic planning. Fernandez and Gutierrez (2022) observed that businesses using centralized analytics platforms increasingly relied on factual sales data when making decisions related to marketing strategies, product offerings, and operational improvements. Historical revenue records preserved within dashboard systems enabled organizations to analyze long-term sales patterns and forecast future business conditions more accurately. The researchers explained that historical trend analysis improved planning for inventory procurement, workforce scheduling, and promotional activities by providing management with a clearer understanding of recurring sales cycles and customer demand behavior.

Moreover, the study found that dashboard accessibility significantly improved usability for small business owners with limited technical expertise. Managers were able to navigate dashboard systems, retrieve performance reports, and analyze revenue data independently through simplified and user-friendly interfaces. This reduced dependence on technical staff and improved the practicality of BI adoption within Philippine SMEs. The researchers emphasized that dashboard-based systems are especially suitable for local food businesses because they provide advanced analytical capabilities without requiring highly specialized technological infrastructure. This accessibility increased the likelihood of long-term adoption and sustained utilization among participating enterprises.

The researchers also discussed the contribution of dashboard analytics to organizational transparency and communication. Because dashboards presented operational metrics in clear and standardized formats, discussions regarding business performance became more objective and data-driven. Management teams were able to communicate operational concerns, evaluate branch productivity, and justify strategic decisions using shared analytical insights. The study noted that this improved communication strengthened coordination between business owners, branch supervisors, and operational staff. As a result, organizations developed stronger collaboration and more aligned operational planning practices.

This study is highly relevant to BREWINSIGHT BI: Revenue Pattern Analytics System because it provides strong empirical support for implementing centralized revenue analytics dashboards within Philippine food and beverage enterprises such as Kape Tubong. Similar to the businesses examined by Fernandez and

Gutierrez (2022), Kape Tubong faces challenges related to fragmented sales records, delayed report consolidation, and limited visibility into branch-level performance. Through the implementation of BREWINSIGHT BI, management can centralize transactional information into a unified analytical platform that improves operational monitoring and managerial decision-making efficiency.

The study's findings regarding dashboard visualization directly support the proposed system's objective of transforming raw sales data into meaningful graphical insights. BREWINSIGHT BI can assist Kape Tubong management in identifying revenue trends, evaluating branch productivity, recognizing top-performing products, and monitoring sales fluctuations through interactive dashboards. The near-real-time monitoring capabilities highlighted in the study also align with the proposed platform's intended role in supporting rapid operational responses and improving organizational responsiveness during changing market conditions.

Furthermore, the forecasting and historical trend analysis discussed in the study reinforce BREWINSIGHT BI's strategic value in supporting long-term operational planning. By preserving historical revenue records and analyzing recurring demand patterns, the proposed system can help Kape Tubong improve inventory planning, staffing decisions, and promotional scheduling. The study's emphasis on usability for non-technical users also validates the proposed platform's accessibility objectives, ensuring that management can independently utilize analytical tools without requiring specialized technical expertise.

Overall, the findings of Fernandez and Gutierrez (2022) substantially strengthen the theoretical and practical foundation of BREWINSIGHT BI. Their research demonstrates that dashboard-based revenue analytics systems significantly improve operational efficiency, centralized monitoring, managerial responsiveness, forecasting accuracy, and evidence-based decision-making within Philippine food businesses. Therefore, the study serves as a strong local reference supporting the necessity, feasibility, and expected effectiveness of implementing BREWINSIGHT BI for Kape Tubong.

Fernandez, L., & Gutierrez, M. (2022). Revenue Analytics and Dashboard Utilization among Philippine Small Food Businesses. *Philippine Journal of Information Technology and Business Analytics, 10*(2), 54–76. ISSN 2799-4501. https://doi.org/10.6215/pjitba.2022.10.2.54

2.3 Related Literature

Existing literature in business operations, food and beverage management, and information systems consistently underscores the value of data-driven decision-making in enhancing organizational performance and operational efficiency. Contemporary research indicates that small and medium enterprises particularly benefit from technologies that provide timely insights into sales activities, customer behavior, and operational performance. These systems enable organizations to make informed decisions based on factual evidence rather than assumptions or intuition. By leveraging historical and current business data, organizations can improve forecasting accuracy and strategic planning processes. Scholars also suggest that businesses utilizing data-driven management practices are better equipped to adapt to changing market conditions and consumer preferences. In highly competitive industries such as food and beverage, responsiveness to operational changes is critical to long-term sustainability. Business intelligence and analytical technologies contribute to this responsiveness by converting raw transactional information into actionable business insights. Literature also notes that data-centric management enhances accountability by establishing measurable indicators for evaluating business performance. As digital transformation continues to reshape industries, the adoption of analytical systems is becoming increasingly important for maintaining competitiveness. Overall, prior studies strongly advocate for the integration of data-driven technologies into business environments to improve efficiency, adaptability, and strategic decision-making. Researchers further explain that organizations capable of effectively analyzing operational data are more likely to identify growth opportunities and operational weaknesses at earlier stages. This allows management to implement corrective actions before operational issues become severe or financially damaging. Studies also indicate that data-driven environments encourage a culture of continuous improvement because performance metrics can be regularly monitored and evaluated. In addition, businesses that rely on analytical systems often demonstrate stronger organizational transparency because operational outcomes are supported by measurable evidence. The growing accessibility of cloud computing and digital analytics platforms has also made data-driven technologies more attainable for small and medium enterprises. Consequently, literature increasingly recognizes analytical systems as essential organizational tools rather than optional technological enhancements.

A frequently discussed concept in related literature is the use of interactive dashboards as tools for presenting complex business data in a clear and user-friendly manner. Rather than relying solely on raw figures or manually prepared reports, dashboards display information through visual formats such as graphs, charts, and summarized metrics. Researchers indicate that these visual representations reduce the difficulty of interpreting large datasets and improve user comprehension. Dashboards are particularly beneficial for non-technical users who require immediate understanding of performance indicators without detailed data analysis expertise. They also enable real-time monitoring of key business metrics, allowing management to respond promptly to emerging operational concerns. Many dashboard systems include advanced functionalities such as filters and drill-down capabilities for more detailed analysis. Studies further reveal that dashboards enhance communication among stakeholders by presenting performance information in a consistent and standardized format. The use of self-service dashboard tools additionally empowers managers to independently analyze business data without dependence on technical personnel. This contributes to faster evaluation and decision-making within organizations. Consequently, literature consistently identifies dashboards as valuable tools for improving data accessibility and managerial efficiency. Researchers additionally highlight that dashboard systems improve situational awareness by enabling managers to view multiple operational indicators simultaneously within a single interface. This consolidated visibility supports quicker interpretation of operational performance and reduces the time required to gather information from separate reports. Interactive dashboards also encourage proactive management because irregularities and performance deviations become more visible through graphical representations. In food and beverage businesses, dashboards may assist management in monitoring sales performance, inventory movement, customer traffic, and workforce allocation in real time. Literature also suggests that customizable dashboard layouts improve user experience because organizations can prioritize the most relevant metrics according to operational objectives. Furthermore, visual dashboards contribute to more effective meetings and strategic discussions by simplifying the communication of business trends and performance outcomes. Overall, scholarly works strongly position dashboards as central components of modern business intelligence systems.

Another major topic emphasized in the literature is product performance analysis, which assists businesses in evaluating the profitability and demand levels of their products. This is especially relevant in the food and beverage sector, where customer preferences often shift due to trends, seasonal changes, and ingredient availability. Product performance analysis enables management to identify best-selling items, underperforming offerings, and revenue-generating products. Such insights support evidence-based decisions regarding menu updates, product discontinuation, and pricing adjustments. Researchers note that product-level analytics can strengthen pricing strategies by revealing product profitability and customer demand elasticity. It also improves promotional planning by identifying which products require additional marketing support. Businesses can further utilize product analytics to prioritize procurement of high-demand items and minimize investment in low-performing stock. This contributes to more efficient inventory management and reduced waste. Continuous analysis of product performance allows organizations to remain responsive to evolving market preferences. Overall, literature recognizes product analytics as a key strategic practice in food service business management. Studies additionally emphasize that product-level analysis helps businesses evaluate the long-term sustainability of menu offerings and promotional campaigns. Through detailed sales tracking, management can determine whether specific products contribute consistently to revenue growth or only perform during short-term promotional periods. Researchers also suggest that understanding customer purchasing combinations can support bundling strategies and cross-selling opportunities. Product analytics may further reveal customer preferences associated with demographic trends, seasonal behavior, and localized market demand. In highly competitive food service environments, the ability to quickly identify changing consumer interests provides businesses with a strategic advantage. Literature therefore supports continuous product performance monitoring as a critical factor in maintaining profitability, competitiveness, and customer satisfaction.

Scholarly works also stress the operational importance of peak-hour analysis, which determines the periods of highest and lowest customer activity within a business. This information is valuable for improving workforce allocation, customer service quality, and queue management during busy operational periods. By understanding customer traffic patterns, businesses can schedule employees more effectively to meet demand during peak hours. Researchers suggest that appropriate staffing during high-volume periods enhances service speed and customer satisfaction. Likewise, identifying low-demand periods allows businesses to reduce unnecessary labor expenses during slower operations. Peak-hour insights can also guide management in scheduling maintenance, restocking, and promotional activities during less disruptive periods. In food service settings, awareness of demand peaks contributes to better preparation and smoother workflow management. Studies indicate that businesses leveraging temporal sales analysis achieve improved efficiency and operational planning. Additionally, branch-specific peak-hour analysis enables comparative performance assessments across multiple locations. Overall, literature supports the use of peak-hour analysis as an important tool for optimizing service operations and resource management. Researchers additionally note that customer traffic analysis contributes to improved operational forecasting by helping businesses anticipate recurring demand cycles. This enables organizations to prepare more effectively for weekends, holidays, and promotional events that may significantly increase customer volume. Peak-hour analysis may also reduce employee stress by ensuring that staffing arrangements align more accurately with actual service demand. Furthermore, businesses that effectively manage customer flow during busy periods are more likely to maintain service consistency and customer satisfaction. Studies also indicate that temporal analytics contribute to better resource utilization because businesses can allocate equipment, ingredients, and personnel according to expected operational activity. Consequently, literature recognizes peak-hour analysis as both an operational efficiency tool and a customer service enhancement strategy.

Inventory monitoring and management systems are likewise prominent topics in related studies, particularly for their role in minimizing waste and maintaining product availability. Effective inventory tracking helps organizations prevent overstocking, which can lead to spoilage and waste, as well as understocking, which may result in missed sales opportunities. Research indicates that inventory systems improve operational continuity by maintaining balanced stock levels aligned with actual demand. Automated inventory tracking reduces dependence on manual stock counting and lowers the likelihood of human error. It also provides businesses with real-time visibility into stock movement and consumption rates. Researchers highlight that inventory data can support more accurate purchasing decisions and supplier coordination. This is particularly critical in food and beverage businesses where inventory often includes perishable ingredients. Inventory analytics also help identify slow-moving items and purchasing inefficiencies. Improved inventory control directly contributes to cost reduction and profitability enhancement. Consequently, literature consistently positions inventory management systems as essential operational tools in food service environments. Researchers further explain that inventory monitoring systems improve accountability by maintaining detailed records of stock movement and usage patterns. This reduces opportunities for inventory discrepancies, wastage, and unauthorized stock handling. Studies also reveal that businesses utilizing automated inventory systems are better able to maintain operational continuity during supply chain disruptions because inventory visibility supports proactive replenishment planning. In addition, inventory analytics can assist management in identifying seasonal purchasing trends and supplier performance issues. Accurate stock monitoring is especially important in cafés and food kiosks where ingredient freshness directly affects product quality and customer satisfaction. Literature therefore recognizes inventory management systems as significant contributors to operational efficiency, financial control, and service reliability.

Furthermore, literature on automation and cloud-based technologies demonstrates that digital systems reduce administrative workload and improve data reliability. Automated reporting tools eliminate many of the errors associated with manual encoding and spreadsheet preparation. Cloud-based platforms allow businesses to store and access centralized information through internet-enabled devices from various locations. Researchers observe that automation accelerates report generation and decreases the time spent consolidating business data manually. This enables managers to focus more on analysis and strategic activities rather than administrative processing. Cloud-based systems also support scalability by allowing organizations to expand user access and system functions with minimal infrastructure changes. Remote accessibility further allows business owners and managers to monitor operations outside physical business premises. Studies additionally indicate that centralized cloud systems improve collaboration by ensuring that all users access consistent and updated records. Enhanced data security and automated backup mechanisms also reduce the risk of data loss. Overall, literature identifies automation and cloud computing as critical components of modern business information systems. Researchers additionally highlight that cloud technologies improve organizational flexibility because information can be accessed through multiple devices and locations without requiring complex hardware infrastructure. Automated systems also reduce dependency on paper-based documentation, which contributes to improved environmental sustainability and operational organization. Literature suggests that businesses adopting cloud-based solutions are more capable of supporting remote management and multi-branch coordination. This is especially beneficial for growing enterprises that require centralized oversight across geographically distributed operations. Furthermore, automated systems reduce repetitive administrative tasks that often contribute to employee fatigue and operational inefficiencies. Consequently, literature consistently identifies automation and cloud integration as important drivers of organizational productivity, scalability, and digital transformation.

Within the Philippine business environment, scholars also highlight the influence of contextual factors such as holidays, festivals, and local events on customer demand and sales performance. These external variables can significantly alter traffic volume and purchasing behavior in food and beverage establishments. As a result, researchers stress the need for analytical systems capable of adapting to localized market conditions and seasonal trends. Businesses that monitor these contextual influences can better anticipate fluctuations in demand and prepare operationally. This includes adjusting inventory levels, staffing schedules, and promotional strategies in anticipation of event-driven sales increases or decreases. Studies suggest that understanding local demand drivers improves market responsiveness and strategic planning. For community-oriented establishments like neighborhood cafés, localized consumer behavior may strongly affect sales outcomes. Systems capable of analyzing contextual sales patterns provide more meaningful insights than generalized reporting alone. Adaptability to local business conditions is therefore considered a significant advantage in analytical system design. Overall, literature supports the inclusion of localized and seasonal trend analysis within business intelligence platforms. Researchers additionally explain that localized market analysis allows businesses to better understand cultural and social influences affecting customer purchasing behavior. In the Philippine setting, community events, school schedules, tourism patterns, and weather conditions may all contribute to fluctuations in sales activity. Businesses capable of analyzing these localized influences can develop more targeted and effective operational strategies. Literature also indicates that organizations with contextual analytical capabilities are more responsive to sudden market shifts and changing consumer trends. This adaptability strengthens customer engagement and improves organizational competitiveness within local markets. Consequently, researchers strongly recommend integrating contextual and localized analytics into food service business intelligence systems.

In summary, related literature consistently supports the integration of business intelligence tools into food and beverage operations to improve efficiency, accuracy, and responsiveness in business management. Prior studies demonstrate that BI technologies enhance operational visibility and facilitate data-based strategic planning. These systems improve performance monitoring by consolidating fragmented data into centralized and actionable insights. They also strengthen forecasting, inventory planning, sales analysis, and branch evaluation processes. In food service environments, business intelligence solutions contribute to improved service quality and operational effectiveness through timely access to relevant business metrics. Researchers note that organizations utilizing BI systems become more adaptive and competitive in dynamic market environments. These technologies reduce reliance on manual reporting while increasing the reliability and speed of business analysis. They further support long-term growth by enabling continuous performance optimization and strategic refinement. As digital adoption continues to expand, BI implementation is becoming increasingly essential for SMEs and growing enterprises alike. Collectively, the literature provides substantial support for the development and implementation of the BREWINSIGHT BI system for Kape Tubong. Furthermore, the reviewed literature consistently demonstrates that organizations adopting analytical technologies are more capable of sustaining operational consistency and strategic flexibility during periods of business growth. Researchers also emphasize that business intelligence systems strengthen organizational accountability by making operational performance more transparent and measurable. In competitive food service environments, the ability to access timely and accurate information is increasingly recognized as a significant strategic advantage. Literature therefore supports the argument that integrated analytical platforms are no longer optional technologies but essential operational tools for modern business management.

Another significant concept discussed in related literature is centralized data management, which refers to the consolidation of business information from multiple operational points into a single unified repository. Researchers emphasize that centralized data systems improve organizational control by ensuring that all departments and branches operate using the same standardized information source. In businesses with multiple outlets or kiosks, centralized databases eliminate inconsistencies caused by decentralized spreadsheets or manually maintained records. This consolidation improves reporting accuracy and reduces the duplication of information across departments. Centralized data management also simplifies auditing, performance tracking, and historical record retrieval by maintaining organized and structured datasets in one location. Studies suggest that organizations with centralized information systems can perform comparative analysis more efficiently because data from multiple branches can be accessed simultaneously in a consistent format. Furthermore, centralized systems improve collaboration among management personnel by providing a shared and synchronized operational view. This unified data environment strengthens decision-making by reducing discrepancies in reported figures and interpretations. Researchers therefore consider centralized data management a foundational element of effective business intelligence implementation. Its inclusion in analytical platforms is widely recommended for organizations operating across multiple business locations. Researchers additionally note that centralized systems improve data governance by establishing standardized procedures for data entry, validation, and reporting across all operational units. This consistency enhances the reliability of organizational records and reduces confusion caused by varying reporting practices between branches. Centralized repositories also strengthen data security because access permissions and monitoring mechanisms can be managed within a single platform. Studies indicate that organizations utilizing centralized systems experience improved efficiency in generating organization-wide performance reports and executive summaries. In multi-branch businesses, centralized databases also support faster communication between branch managers and administrative personnel because updated operational information becomes immediately accessible. Furthermore, centralized data structures improve scalability by allowing additional branches or operational units to be integrated into the system without major restructuring. Literature therefore recognizes centralized data management as essential for improving operational coordination, reporting consistency, and long-term organizational growth.

Another area emphasized in literature is branch performance evaluation, particularly in businesses managing multiple outlets, franchises, or satellite branches. Branch performance analysis allows organizations to compare operational outputs, revenue generation, and sales efficiency across different locations. This comparative insight helps management identify which branches are outperforming expectations and which require operational intervention. Researchers note that branch-level analytics improve accountability by establishing measurable performance benchmarks for each location. Comparative branch reporting also helps businesses identify location-specific trends, operational inefficiencies, and contextual market differences affecting sales performance. In food and beverage operations, branch performance evaluation can reveal differences in customer traffic, demand behavior, and staff productivity across locations. This enables management to implement targeted interventions based on branch-specific performance issues rather than generalized assumptions. Studies suggest that branch comparison tools contribute to more strategic expansion planning by identifying characteristics of successful locations. Consequently, branch performance monitoring is recognized as a critical analytical practice for multi-branch enterprises. This literature supports the inclusion of comparative branch analytics in the BREWINSIGHT BI platform. Researchers further explain that branch-level analytics contribute to organizational transparency by making operational performance measurable and comparable across all locations. Through comparative dashboards and branch-specific reporting, management can identify operational strengths and weaknesses more effectively. Literature also indicates that branch evaluation systems improve managerial accountability because branch supervisors become more aware of measurable operational expectations. In addition, comparative analysis enables organizations to identify best practices from high-performing branches that can be replicated across other locations. Businesses can also use branch performance metrics to evaluate the effectiveness of staffing strategies, promotional activities, and customer service practices within specific operational contexts. Studies further reveal that branch analytics support evidence-based expansion decisions because management can identify which operational conditions contribute most significantly to branch success. Overall, literature consistently highlights branch performance analysis as an important tool for improving operational consistency, organizational accountability, and strategic planning.

Scholars also highlight the growing relevance of real-time reporting systems in enhancing managerial responsiveness and operational agility. Real-time reporting allows decision-makers to access updated business data immediately after transactions occur, reducing delays associated with batch processing or manual report preparation. Researchers argue that timely access to current operational information improves the speed and quality of managerial responses to performance fluctuations. In dynamic industries such as food and beverage, where demand patterns may shift rapidly within a single day, real-time visibility is especially valuable. Businesses utilizing real-time reporting can respond more quickly to sales declines, product shortages, or unexpected demand surges. This capability supports immediate operational adjustments and reduces the risk of delayed interventions. Studies further note that real-time data access improves managerial confidence by ensuring decisions are based on the most current available information. It also enhances situational awareness across distributed business locations. As a result, real-time reporting is increasingly considered a standard feature of modern business intelligence systems. Its integration strengthens the practical value of analytical platforms intended for operational decision support. Researchers additionally emphasize that real-time systems improve organizational responsiveness by enabling management to monitor ongoing business conditions continuously throughout operational hours. This continuous visibility supports faster identification of anomalies such as unusual sales declines, inventory discrepancies, or sudden increases in customer traffic. Literature also suggests that real-time reporting contributes to improved customer satisfaction because operational adjustments can be implemented immediately when service issues arise. In food service businesses, real-time visibility may help management reduce waiting times, improve product availability, and maintain service consistency during peak demand periods. Furthermore, real-time reporting reduces reliance on end-of-day summaries that may delay critical business decisions. The capability to access live operational information therefore strengthens strategic agility and operational control within fast-paced business environments. Consequently, scholars strongly advocate for the integration of real-time analytics within modern business intelligence platforms.

In addition, literature frequently discusses the importance of historical trend analysis in supporting business forecasting and long-term strategic planning. Historical trend analysis involves examining past performance data to identify recurring patterns, growth trajectories, and seasonal fluctuations over time. Researchers suggest that organizations using historical analytics can better anticipate future operational demands and market conditions. In food and beverage businesses, historical sales patterns can reveal recurring high-demand periods associated with holidays, weekends, local events, or seasonal purchasing behavior. These insights assist businesses in planning inventory procurement, staffing schedules, and promotional activities more effectively. Historical trend analysis also supports long-term performance evaluation by enabling organizations to assess progress over extended operational periods. Comparative historical reporting helps management determine whether business performance is improving, declining, or remaining stable. Studies indicate that organizations with access to historical analytics are better equipped to engage in proactive rather than reactive management. Consequently, literature strongly supports incorporating historical trend analysis into business intelligence platforms. This validates BREWINSIGHT BI’s focus on revenue pattern analysis across multiple time intervals. Researchers additionally explain that historical trend analysis strengthens organizational learning by enabling businesses to evaluate the outcomes of previous operational strategies and management decisions. Through longitudinal data examination, organizations can identify which promotional activities, pricing strategies, or operational adjustments generated positive performance outcomes. Literature also notes that historical analytics support more accurate forecasting because predictive models become more reliable when based on extensive historical datasets. In food service operations, historical analysis may reveal recurring consumer preferences that can guide future menu planning and inventory management. Furthermore, long-term trend analysis assists management in evaluating branch growth trajectories and revenue sustainability across multiple operational periods. This capability improves strategic planning by providing measurable evidence for expansion, investment, and operational refinement initiatives. Overall, researchers recognize historical analytics as essential for strengthening forecasting accuracy, strategic planning quality, and organizational adaptability.

Finally, related literature highlights the strategic role of decision-support systems in reducing managerial uncertainty and improving organizational planning quality. Decision-support systems integrate analytical tools, reporting mechanisms, and structured data presentation to help management evaluate alternatives and make informed operational decisions. Researchers note that these systems reduce reliance on intuition-based management practices by providing objective evidence to support strategic actions. In small business environments, decision-support systems are especially valuable because management often handles multiple responsibilities simultaneously and requires efficient access to summarized insights. Studies indicate that businesses using decision-support platforms demonstrate improved consistency in strategic planning and stronger justification for operational changes. These systems also support transparency by allowing decision-makers to trace conclusions back to measurable performance data. In addition, decision-support tools improve confidence in managerial actions because decisions are grounded in verifiable analytical evidence. This contributes to more disciplined and accountable business management practices. Therefore, literature consistently recognizes decision-support systems as essential tools for modern organizational governance and strategic execution. This further reinforces the relevance of BREWINSIGHT BI as a decision-support and analytical platform for Kape Tubong. Researchers additionally emphasize that decision-support systems improve organizational coordination because managers across different departments can access the same analytical information when making operational decisions. This consistency reduces communication gaps and minimizes conflicting interpretations of business performance. Literature also indicates that decision-support technologies contribute to faster problem-solving because operational concerns can be identified, analyzed, and addressed using integrated analytical reports. In competitive business environments, rapid and evidence-based decision-making is considered critical for maintaining operational efficiency and market responsiveness. Decision-support systems may also improve strategic evaluation by enabling organizations to simulate potential operational outcomes using historical and real-time data insights. Furthermore, these systems strengthen managerial accountability because business decisions can be reviewed and justified using documented analytical evidence. Studies consistently show that organizations utilizing decision-support tools demonstrate stronger adaptability, improved planning discipline, and more effective resource allocation practices. Consequently, literature strongly supports integrating decision-support functionalities within business intelligence platforms intended for operational and strategic management purposes.

2.4 Synthesis of the Reviewed Literature

The reviewed foreign and local studies, together with the related literature, collectively provide a substantial theoretical and practical foundation for the development of **BREWINSIGHT BI: Revenue Pattern Analytics System for Kape Tubong**. Across the examined sources, there is a recurring emphasis on the strategic value of data visualization, automation, and business analytics in improving organizational performance. The literature consistently demonstrates that businesses benefit from transforming raw operational data into meaningful analytical insights through structured business intelligence systems. These findings establish that data accessibility and interpretability are essential factors in effective managerial decision-making. The reviewed studies also indicate that businesses adopting analytical technologies demonstrate stronger operational responsiveness and strategic adaptability. In food and beverage environments, where customer behavior and demand patterns can fluctuate rapidly, the ability to interpret business data efficiently is especially important. The reviewed literature further confirms that analytical systems improve not only strategic planning but also day-to-day operational management. Collectively, these sources reinforce the importance of integrating business intelligence tools into modern F&B business operations. The convergence of findings from multiple authors and contexts provides strong justification for implementing a revenue analytics platform in Kape Tubong's operational environment. Therefore, the body of reviewed literature strongly supports the conceptual and technical foundation of the proposed BREWINSIGHT BI system.

Foreign studies highlight that interactive dashboards significantly improve decision-making speed and accuracy by transforming raw data into meaningful visual insights. These studies show that dashboards reduce the complexity of interpreting large transactional datasets through visual representations such as graphs, charts, and trend indicators. Researchers consistently report that visual analytics improve managerial comprehension of key performance metrics and operational patterns. Dashboard systems also enable organizations to detect anomalies, performance declines, and revenue fluctuations more efficiently than traditional reporting methods. In addition, foreign literature indicates that dashboard-based environments improve collaboration by providing a shared visual framework for discussing business performance. Interactive filtering and drill-down functionalities further enhance analytical depth and allow users to investigate performance at granular levels. These capabilities make dashboards particularly effective for businesses requiring fast, evidence-based decision-making. The reviewed foreign studies therefore validate the use of dashboard interfaces as central components of business intelligence systems. Their findings directly support the inclusion of visual analytics and reporting dashboards within BREWINSIGHT BI. This reinforces the proposed system's objective of delivering clear and actionable insights to

Kape Tubong management.

Local studies further strengthen this perspective by demonstrating that Philippine small and medium enterprises experience measurable improvements when adopting business intelligence and digital monitoring tools. Research involving local F&B and retail businesses shows that dashboard systems reduce report preparation time, improve sales analysis speed, and enhance management responsiveness. Local studies also reveal that BI adoption contributes to better data organization and more accurate monitoring of business performance. These findings indicate that analytical systems are not limited to large corporations but are equally valuable for smaller enterprises operating in localized markets. In the Philippine context, SMEs often face operational limitations such as fragmented data, manual reporting, and resource constraints. The literature suggests that business intelligence tools address these issues by automating data processing and centralizing business information. This demonstrates the practicality and relevance of implementing BI solutions within small-scale business environments like Kape Tubong. Furthermore, local adoption studies confirm that Philippine businesses are increasingly embracing digital transformation for competitive advantage. The applicability of BI systems in local SME operations validates the feasibility of BREWINSIGHT BI's implementation context. Thus, local literature provides contextual support for the proposed system's adoption within a Philippine food and beverage business setting.

The reviewed literature also reveals that combining multiple business intelligence components into a single platform produces more comprehensive and effective decision-support capabilities. Studies indicate that integrating features such as sales tracking, revenue trend analysis, product performance evaluation, and peak-hour identification enhances analytical depth and strategic value. Rather than examining isolated reports, integrated BI systems allow management to understand the relationships between multiple operational variables simultaneously. This holistic visibility improves strategic planning and operational diagnosis by providing a broader perspective on business performance. Integrated systems also improve workflow efficiency by reducing the need to access multiple separate tools or reports. Researchers note that centralized BI platforms promote consistency in data interpretation and reporting standards across organizations. In addition, integration supports better comparative analysis across branches, products, and operational periods. Businesses benefit from improved decision quality when all relevant metrics are available within a unified analytical environment. These findings support the design approach of BREWINSIGHT BI as a consolidated platform rather than a fragmented reporting solution. Therefore, literature strongly validates the system's integrated architecture and multifunctional analytics framework.

In the context of **Kape Tubong**, the reviewed findings are highly applicable due to the alignment between the literature-identified challenges and the organization's current operational difficulties. Kape Tubong experiences fragmented sales records, delayed manual reporting, and limited visibility into kiosk-level performance across multiple locations. These issues reflect the same inefficiencies and informational gaps identified in both local and foreign studies regarding manual and decentralized business monitoring. Literature suggests that such limitations hinder strategic planning, reduce operational responsiveness, and weaken performance evaluation. BREWINSIGHT BI directly addresses these challenges by centralizing transactional data into a unified analytical dashboard system. Through this approach, management can monitor sales performance, compare kiosk outputs, evaluate product demand, and identify revenue trends in real time. The proposed system thereby resolves the information fragmentation and analysis delays currently affecting Kape Tubong's operations. Furthermore, its web-based accessibility ensures that authorized personnel can review business insights remotely and conveniently. This operational alignment demonstrates the practical necessity of the proposed system within Kape Tubong's business environment. Thus, the literature confirms that BREWINSIGHT BI is both relevant and responsive to the organization's identified needs.

Ultimately, the synthesis of the reviewed literature confirms a clear and substantial need for an integrated, dashboard-based business intelligence platform tailored to the operational realities of small food and beverage enterprises in the Philippines. Existing research consistently supports the value of combining data visualization, automated reporting, centralized sales tracking, and analytical decision-support tools into one unified system. The literature demonstrates that such systems improve efficiency, reduce manual workload, and enhance strategic decision-making across business functions. It also confirms that SMEs increasingly require scalable and accessible digital tools to remain competitive in dynamic market environments. Despite this, many smaller businesses continue to operate using fragmented or manual sales monitoring practices, indicating an implementation gap in practical BI adoption. BREWINSIGHT BI addresses this gap by offering a specialized revenue analytics solution designed specifically for Kape Tubong's operational needs. The proposed platform consolidates essential BI functionalities into a centralized, web-based system appropriate for multi-kiosk F&B management. Its development is strongly justified by the theoretical, empirical, and contextual evidence presented throughout the reviewed literature. As such, BREWINSIGHT BI is positioned not only as a technological enhancement but also as a strategic operational solution. Overall, the reviewed literature provides comprehensive support for the necessity, design, and implementation of the proposed system.

Moreover, the reviewed literature collectively suggests that the successful implementation of business intelligence systems depends not only on the availability of analytical tools but also on how effectively these tools are aligned with organizational workflows and decision-making processes. Researchers emphasize that BI platforms generate the greatest value when their functionalities are tailored to the specific operational requirements of the organization rather than implemented as generic reporting tools. This highlights the importance of designing BREWINSIGHT BI according to the unique structure, workflow, and sales monitoring needs of Kape Tubong. By aligning system features with the business's kiosk-based operational setup, branch monitoring requirements, and revenue analysis objectives, the proposed platform increases its practical relevance and expected effectiveness. This literature-based principle supports the study's focus on developing a customized analytics solution rather than adopting a generalized business reporting system. Tailored system design improves usability, operational fit, and long-term adoption among intended users. Consequently, the customization approach applied in BREWINSIGHT BI is consistent with best practices identified in prior business intelligence research. This strengthens the system's expected practical value and implementation success.

In addition, several reviewed sources emphasize that the adoption of digital analytical systems contributes significantly to organizational digital transformation and technological maturity. Businesses that implement BI platforms often develop stronger data management practices, more structured reporting processes, and improved organizational discipline in performance monitoring. These improvements extend beyond analytics alone and influence broader operational behavior within the organization. For Kape Tubong, the implementation of BREWINSIGHT BI may therefore contribute not only to improved sales analysis but also to the establishment of more systematic and technology-oriented business management practices. The system may encourage greater accountability in branch reporting, improved standardization of sales documentation, and more consistent monitoring routines across kiosk locations. Over time, this can support the business's transition toward a more digitally mature operational model. The literature indicates that such transformation strengthens organizational adaptability and competitiveness in increasingly data-driven markets. Therefore, the benefits of BREWINSIGHT BI may extend beyond immediate analytical improvements and contribute to broader organizational development.

Finally, the reviewed literature indicates that centralized business intelligence systems provide long-term strategic advantages by creating scalable digital infrastructures that can support future enhancements and business expansion. Organizations that adopt centralized analytics platforms establish a technological foundation upon which more advanced capabilities—such as predictive analytics, automated forecasting, inventory optimization, and customer behavior modeling—can later be built. This suggests that BREWINSIGHT BI should not be viewed solely as a short-term operational tool, but also as an initial step toward broader digital transformation and future enterprise system development within Kape Tubong. By centralizing transactional and analytical processes in one structured platform, the business gains a scalable framework capable of supporting future innovation and expansion. This scalability is especially important if Kape Tubong continues to increase the number of kiosks or diversify its operational activities. The literature therefore supports the strategic value of implementing foundational BI systems even in smaller enterprises with current limited digital infrastructure. In this regard, BREWINSIGHT BI represents both an immediate operational solution and a strategic investment in future business capability. Accordingly, the proposed system aligns with long-term technological and organizational development goals identified in contemporary business intelligence literature.

CHAPTER 3

METHODOLOGY

3.1 Research Design

This study employs a **Developmental Research Design**, which is appropriate because the primary objective of the research is to systematically design, develop, and evaluate the **BREWINSIGHT BI: Revenue Pattern Analytics System** as a practical solution to the operational challenges currently experienced by Kape Tubong. Developmental research is particularly suitable for studies involving the creation of technological systems, software applications, or process innovations intended to solve real-world organizational problems. Unlike purely descriptive or experimental research, developmental research focuses on both the systematic investigation of existing issues and the structured development of a corresponding intervention or solution. This design enables the researchers to move beyond problem identification by producing a tangible and functional output that addresses identified business needs. It is commonly applied in information technology and systems development studies where the objective is to create and assess software-based solutions. Through this approach, the study ensures that the resulting system is grounded in actual operational requirements rather than theoretical assumptions alone. The developmental design also supports iterative refinement, allowing the system to evolve through continuous testing and feedback. Because the study aims to generate a deployable analytics platform rather than merely analyze existing phenomena, this research design is highly appropriate. Furthermore, developmental research promotes alignment between user needs, technical implementation, and practical usability. Therefore, the use of developmental research design provides a strong methodological framework for the systematic creation and validation of the BREWINSIGHT BI platform.

In addition, developmental research supports a structured process for translating identified organizational problems into measurable technical objectives and corresponding system functionalities. This ensures that each developed component of the system is purposefully aligned with a documented operational need. By following this design, the researchers are able to maintain a systematic relationship between problem analysis, system development, and performance evaluation throughout the study. The methodology also allows for continuous validation at each development stage to ensure that system outputs remain aligned with project goals. Such an approach is essential when developing decision-support technologies intended for actual business implementation. The research design therefore strengthens both the practical relevance and methodological rigor of the project.

During the initial stage of the study, a descriptive research component is incorporated to analyze and understand the current operational practices of Kape Tubong. This phase focuses on documenting the existing methods used by the business for recording sales, consolidating reports, and monitoring branch performance. The researchers examine how transactional data is currently collected from kiosk operations and how this information is processed for managerial review. Particular attention is given to identifying inefficiencies, redundancies, and limitations in the present workflow. Existing challenges such as fragmented sales records, delayed report preparation, and lack of centralized data visibility are carefully assessed during this stage. In addition, the descriptive phase includes observing current reporting procedures and evaluating the accessibility and usefulness of available sales information. Interviews and consultations with the business owner and operational staff are also conducted to gather qualitative insights regarding recurring operational difficulties. These activities help establish a detailed understanding of the business context and define the specific problems that the proposed system must address. By thoroughly examining the current operational environment, the study ensures that the development process is informed by real business conditions. This descriptive analysis provides the foundational basis for identifying system requirements and guiding the development of relevant functionalities.

The descriptive component also assists in identifying user expectations, workflow preferences, and practical limitations that may affect system adoption. Understanding these factors is important in ensuring that the proposed platform remains compatible with the organization’s daily operational practices. This phase helps minimize the risk of developing features that are technically functional but operationally impractical. It also ensures that the system addresses not only technical requirements but also usability and workflow considerations. As a result, the descriptive phase contributes significantly to the relevance and practicality of the final system design.

Following the identification of operational issues and user requirements, the developmental phase of the study focuses on the design and construction of the BREWINSIGHT BI system. During this phase, the researchers translate the identified requirements into system specifications, interface designs, database structures, and functional modules. The system is developed to include a centralized dashboard capable of aggregating and presenting sales data from multiple Kape Tubong kiosks in a unified platform. Core analytical functionalities such as revenue trend analysis, product performance tracking, branch comparison, and peak-hour identification are integrated into the system architecture. Automated reporting features are also implemented to reduce manual report generation and improve data accessibility for management. The system design prioritizes usability, responsiveness, and accessibility through a web-based deployment model. Security mechanisms such as user authentication and role-based access controls are incorporated to ensure that only authorized personnel can access sensitive business information. Throughout development, iterative refinement is applied to ensure alignment between system features and user expectations. Technical considerations such as scalability, reliability, and maintainability are also incorporated into the development process. This phase ultimately results in the creation of a functional prototype of the BREWINSIGHT BI platform tailored to Kape Tubong's operational needs.

Moreover, the developmental process includes repeated internal assessments of system progress to ensure that design outputs remain consistent with technical specifications and business requirements. Prototype evaluations may be conducted during intermediate development stages to verify interface design, workflow logic, and analytical accuracy before final implementation. This iterative approach allows developers to identify issues early and implement improvements before full deployment. By continuously refining the prototype throughout development, the researchers improve the quality and usability of the final product. This iterative construction strategy contributes to a more stable and user-centered software solution.

After development, the system undergoes testing and evaluation to determine its functionality, reliability, usability, and overall effectiveness. Functional testing is conducted to verify that all system modules perform according to their intended specifications and process data accurately. Reliability testing ensures that the system operates consistently under normal usage conditions without significant performance issues or failures. Usability evaluation is performed to assess the system's ease of use, interface clarity, and user-friendliness from the perspective of intended users. Selected Kape Tubong personnel are involved in user acceptance testing to provide direct feedback regarding the practicality and effectiveness of the system in real operational scenarios. Their observations and recommendations are documented and analyzed to identify areas requiring revision or enhancement. Based on the evaluation results, the researchers refine and improve the system to address identified issues and optimize user experience. This iterative testing and feedback process helps ensure that the final system is both technically sound and operationally relevant. The evaluation stage also confirms whether the developed system successfully addresses the identified business problems and achieves its intended objectives. Through comprehensive testing and refinement, the study ensures that BREWINSIGHT BI meets acceptable standards of quality, functionality, and usability before final implementation.

The evaluation process also helps determine the extent to which the system satisfies stakeholder expectations and business requirements. User feedback gathered during testing serves as a practical basis for validating the system's real-world applicability. This ensures that the platform performs not only according to technical design standards but also according to user experience and operational effectiveness criteria. Testing outcomes may further reveal opportunities for enhancement that can improve long-term maintainability and system scalability. Therefore, the evaluation phase plays a critical role in validating both the technical and practical success of the developed platform.

Overall, the use of a Developmental Research Design enables the study to effectively bridge the gap between theoretical knowledge and practical application by producing a research-based yet application-oriented technological solution. Rather than limiting the study to conceptual analysis, this design supports the development of a fully functional business intelligence system that directly addresses real operational concerns. It allows the researchers to systematically investigate business challenges, gather user requirements, and translate findings into practical software features. The iterative nature of developmental research further ensures that the system evolves through continuous validation and refinement. This approach enhances the relevance and effectiveness of the resulting platform by grounding development decisions in empirical findings and user feedback. Additionally, developmental research promotes accountability in system design by requiring that the developed solution be tested and evaluated against established objectives. The methodology therefore ensures that BREWINSIGHT BI is not only technically feasible but also operationally valuable to its intended users. By integrating analysis, development, testing, and refinement into a structured research process, the study achieves both academic rigor and practical utility. This makes developmental research particularly suitable for technology-driven capstone projects focused on software innovation and applied system development. Consequently, the chosen research design provides a comprehensive methodological framework for the successful completion of the study.

Furthermore, the selected methodology contributes to the credibility of the study by ensuring that each stage of system creation is systematically documented, justified, and evaluated. This strengthens the academic validity of the research while also enhancing the reproducibility of the development process for future studies. The structured nature of developmental research also ensures transparency in how design decisions were made throughout the project lifecycle. As a result, the methodology supports both scholarly rigor and practical software engineering standards. This comprehensive research design ultimately positions the study to produce a meaningful technological contribution to both academic and business contexts.

Moreover, the developmental research design provides the study with the flexibility to adapt throughout the software development process while still following a systematic methodological structure. This adaptability is essential because system requirements may change as users interact with the prototype and share feedback regarding its functionality and usability. The methodology allows the researchers to integrate necessary improvements without affecting the consistency of the research process. Through continuous collaboration between the developers and intended users, the system becomes more responsive to actual organizational workflows and operational requirements. This cooperative process enhances both user satisfaction and the likelihood of successful system implementation.

The selected methodology also promotes the consistent preparation of technical documentation during every stage of development. Records such as system specifications, database structures, workflow diagrams, and testing procedures help maintain organization and continuity throughout the project. These documents are valuable not only during the completion of the study but also for future system maintenance, scalability, and enhancement. Proper documentation ensures transparency in the development process and contributes to the long-term sustainability of the BREWINSIGHT BI platform.

In addition, the methodology emphasizes evidence-based decision-making throughout the development cycle. Design revisions and system enhancements are guided by collected data, evaluation findings, and user feedback rather than personal assumptions. This approach improves the objectivity of the study and reduces the risk of implementing unnecessary features or ineffective functionalities. Each modification introduced into the system is therefore connected to identified operational concerns and practical business requirements. As a result, the final platform becomes more efficient, relevant, and aligned with organizational needs.

Another advantage of developmental research is its ability to combine qualitative and quantitative methods of evaluation. Qualitative feedback from users helps assess usability, accessibility, and overall user experience, while quantitative testing measures system performance, processing speed, and functional accuracy. The integration of these evaluation methods allows the researchers to obtain a broader and more balanced assessment of the system’s effectiveness. This comprehensive evaluation process strengthens the reliability and credibility of the study’s findings.

Furthermore, the chosen methodology allows the researchers to identify potential future improvements that may expand the capabilities of the BREWINSIGHT BI platform beyond its initial scope. During testing and evaluation, users may suggest additional analytical features, reporting tools, or operational enhancements that can guide future upgrades. This ensures that the system remains adaptable to organizational expansion and evolving technological demands. Consequently, the methodology contributes not only to the current development of the platform but also to its future sustainability and continued relevance.

The study also benefits from the problem-solving orientation of developmental research.

Rather than simply describing operational difficulties, the methodology directs the researchers toward creating practical technological solutions that address identified issues. This solution-focused approach is particularly valuable in information technology studies where the primary goal is to improve organizational efficiency through system innovation. Through systematic planning, implementation, testing, and refinement, the methodology ensures that the developed platform effectively supports both business operations and managerial decision-making.

Additionally, the methodology highlights the importance of user-centered system development. By involving Kape Tubong personnel during analysis, testing, and evaluation activities, the researchers ensure that the system reflects the actual experiences, preferences, and operational needs of its intended users. This participation contributes to improved interface design, workflow compatibility, and functional relevance. User involvement also increases the possibility of successful adoption because employees become more familiar and comfortable with the system during the development process.

Overall, the selected research design establishes a comprehensive framework that integrates analysis, development, and evaluation into a single organized process. Each stage contributes directly to achieving the objectives of the study, from identifying operational challenges to validating the effectiveness of the completed system. The structured progression of activities ensures that BREWINSIGHT BI is developed according to academic standards while remaining practical and responsive to real business conditions. Through this methodological approach, the study demonstrates how developmental research can support the creation of technology-based solutions that improve operational efficiency, strategic decision-making, and organizational growth in local business environments.

3.2 System Development Methodology

Agile is a software development and project management methodology that emphasizes flexibility, collaboration, and continuous improvement throughout the development process. It focuses on delivering work in small, manageable increments rather than completing the entire project at once. Agile allows development teams to adapt quickly to changes in requirements or user feedback during the project lifecycle. This methodology encourages frequent communication and collaboration among developers, stakeholders, and end users. Each development cycle, often called a sprint or iteration, produces a working portion of the system for review and evaluation. Agile promotes regular testing and refinement to identify issues early and improve system quality continuously. It prioritizes customer satisfaction by involving users throughout the development process and incorporating their feedback into ongoing improvements. Compared to traditional development models, Agile is more adaptable to changing project conditions and evolving business needs. It is widely used in software development because it reduces project risks and improves development efficiency. Overall, Agile provides a structured yet flexible approach for creating high-quality systems through iterative progress and collaboration.

Additionally, Agile supports incremental delivery of system features, allowing stakeholders to evaluate partial outputs throughout development rather than waiting until project completion. This approach improves transparency by making development progress visible at every stage of the project lifecycle. It also reduces the risk of large-scale rework because errors and design issues can be identified earlier during iteration reviews. Through continuous collaboration and evaluation, Agile ensures that the developed system remains aligned with stakeholder expectations and operational requirements. For a project such as BREWINSIGHT BI, where business requirements may evolve during development, Agile provides an appropriate framework for accommodating changes efficiently. The methodology also encourages continuous interaction between researchers, system developers, and users to ensure that the platform addresses actual operational challenges encountered by Kape Tubong. Frequent communication enables the development team to clarify requirements immediately and respond to operational concerns without causing major delays in development activities.

Agile further emphasizes adaptability and responsiveness, which are important in projects involving dynamic business environments and evolving operational needs. Since BREWINSIGHT BI focuses on sales monitoring, analytics, and business intelligence functions, management requirements may change as additional insights and operational expectations emerge during development. Through Agile, modifications to dashboard layouts, reporting structures, and analytical functionalities can be implemented gradually without disrupting the entire development process. This flexibility enables the researchers to continuously refine the system according to stakeholder recommendations and usability feedback. Agile also supports the prioritization of critical system features, ensuring that the most essential components are developed and tested first before proceeding to more advanced functionalities. As a result, the development process becomes more organized, manageable, and responsive to business priorities.

Another important characteristic of Agile is its strong emphasis on iterative testing and continuous quality assurance. Instead of postponing testing activities until the end of development, Agile incorporates testing within every sprint or iteration cycle. This allows the researchers to identify software defects, interface inconsistencies, and functional errors at earlier stages of development. Early detection of issues minimizes the likelihood of major system failures during deployment and reduces the cost of correcting errors later in the project lifecycle. Continuous testing also improves system stability and reliability because each feature undergoes repeated evaluation and refinement before integration into the complete platform. In the context of BREWINSIGHT BI, iterative testing is essential to ensure the accuracy of sales analytics, dashboard visualizations, and reporting functionalities. The reliability of data processing and visualization features is critical because management decisions will depend on the information generated by the system.

Agile methodology also promotes user-centered development by involving stakeholders actively throughout the project lifecycle. Rather than limiting stakeholder participation to the initial planning phase, Agile encourages continuous feedback and evaluation during every development iteration. This ensures that the developed system remains aligned with actual user expectations and operational workflows. For BREWINSIGHT BI, management and staff members of Kape Tubong can evaluate prototype dashboards, reporting modules, and analytical features during development and provide recommendations for improvement. User participation strengthens system usability because suggestions from actual end users can guide interface design, functionality enhancement, and workflow optimization. This collaborative process increases the likelihood that the final system will effectively support operational decision-making and business management activities.

Furthermore, Agile supports efficient project management by dividing the development process into smaller and more achievable tasks. These smaller development activities allow the researchers to monitor progress more effectively and evaluate whether project objectives are being achieved within the intended timeframe. Sprint-based development also improves productivity because the development team can focus on completing prioritized objectives during each iteration. The methodology additionally promotes accountability among developers because tasks, deliverables, and timelines are clearly defined within every sprint cycle. This structured approach helps maintain consistent development progress while still allowing flexibility for revisions and feature enhancements. Agile therefore combines disciplined project organization with adaptability, making it highly suitable for information system development projects.

Another advantage of Agile is its capability to support continuous improvement throughout system development. At the end of every sprint, developers and stakeholders review completed functionalities and identify areas requiring refinement or enhancement. This reflective process allows the development team to improve both the system and the development strategy itself. Lessons learned from earlier iterations can be applied immediately in succeeding development cycles, improving overall efficiency and system quality. Continuous improvement also ensures that system components evolve according to actual operational requirements rather than fixed assumptions established at the beginning of the project. In the case of BREWINSIGHT BI, iterative enhancement is important because analytical requirements and dashboard preferences may become more specific as stakeholders interact with early system prototypes.

Agile methodology is also beneficial for reducing development risks associated with changing requirements, technical uncertainties, and user dissatisfaction. Because the system is developed incrementally, stakeholders can identify concerns early and recommend corrective actions before the project reaches advanced stages. This reduces the possibility of developing features that may not align with actual organizational needs. Frequent reviews and demonstrations additionally improve stakeholder confidence because project progress is continuously visible and measurable. Agile therefore minimizes uncertainty by maintaining regular communication and ensuring continuous alignment between technical development and business objectives. For BREWINSIGHT BI, this is particularly valuable because the project integrates operational monitoring, analytics, and reporting functionalities that must remain responsive to Kape Tubong's evolving operational environment.

Moreover, Agile supports scalability and long-term maintainability of the system. Since the methodology encourages modular development, new features and enhancements can be integrated into the system more efficiently even after initial deployment. This is advantageous for BREWINSIGHT BI because Kape Tubong may eventually expand its operations and require additional analytical capabilities, branch integrations, or reporting functionalities. Agile's modular and iterative structure allows future system upgrades to be implemented gradually without requiring complete redevelopment of the platform. This ensures that the system can continue evolving alongside organizational growth and technological advancements.

Overall, Agile methodology provides a highly appropriate framework for the development of BREWINSIGHT BI because it combines flexibility, collaboration, iterative testing, and continuous improvement within a structured development process. The methodology enables the researchers to create a system that is adaptable to changing business requirements while maintaining development efficiency and system quality. Through stakeholder collaboration, iterative refinement, and continuous evaluation, Agile supports the creation of a reliable and user-centered Business Intelligence platform capable of addressing Kape Tubong's operational and analytical needs.The **Agile Development Model** will be used, consisting of the following phases:

1.Planning Phase : The first stage of the Agile Development Model is the Planning Phase, where the researchers establish the overall scope, objectives, deliverables, timelines, and development strategies for the BREWINSIGHT BI: Revenue Pattern Analytics System. This phase serves as the foundation of the entire project because it defines the direction, priorities, and structure that will guide all succeeding development activities. During this stage, the researchers carefully analyze the organizational needs of Kape Tubong and identify the specific operational problems that the proposed system must address. Proper planning is essential because it ensures that the project remains organized, goal-oriented, and aligned with both technical requirements and business expectations throughout the development lifecycle.

In this phase, meetings, consultations, and discussions were conducted with the management of Kape Tubong, including the business owner, Bryan Tacayon, together with selected operational personnel and branch representatives. These consultations were intended to gather detailed information regarding the organization's business processes, operational workflows, reporting procedures, data management practices, and long-term business objectives. Through these interactions, the researchers were able to identify the key challenges currently affecting the organization, such as scattered sales records, decentralized reporting systems, delayed access to business information, difficulty in monitoring branch performance, and limited analytical visibility across kiosk branches.

The planning discussions also focused on understanding how management currently monitors sales performance, tracks product demand, evaluates branch operations, and makes strategic decisions. By analyzing the organization's current operational environment, the researchers gained valuable insights into the types of information, reports, and analytical tools that management considered most important for improving business performance and operational efficiency. This allowed the research team to ensure that the planned system functionalities directly addressed verified organizational needs rather than relying on assumptions or generalized solutions.

Based on the information gathered during consultations and preliminary assessments, the researchers formally defined the project goals, objectives, expected outputs, and intended system capabilities. The primary objective established during this phase was the development of a centralized Business Intelligence platform capable of consolidating sales and operational data, generating analytical dashboards, producing automated reports, and identifying meaningful revenue patterns to support data-driven decision-making within Kape Tubong. The researchers also defined measurable project outcomes to evaluate whether the developed system successfully achieved its intended purpose upon completion.

Functional requirements were identified and documented during the planning phase to serve as the basis for system design and development. These functional requirements included dashboard visualization for sales monitoring, branch comparison tools, product performance analysis, automated report generation, filtering mechanisms, revenue trend analysis, user account management, data export capabilities, and real-time or near real-time data presentation. The researchers also identified the need for analytical features capable of presenting business insights through charts, graphs, tables, and summary indicators that can assist management in evaluating operational performance and making informed decisions.

In addition to functional requirements, the researchers identified important non-functional requirements necessary to ensure the quality and effectiveness of the system. These included usability, accessibility, scalability, reliability, maintainability, responsiveness, security, and compatibility across different devices and web browsers. The planning phase emphasized the importance of designing a user-friendly system that can be easily operated by employees regardless of their technical expertise or familiarity with business intelligence platforms. Security considerations were also discussed to ensure that sensitive business information would be protected through appropriate authentication and access control mechanisms.

The researchers further determined the technical resources, programming languages, software tools, development frameworks, database technologies, and hosting environments needed for the successful implementation of the system. This included evaluating available technologies based on factors such as compatibility, scalability, development efficiency, system performance, maintainability, and cost-effectiveness. The selection of appropriate technologies during this phase was important to ensure that the system could operate efficiently while remaining adaptable to future organizational growth and expansion.

A project timeline was also created during the Planning Phase to organize development activities into manageable Agile iterations or sprints. The researchers divided the project into smaller development cycles to support continuous progress evaluation, incremental feature delivery, and iterative system improvement. Each sprint was assigned specific objectives, tasks, and deliverables to ensure that development activities remained structured and measurable throughout implementation. This iterative approach allowed the researchers to monitor progress more effectively and make adjustments whenever necessary based on stakeholder feedback or emerging requirements.

Feature prioritization was another important activity conducted during planning. The researchers identified which system functionalities were most critical to the operational needs of Kape Tubong and determined the order in which these features would be developed. Essential modules such as user authentication, dashboard analytics, sales data management, and report generation were prioritized to ensure that the core functionalities of the platform would be available early in the development process. This prioritization strategy allowed the researchers to deliver functional system components incrementally while continuously evaluating and improving the platform throughout the Agile lifecycle.

The planning phase also involved identifying possible project risks, technical limitations, operational constraints, and potential implementation challenges that could affect development progress or system performance. These risks included issues related to data availability, integration complexity, time limitations, resource constraints, compatibility concerns, and possible changes in stakeholder requirements during development. By identifying these risks early, the researchers were able to prepare mitigation strategies and contingency plans to reduce the likelihood of delays, technical failures, or operational disruptions during subsequent phases of the project.

Furthermore, the planning phase enabled the researchers to determine appropriate methodologies for data gathering, stakeholder feedback collection, testing procedures, and iterative evaluation activities throughout development. The researchers established methods for communicating with stakeholders, collecting user input, documenting requirements, and evaluating completed features after each sprint cycle. This ensured that stakeholder feedback could be continuously incorporated into the system development process, promoting flexibility and responsiveness to changing organizational needs.

Acceptance criteria for each planned feature and system module were also defined during this stage. These criteria established clear standards and expectations that each functionality must satisfy before being considered complete and acceptable for deployment. Defining acceptance criteria helped improve quality assurance by ensuring that completed outputs aligned with stakeholder expectations, operational requirements, and technical specifications.

The planning phase additionally improved project coordination and collaboration among the researchers by clearly defining roles, responsibilities, and development priorities. Team members were assigned specific tasks and responsibilities based on their expertise and involvement within the project. Communication protocols, collaboration workflows, documentation procedures, and reporting mechanisms were also established to support efficient coordination throughout development. These preparations minimized confusion, reduced task overlap, and improved accountability among the members of the development team.

Another important contribution of the planning phase was the identification of dependencies between different system modules and functionalities. The researchers analyzed how each module would interact with other components of the platform and scheduled development activities accordingly. This helped prevent integration conflicts and ensured that foundational modules were completed before dependent functionalities were developed. Proper scheduling of module dependencies contributed to smoother integration and more efficient development progression.

Budget considerations and resource allocation were likewise reviewed during this stage. The researchers evaluated the availability of development tools, hosting services, software licenses, hardware requirements, and other technical resources necessary to support implementation. Efficient resource planning helped ensure that the project could be completed within the available limitations while maintaining acceptable quality standards.

The planning phase also served as a communication and documentation framework for the project. All agreed objectives, requirements, priorities, schedules, and deliverables were properly documented to provide a clear reference for stakeholders and developers throughout the Agile development lifecycle. This documentation helped maintain consistency in project direction and minimized misunderstandings regarding system expectations and development scope.

Additionally, the planning process established measurable milestones and performance indicators that could be used to monitor project progress and evaluate development efficiency throughout implementation. These milestones allowed the researchers to track whether development activities remained aligned with project timelines and objectives. Regular monitoring against these planned targets improved project manageability and supported timely identification of delays or deviations that may require corrective action.

Through comprehensive and structured planning, the researchers created a detailed roadmap that guided all succeeding Agile development activities for the BREWINSIGHT BI: Revenue Pattern Analytics System. The Planning Phase significantly contributed to reducing uncertainty, improving resource utilization, enhancing team coordination, and increasing the likelihood of successful project completion. By establishing clear objectives, development priorities, technical strategies, and operational expectations at the beginning of the project, the researchers ensured that the system development process remained organized, efficient, and aligned with the needs of Kape Tubong.

Ultimately, the Planning Phase played a critical role in ensuring the successful implementation of the proposed Business Intelligence platform. It provided the strategic direction, organizational structure, and operational framework necessary for developing a reliable, scalable, and user-centered system capable of supporting data-driven decision-making, improving operational efficiency, and contributing to the long-term growth and success of Kape Tubong.

Analysis Phase : The Analysis Phase focuses on gathering, examining, and interpreting the detailed operational, technical, and organizational information necessary for the successful development of the BREWINSIGHT BI: Revenue Pattern Analytics System. This phase serves as a critical step in the Agile Development Model because it allows the researchers to develop a deep understanding of Kape Tubong's existing business processes, operational challenges, data management practices, and analytical requirements before proceeding to system design and implementation. Through comprehensive analysis, the researchers ensure that the proposed system is based on actual operational conditions and stakeholder needs rather than assumptions or generalized solutions.

During this stage, the researchers carefully assessed Kape Tubong's current workflow for recording sales transactions, consolidating reports, monitoring kiosk performance, tracking product sales, and evaluating branch operations. Existing operational procedures were thoroughly reviewed to understand how business information flows within the organization, how data is collected and processed, and how reports are prepared and communicated among employees and management personnel. This analysis enabled the researchers to identify inefficiencies and limitations in the current system, including delayed report consolidation, scattered sales records, inconsistent data formatting, duplicate entries, limited analytical visibility, and the absence of real-time monitoring and centralized reporting capabilities.

The researchers also examined how employees and management currently access, interpret, and utilize business data for operational monitoring and decision-making. Particular attention was given to identifying how sales information is encoded, stored, retrieved, and analyzed across different kiosk branches. By understanding these operational processes, the researchers were able to identify bottlenecks, repetitive manual tasks, and areas where automation and business intelligence functionalities could improve efficiency, accuracy, and productivity within the organization.

To support the analysis process, interviews, consultations, and discussions were conducted with the business owner, Bryan Tacayon, together with branch managers, administrative personnel, and selected staff members directly involved in sales recording, reporting, inventory management, and operational monitoring. These interactions provided valuable insights regarding user expectations, operational difficulties, reporting requirements, and the specific analytical information needed by management to support business decisions. Participants were encouraged to discuss the problems they encounter when managing business records, generating reports, monitoring sales performance, and analyzing operational trends.

The consultations also focused on identifying the types of information considered most important for evaluating business performance and guiding strategic planning. Management provided input regarding desired dashboard features, preferred reporting formats, revenue monitoring tools, branch comparison functions, and performance indicators necessary for effective decision-making. Through these discussions, the researchers gained a clearer understanding of the practical needs and expectations that the BREWINSIGHT BI system must fulfill in order to become a useful and valuable operational tool for the organization.

During the Analysis Phase, the researchers determined the specific types of data required for the system's operation and analytics processing. These data categories included sales transaction records, product information, inventory details, branch identifiers, timestamps, customer order information, revenue values, product categories, and operational performance metrics. The researchers analyzed how these data elements should be organized, stored, validated, and processed within the system to ensure accurate reporting and reliable analytical outputs.

The researchers also examined the structure and quality of existing business records to identify issues related to missing data, inconsistent formats, incomplete entries, and duplicate information. This analysis was necessary to establish appropriate data validation rules, formatting standards, and organizational structures that would support reliable business intelligence processing. Data standardization procedures were likewise identified during this phase to ensure that information collected from different branches and operational sources could be consolidated and analyzed consistently across the platform.

User roles, responsibilities, and access permissions were also analyzed in detail to determine appropriate levels of system access for administrators, managers, branch personnel, and authorized users. The researchers identified which users should have permission to access sensitive reports, modify data, generate analytics, manage accounts, or perform administrative functions within the system. This analysis supported the development of access control mechanisms that protect confidential business information while ensuring that users can access the tools and reports necessary for their respective responsibilities.

Functional requirements identified during the analysis process were translated into detailed requirement specifications that served as references for the succeeding design and development stages. These requirements included system functionalities related to dashboard analytics, report generation, sales filtering, branch performance monitoring, data visualization, user authentication, revenue tracking, and automated business reporting. Non-functional requirements such as usability, responsiveness, scalability, compatibility, maintainability, and system security were also documented to guide the technical development of the platform.

To better understand the organization’s operational structure and workflow processes, the researchers created workflow diagrams, process models, and operational flowcharts representing both the existing business procedures and the proposed future system environment. These visual representations helped clarify how information moves throughout the organization and how the new system would improve operational efficiency and data accessibility. Documenting current-state and future-state workflows also enabled the researchers to identify operational improvements that could be achieved through automation and centralized analytics.

The Analysis Phase additionally allowed the researchers to validate the feasibility of proposed system features within the technical and operational environment of Kape Tubong. Factors such as internet reliability, device compatibility, branch connectivity, existing technological infrastructure, and employee readiness for digital system adoption were carefully assessed during this stage. By identifying these limitations early, the researchers were able to make informed adjustments to development priorities, implementation strategies, and system architecture before entering the design phase.

This stage also enabled the researchers to identify operational constraints and dependencies that could influence system implementation and long-term maintenance. The analysis considered how organizational policies, approval procedures, reporting hierarchies, and branch management structures might affect access control, reporting workflows, and data management processes within the platform. Understanding these organizational factors helped ensure that the system would align with the actual structure and operational culture of the business.

Requirement prioritization was another important activity conducted during the Analysis Phase. The researchers categorized and prioritized system requirements based on operational importance, stakeholder urgency, implementation complexity, and overall business impact. Essential functionalities such as sales monitoring dashboards, revenue analytics, report generation, and centralized data management were given high priority to ensure that the core capabilities of the platform would be addressed during the early stages of Agile development. This prioritization supported more efficient sprint planning and incremental system delivery throughout the development lifecycle.

Furthermore, the Analysis Phase provided an opportunity to identify future integration possibilities and expansion requirements for the BREWINSIGHT BI system. The researchers examined whether the platform may eventually need compatibility with additional digital business tools, inventory systems, accounting platforms, or customer management applications as the organization grows and expands. Considering future scalability during analysis helped ensure that the proposed system architecture would remain flexible and adaptable to future operational demands and technological developments.

The researchers also analyzed the preferred reporting frequency, dashboard update intervals, and data visualization formats expected by management and operational users. Some users required daily sales summaries, while others preferred weekly or monthly trend analyses. Preferences regarding graphs, tables, charts, color indicators, and summary presentations were gathered to ensure that the final system would present analytical information in a manner that supports fast interpretation and effective managerial decision-making.

Comprehensive documentation was maintained throughout the Analysis Phase to record identified requirements, operational findings, stakeholder feedback, workflow observations, technical constraints, and proposed solutions. This documentation became an essential reference for the succeeding phases of system design, coding, testing, and implementation. Maintaining accurate and organized records also improved communication among researchers and stakeholders by ensuring that all agreed-upon requirements and project objectives were clearly documented and understood.

By thoroughly analyzing both operational and technical aspects of the organization, the researchers were able to reduce the likelihood of missing critical requirements or developing unnecessary system features. The Analysis Phase strengthened the overall relevance, practicality, and strategic alignment of the BREWINSIGHT BI platform by ensuring that development decisions were grounded in actual business conditions and verified organizational needs.

Ultimately, the Analysis Phase played a vital role in establishing a strong and reliable foundation for the design and development of the BREWINSIGHT BI: Revenue Pattern Analytics System. Through detailed operational assessment, stakeholder consultation, workflow evaluation, and requirement analysis, the researchers ensured that the proposed platform would be feasible, user-centered, technically appropriate, and highly relevant to the business environment of Kape Tubong. This phase significantly contributed to improving development accuracy, reducing implementation risks, and increasing the likelihood of delivering a successful and effective Business Intelligence solution capable of supporting informed decision-making, operational efficiency, and long-term organizational growth.

Design Phase: In the Design Phase, the researchers transform the analyzed business requirements, operational findings, and technical specifications into detailed system blueprints, architectural plans, and interface designs that will guide the actual development of the BREWINSIGHT BI: Revenue Pattern Analytics System. This phase serves as the bridge between system analysis and implementation because it translates identified needs and requirements into structured technical solutions and visual representations. Through careful design planning, the researchers ensure that the proposed platform will be functional, efficient, scalable, visually organized, and aligned with the operational expectations of Kape Tubong before coding activities begin.

During this stage, the researchers design the overall system architecture, which defines how the major components of the platform interact with one another. This includes establishing the relationships between the front-end user interface, back-end application logic, database management system, analytics processing engine, and reporting modules. The architectural design outlines how data will move through the system, how requests will be processed, and how information will be retrieved, analyzed, and presented to users. By defining these structural relationships early, the researchers ensure that the platform will maintain smooth communication between components while supporting efficient and stable system operation.

The researchers also develop the database schema for the BREWINSIGHT BI platform. This process involves designing the structure of database tables, relationships, attributes, keys, and constraints needed to store and manage business information effectively. Database entities such as sales transactions, product records, branch details, user accounts, inventory information, timestamps, and revenue reports are carefully organized to support accurate data storage, retrieval, and analytics processing. Relationships between tables are established to ensure data consistency, minimize redundancy, and improve query efficiency throughout the system.

Entity-Relationship Diagrams (ERDs) and database relationship models may also be created during this phase to visually represent how information is connected within the platform. These diagrams help the researchers verify that all required data relationships are properly structured and capable of supporting the analytical and reporting requirements of the business. Data Flow Diagrams (DFDs) may likewise be developed to illustrate how information moves between users, processes, databases, and system modules. These visual tools improve understanding of the system's internal operations and guide developers during implementation.

Another major activity during the Design Phase involves creating wireframes, prototypes, and interface mockups for the system dashboard and user environment. The researchers design the layout and visual structure of the platform to ensure that information is presented clearly, logically, and efficiently to users. Dashboard interfaces are organized using charts, graphs, tables, summary cards, navigation menus, filters, and data visualization panels that support easy interpretation of business analytics and operational reports. Particular attention is given to arranging interface elements in a way that reduces confusion, improves readability, and supports quick access to important information.

The dashboard design is specifically tailored to the operational and analytical needs of Kape Tubong. Revenue trends, branch comparisons, product performance indicators, and sales summaries are visualized using appropriate graphical representations such as line graphs, bar charts, pie charts, tables, and statistical summaries. The researchers carefully select visualization methods based on the nature of the data and the type of insights required by management. For example, line graphs may be used to display changes in sales over time, while bar charts may be used to compare product performance across branches. Summary cards and indicators are also designed to highlight important business metrics such as total revenue, best-selling products, and branch performance rankings.

Navigation flow and user interaction pathways are planned extensively during this stage to ensure that the platform remains intuitive, user-friendly, and easy to operate. The researchers determine how users will move between pages, access reports, apply filters, retrieve analytics, and interact with dashboard components. Menu structures, button placements, search functions, and navigation hierarchies are designed to minimize unnecessary complexity and improve overall usability. This planning helps ensure that users can complete tasks efficiently without requiring advanced technical knowledge or extensive training.

Responsiveness and accessibility are also major considerations during the Design Phase. The researchers design the platform to function effectively across different screen sizes, devices, and operating environments, including desktop computers, laptops, tablets, and mobile devices. Responsive layouts are planned to ensure that dashboard components, reports, and interface elements automatically adjust according to screen dimensions while maintaining readability and usability. Accessibility principles are likewise considered to ensure that the platform remains usable for a broad range of users with varying levels of technical expertise and interaction preferences.

Security features are integrated into the system design to protect sensitive business information and regulate user access throughout the platform. The researchers design authentication mechanisms, login interfaces, role-based access controls, protected routes, and permission management structures to ensure that only authorized personnel can access specific system functions and reports. User roles such as administrators, managers, and staff members are assigned appropriate levels of access according to their responsibilities within the organization. Additional security considerations such as password validation, session management, and data protection mechanisms are also incorporated into the system design.

The researchers further design the logic flow for analytics processing, report generation, and automated calculations within the platform. This includes planning how transaction data will be processed, aggregated, filtered, and transformed into meaningful analytical outputs. Logic structures for calculating sales trends, branch comparisons, revenue summaries, and product performance indicators are carefully defined to ensure accuracy and efficiency during system operation. Automated report generation workflows are also planned to reduce manual reporting effort and improve the speed of accessing business insights.

During the Design Phase, the researchers may also prepare technical documentation for system modules, API structures, database interactions, and component behaviors to guide development activities during implementation. These technical references help ensure consistency in coding practices, module integration, and system maintenance. Naming conventions, coding standards, file organization structures, and component hierarchies are likewise established to support scalable and maintainable software development practices.

Performance optimization strategies are also considered during system design. The researchers plan efficient database query structures, caching mechanisms, data retrieval procedures, and processing workflows to improve system responsiveness and reduce unnecessary resource consumption. By considering performance requirements early in the design stage, the researchers help ensure that the platform can handle growing amounts of data and increasing user activity without significant decreases in speed or operational stability.

Design validation activities may also be conducted during this phase by presenting mockups, wireframes, prototypes, and workflow models to stakeholders and intended users for preliminary feedback. The business owner, Bryan Tacayon, together with selected staff members and operational personnel, may review the proposed interface layouts, navigation structures, and dashboard designs to determine whether they align with operational expectations and user preferences. Feedback gathered during these validation sessions helps the researchers identify areas where interface layouts, workflows, or visual elements may need improvement before actual coding begins.

This iterative validation process reduces the likelihood of major redesigns during implementation by identifying usability concerns and requirement mismatches early in the project lifecycle. Based on stakeholder recommendations, the researchers may revise dashboard layouts, navigation flows, color schemes, data presentation formats, and workflow structures to improve clarity, accessibility, and overall user experience. This collaborative approach ensures that the platform remains user-centered and operationally practical throughout development.

The Design Phase also includes planning standardized visual themes, dashboard styling, typography, icons, and interface consistency to create a professional and cohesive appearance across all system modules. Error handling interfaces, warning messages, notifications, and confirmation prompts are likewise conceptualized to improve communication between the system and its users during operation. These design elements contribute to improving usability, reducing user confusion, and enhancing the overall quality of the user experience.

Additionally, the researchers consider future scalability and system expansion during the design process. The architecture is planned in a modular and flexible manner to allow additional features, branches, reporting tools, or integrations with other business systems to be incorporated in the future without requiring major structural redesign. This forward-looking approach helps ensure that the BREWINSIGHT BI platform can continue evolving alongside the operational growth and changing requirements of Kape Tubong.

Comprehensive design documentation is maintained throughout the phase to record interface specifications, workflow models, technical diagrams, database structures, security plans, and system architecture decisions. These documents serve as important references for developers during coding, testing, implementation, and future maintenance activities. Proper documentation also supports clearer communication among team members and stakeholders by ensuring that all design decisions and technical specifications are accurately recorded and understood.

Through comprehensive and structured design planning, the researchers establish a detailed implementation roadmap that supports organized, efficient, and high-quality system development. The Design Phase significantly contributes to reducing development errors, minimizing implementation inefficiencies, improving usability, and ensuring architectural stability before coding begins. By carefully designing both the technical infrastructure and user interface of the BREWINSIGHT BI platform, the researchers ensure that the proposed system will be technically robust, visually coherent, operationally effective, and aligned with the analytical and managerial needs of Kape Tubong.

Ultimately, the Design Phase plays a critical role in preparing the BREWINSIGHT BI: Revenue Pattern Analytics System for successful implementation by establishing clear technical blueprints, organized workflows, and user-centered interface designs. This phase ensures that the platform is not only capable of delivering accurate and meaningful business intelligence analytics but is also practical, scalable, secure, and easy to use within the real operational environment of Kape Tubong.

Implementation: The Implementation Phase involves the actual construction, coding, integration, and configuration of the BREWINSIGHT BI: Revenue Pattern Analytics System based on the approved designs, technical specifications, and requirements established during the previous phases of the Agile Development Model. This phase serves as the stage where conceptual plans and system blueprints are transformed into a fully operational Business Intelligence platform capable of supporting the analytical and operational needs of Kape Tubong. Through systematic development activities, the researchers convert the designed architecture, workflows, and interface layouts into functional software components that work together as an integrated web-based application.

During this stage, the researchers develop the platform incrementally using Agile sprint cycles, allowing functionalities to be completed, reviewed, tested, and improved continuously throughout development. The iterative nature of Agile implementation supports flexibility, progressive refinement, and continuous feedback integration, ensuring that the developed system remains aligned with stakeholder expectations and operational requirements as development progresses. Each sprint focuses on specific modules or functionalities, enabling the researchers to prioritize essential system components while maintaining manageable development tasks and timelines.

The implementation process begins with the development of the front-end components of the BREWINSIGHT BI platform. These components include the dashboard interface, navigation menus, data filtering controls, user forms, report views, graphical charts, summary cards, and visualization panels. The researchers code and style these interface elements to ensure that they are visually organized, responsive, and easy to use across different devices and screen sizes. Emphasis is placed on creating an intuitive and user-friendly environment that allows users to navigate through the system efficiently and interpret business information clearly.

The dashboard interfaces are programmed to display analytical information using dynamic charts, graphs, tables, and statistical summaries. Visualization components are integrated to present revenue trends, product performance metrics, branch comparisons, sales distributions, and operational summaries in a clear and meaningful format. Interactive features such as filtering tools, date selectors, search functions, and sortable reports are also implemented to allow users to customize and analyze data according to their specific needs and preferences.

Simultaneously, the researchers develop the back-end functionalities responsible for handling data processing, system logic, database communication, and business rule execution. Back-end implementation includes programming user authentication processes, access control mechanisms, session management, report generation functions, analytics computations, and database operations. The researchers create secure and structured server-side processes that manage the retrieval, storage, updating, and analysis of business information within the platform.

One of the major components implemented during this phase is the analytics processing and data aggregation logic of the BREWINSIGHT BI system. The researchers develop algorithms and processing workflows that transform raw sales records and operational data into meaningful business intelligence outputs. These processes include calculating revenue trends, identifying top-performing products, generating branch comparison metrics, analyzing sales patterns, monitoring operational performance, and identifying peak business hours. The implementation of these analytical functions allows the system to generate insights that support managerial decision-making and operational evaluation.

Revenue trend analysis modules are programmed to evaluate changes in sales performance over specific periods of time, allowing management to monitor business growth, seasonal patterns, and fluctuations in revenue. Product performance metrics are implemented to identify best-selling items, low-performing products, and customer purchasing patterns across different kiosk branches. Branch comparison reports are likewise developed to evaluate operational performance between locations, enabling management to identify strengths, weaknesses, and opportunities for improvement within the organization.

The researchers also implement automated reporting modules designed to minimize manual data compilation and simplify operational monitoring. These modules allow the system to automatically generate summaries, sales reports, and analytical outputs based on updated transaction records stored within the database. Automated reporting significantly improves efficiency by reducing the time and effort required to prepare business reports manually while ensuring that information remains accurate, consistent, and readily accessible to authorized users.

Database integration is another critical activity performed during the Implementation Phase. The researchers connect the platform to the designed database structure to support real-time or near real-time data storage, retrieval, and processing. Database connectivity functions are implemented to ensure that information entered into the system is properly validated, securely stored, and accurately reflected within dashboards and reports. The researchers also optimize database queries and retrieval operations to improve processing efficiency and reduce loading times during system use.

Each module and functionality is developed incrementally following Agile sprint cycles. After completing a specific component or feature, continuous internal testing is performed to verify correctness, functionality, and compatibility before proceeding to the next development task. This ongoing validation process helps identify programming errors, logical inconsistencies, interface issues, and integration problems at an early stage of development. Any detected issues are documented, corrected, and retested to ensure that the implemented solutions function properly and do not negatively affect other parts of the platform.

Incremental implementation also allows stakeholders, including the business owner, Bryan Tacayon, and selected operational personnel, to review partially completed modules and provide feedback before the entire system is finalized. This collaborative process helps improve alignment between technical outputs and actual business expectations. Stakeholder feedback gathered during implementation enables the researchers to make adjustments efficiently without requiring major redevelopment later in the project lifecycle. Sprint review meetings may also be conducted after each iteration to assess completed functionalities, discuss identified concerns, and prioritize tasks for succeeding development cycles.

Code optimization and refactoring activities are likewise performed throughout implementation to improve system efficiency, maintainability, and long-term scalability. The researchers review and reorganize program structures to simplify code management, eliminate unnecessary redundancy, and improve overall software performance. Modular programming techniques and reusable software components are utilized to support future system enhancements and simplify maintenance activities after deployment.

Version control practices are implemented during development to manage code changes, monitor revisions, and support collaborative programming among the researchers. Version control systems allow developers to track modifications, restore previous code versions when necessary, and coordinate development activities more effectively. This process helps maintain development organization, reduce conflicts between code updates, and improve overall project management throughout implementation.

The researchers also integrate error logging and debugging mechanisms into the platform to support efficient identification and resolution of system issues during development and future maintenance. Logging functions record system activities, operational errors, failed requests, and unexpected behaviors, allowing developers to analyze problems more effectively and implement corrective actions efficiently.

Additional implementation activities include configuring environment settings, server connections, deployment dependencies, and software libraries required for system operation. Third-party libraries, charting frameworks, development tools, and interface components may be integrated into the platform to enhance analytical visualization, system responsiveness, and user experience. These external resources help improve development efficiency while expanding the capabilities and appearance of the BREWINSIGHT BI platform.

Security measures are carefully implemented throughout the development process to protect system integrity, user accounts, and sensitive business information. Password encryption mechanisms, input validation procedures, authentication controls, session handling protocols, and protected routes are integrated to prevent unauthorized access and reduce potential security vulnerabilities. Data validation processes are also implemented to ensure that only properly formatted and authorized information can be stored or processed within the system.

Performance testing and optimization activities may also be conducted during implementation to evaluate system responsiveness, dashboard loading speed, database query efficiency, and analytics processing performance under expected operational conditions. These evaluations help ensure that the platform remains stable, responsive, and capable of supporting regular business activities without excessive delays or performance degradation.

Comprehensive documentation is maintained throughout the Implementation Phase to record completed functionalities, coding structures, configuration settings, technical changes, system dependencies, and development decisions. This documentation serves as an important reference for testing, maintenance, troubleshooting, future updates, and additional feature development after deployment. Proper documentation also supports better collaboration among researchers and improves long-term maintainability of the system.

The iterative and incremental nature of the Implementation Phase allows the researchers to continuously evaluate, refine, and improve the BREWINSIGHT BI platform throughout development. By integrating coding, testing, feedback collection, optimization, and validation within each development cycle, the researchers ensure that the platform progressively evolves into a reliable, efficient, and user-centered Business Intelligence solution.

Ultimately, the Implementation Phase transforms the approved system designs and technical specifications into a fully functional, secure, responsive, and operational Business Intelligence platform ready for evaluation, testing, and deployment within Kape Tubong. This phase plays a critical role in ensuring that the BREWINSIGHT BI: Revenue Pattern Analytics System not only meets technical standards but also delivers meaningful analytical capabilities, operational efficiency, and practical value that support informed decision-making and long-term organizational growth.

Testing Phase: The Testing Phase is conducted to evaluate the accuracy, reliability, functionality, security, usability, and overall readiness of the BREWINSIGHT BI: Revenue Pattern Analytics System before final deployment within Kape Tubong. This phase serves as a critical quality assurance process that ensures the developed platform operates according to its intended design, performs efficiently under various operational conditions, and satisfies both technical standards and user expectations. Through systematic and repeated evaluation procedures, the researchers verify that all system components function correctly, interact properly, and provide dependable analytical outputs suitable for real-world business operations.

During this stage, the researchers perform multiple forms of testing to assess different aspects of the platform's behavior and performance. Functional testing is conducted to ensure that every feature, module, and process within the system performs according to its intended purpose and established requirements. Individual functionalities such as user login, dashboard visualization, report generation, data filtering, branch comparison, sales analytics, revenue tracking, and product performance monitoring are carefully examined to confirm that they produce accurate and expected results.

The researchers specifically verify that dashboard visualizations display updated and accurate analytical data retrieved from the database. Revenue calculations, sales summaries, filtering operations, chart updates, and automated report generation processes are tested thoroughly to ensure correctness, consistency, and reliability. These evaluations help confirm that the platform can accurately transform raw transactional records into meaningful business insights that support managerial analysis and decision-making.

Integration testing is also performed during this phase to assess whether the different modules and components of the BREWINSIGHT BI platform interact properly as a complete and unified system. The researchers evaluate the communication between the front-end interface, back-end logic, analytics engine, database management system, and reporting modules to ensure smooth and synchronized operation. This testing process helps identify issues related to data flow, module compatibility, synchronization delays, broken functionalities, or communication failures between system components.

The platform is further tested under various operational and usage scenarios to evaluate system responsiveness, stability, and consistency. The researchers simulate common business activities such as multiple dashboard requests, repeated filtering operations, simultaneous user access, report generation, and continuous data retrieval processes to determine whether the platform remains stable and responsive during regular use. These tests help assess whether the system can maintain acceptable performance levels under both normal and demanding operational conditions.

Performance testing is conducted to evaluate the speed and efficiency of data processing, dashboard rendering, analytics computation, and report generation within the system. The researchers measure how quickly dashboards load, how efficiently reports are generated, and how responsive the platform remains when processing larger datasets or multiple simultaneous requests. These evaluations are important to ensure that users can interact with the platform efficiently without experiencing delays or interruptions that could negatively affect operational productivity.

Compatibility testing is likewise carried out to verify that the BREWINSIGHT BI platform functions consistently across different web browsers, operating systems, devices, and hardware environments. The researchers assess whether users accessing the system through desktop computers, laptops, tablets, or mobile devices experience consistent interface behavior, responsiveness, and functionality. Browser compatibility testing ensures that visual elements, navigation structures, and analytical components display correctly regardless of the software environment being used. This broadens system accessibility and improves deployment readiness across different operational settings within Kape Tubong.

Security testing is another important activity conducted during the Testing Phase. The researchers evaluate the effectiveness of authentication mechanisms, password protection, session management, role-based access controls, and input validation procedures implemented within the system. Security testing helps identify vulnerabilities related to unauthorized access, data manipulation, improper authentication handling, and potential threats to sensitive business information. The researchers verify that users can only access the features, reports, and system functions appropriate to their assigned roles and permissions.

The testing process also includes validating the platform’s ability to handle incorrect, incomplete, or invalid user inputs appropriately. The researchers intentionally simulate erroneous entries, invalid credentials, incomplete form submissions, and unexpected user actions to assess how the system responds to abnormal situations. Error handling mechanisms, warning messages, validation prompts, and exception handling routines are examined to ensure that the platform prevents data corruption, maintains operational stability, and provides clear guidance to users whenever issues occur.

Stress testing may additionally be conducted to determine how the system behaves under unusually high workloads, increased transaction volumes, or large datasets. This form of testing helps identify performance limitations, system bottlenecks, memory issues, or processing constraints that may emerge during peak operational conditions. By evaluating the system under demanding scenarios, the researchers can assess whether the platform remains stable and capable of supporting future business growth and expanded operational requirements.

Another significant aspect of the Testing Phase is verifying the accuracy and reliability of the analytical outputs generated by the platform. The researchers compare dashboard analytics, revenue calculations, sales trends, and generated reports against actual raw transactional records and business data to ensure that the system produces valid and trustworthy results. This validation process is particularly important because management decisions may rely heavily on the accuracy of the information generated by the platform.

User Acceptance Testing (UAT) is also conducted with selected personnel from Kape Tubong, including management representatives, operational staff, and authorized users. During this testing activity, participants interact with the system in realistic operational scenarios to evaluate the platform's usability, practicality, and effectiveness in supporting daily business activities. Users perform tasks such as accessing dashboards, reviewing reports, filtering sales information, and analyzing branch performance to determine whether the system is intuitive, user-friendly, and beneficial for operational monitoring and decision-making.

Feedback gathered during User Acceptance Testing provides valuable insights regarding interface clarity, navigation efficiency, report readability, dashboard organization, and overall user experience. Participants may identify usability concerns, suggest interface improvements, recommend additional features, or provide observations regarding system convenience and effectiveness. The researchers carefully document all feedback and use the findings to refine the platform before final deployment.

Regression testing is also conducted whenever modifications, bug fixes, or revisions are implemented during the testing process. This form of testing ensures that newly applied corrections or feature updates do not negatively affect existing functionalities or introduce additional issues into the system. Regression testing helps maintain overall platform stability while supporting continuous improvement throughout the development lifecycle.

The Testing Phase additionally includes documenting all testing procedures, identified issues, corrective actions, revisions, and evaluation results. Comprehensive testing documentation is maintained to provide transparency regarding the system evaluation process and to support future troubleshooting, maintenance, upgrades, and system enhancement activities. These records also serve as evidence that the platform has undergone thorough validation prior to deployment.

Continuous testing and iterative correction cycles are performed until the BREWINSIGHT BI platform achieves acceptable performance standards and satisfies the defined functional and operational requirements established during earlier phases of development. Problems identified during testing are analyzed, corrected, and retested to verify that implemented solutions effectively resolve the issues without creating additional complications.

Through systematic and repeated evaluation procedures, the researchers ensure that the BREWINSIGHT BI: Revenue Pattern Analytics System is technically functional, operationally stable, secure, responsive, accurate, and user-friendly before deployment within Kape Tubong. The Testing Phase significantly contributes to minimizing operational risks, improving system reliability, enhancing user confidence, and validating the overall effectiveness of the platform as a Business Intelligence solution.

Ultimately, the Testing Phase confirms that the BREWINSIGHT BI platform is capable of delivering meaningful analytical insights, supporting informed managerial decision-making, improving operational efficiency, and meeting the real-world operational requirements of Kape Tubong. By thoroughly evaluating the platform from technical, operational, and user-centered perspectives, the researchers ensure that the developed system is fully prepared for practical deployment and long-term organizational use.

Deployment Phase: The Deployment Phase involves installing and launching the completed BREWINSIGHT BI platform in Kape Tubong's operational environment. During this phase, the system is deployed to a live web server or hosting platform for actual business use. Necessary server configurations, domain setup, and database deployment are completed to support online accessibility. The researchers verify that the deployed system functions correctly in the production environment. Historical sales records may be migrated into the system if needed for initial reporting and analytics purposes. User accounts are configured for authorized Kape Tubong personnel based on assigned access roles. Training sessions are conducted to orient users on system navigation, dashboard interpretation, and report generation. User manuals or instructional documentation may also be provided for reference. Initial operational monitoring is performed to ensure successful adoption during live use. The deployment phase marks the transition of BREWINSIGHT BI from development to practical business implementation. It ensures that the system is fully operational and accessible to intended users. This phase is essential for enabling the real-world application of the developed platform.

A successful deployment process helps minimize operational disruption during system transition. Careful rollout planning ensures that users can begin utilizing the platform efficiently with minimal downtime or confusion. The researchers may conduct final pre-launch checks to verify server stability, database integrity, and connectivity across all system modules before full deployment. Backup and recovery procedures may also be configured during this phase to protect organizational data against accidental loss or technical failures. Deployment logs and monitoring tools may be implemented to track system health and identify issues immediately after launch. Security settings such as SSL certificates, firewall rules, and production authentication configurations are finalized to protect the live environment. Initial feedback from users during early deployment may be collected to identify minor usability adjustments or support needs.

Furthermore, the deployment phase includes validating that all integrated analytics and reporting features perform accurately under actual operational conditions. The researchers may observe how end users interact with the system in real workflows to ensure successful adoption and proper utilization. Post-deployment support may be temporarily provided to assist users in resolving initial technical concerns or operational questions. Performance metrics from the live environment may also be monitored to evaluate system responsiveness and scalability after launch. Documentation of deployment procedures and configuration settings is maintained to support future updates, maintenance, or migration activities. Through careful implementation and monitoring, the deployment phase ensures that BREWINSIGHT BI transitions smoothly from development into active business use. Ultimately, this phase operationalizes the proposed system and enables Kape Tubong to begin leveraging data-driven analytics for informed decision-making and performance management.

Maintenance: The final stage is the Maintenance Phase, which focuses on sustaining and improving the system after deployment. During this phase, the researchers provide technical support to address issues encountered during real-world use. Software bugs, system errors, and user-reported concerns are investigated and resolved promptly. Performance monitoring is conducted to ensure that the platform remains stable, responsive, and efficient. Security patches and software updates are applied when necessary to maintain system integrity. The researchers also gather ongoing feedback from users regarding additional feature requests or usability improvements. Minor enhancements and refinements may be implemented based on evolving business needs. Compatibility with updated browsers, operating systems, and hosting environments is regularly checked. Backup and recovery procedures are monitored to protect against data loss or corruption. Maintenance ensures that the platform remains functional and relevant over time. This phase supports the long-term sustainability of the BREWINSIGHT BI platform. Through continuous improvement and support, the system can adapt to future operational and technological changes. Therefore, maintenance is critical to preserving the effectiveness and usability of the developed system.

Maintenance also provides opportunities for future expansion of the platform as Kape Tubong's operational needs evolve. As new branches are added or reporting requirements change, the system can be enhanced accordingly. This ensures that BREWINSIGHT BI remains scalable and sustainable for long-term organizational use. The maintenance phase may include periodic system audits to evaluate whether the platform continues to meet organizational objectives and performance expectations. Database optimization activities may be performed to maintain efficient data retrieval as transaction volumes increase over time. The researchers may also review system logs and analytics to identify recurring issues or opportunities for technical improvement. Preventive maintenance measures can be scheduled to reduce the likelihood of unexpected downtime or performance degradation. User training materials may be updated as new features or interface changes are introduced.

Furthermore, the maintenance phase supports the continuous alignment of the system with changing business strategies and operational workflows. As Kape Tubong adopts new products, promotions, or reporting standards, the platform may be modified to reflect these evolving requirements. Long-term monitoring of user feedback can help identify emerging needs that were not evident during initial deployment. The researchers may also evaluate opportunities for integrating advanced features such as predictive analytics, forecasting tools, or machine learning enhancements in future updates. Documentation of maintenance activities, revisions, and system updates is maintained to ensure traceability and facilitate future technical support. Through proactive maintenance and iterative enhancement, the BREWINSIGHT BI platform remains adaptable, reliable, and strategically valuable to the organization over time. Ultimately, the maintenance phase ensures the continued operational effectiveness and long-term success of the deployed Business Intelligence system.

3.3 System Architecture

The BREWINSIGHT BI: Revenue Pattern Analytics System is designed using a client-server architecture, which serves as the core framework for the system's web-based operation. This type of architecture separates the platform into client-side and server-side components to improve efficiency, organization, and scalability. The client side refers to the interface that authorized users access through web browsers such as Google Chrome, Mozilla Firefox, and Microsoft Edge. Through these browsers, users can interact with the system by logging in, viewing dashboards, generating reports, and analyzing business performance data. The browser functions as the presentation layer of the platform, displaying visual analytics such as charts, graphs, tables, and summary metrics. Since the system is web-based, users are not required to install any software on their individual devices. This makes the platform more accessible and convenient for users across different locations and devices. The interface is also designed to be responsive so that it can adapt to various screen sizes and device types. As a result, Kape Tubong management can monitor business performance more flexibly and efficiently. This client-side accessibility improves convenience and supports timely managerial decision-making.In addition, the client-side design prioritizes usability and user experience to ensure that management can navigate the platform with minimal technical difficulty. The interface is structured to present information clearly and logically so that users can interpret business metrics without requiring specialized analytical training. Interactive dashboard controls such as filters, date selectors, and comparison tools are integrated to improve user engagement and analytical flexibility. These features enable management to customize data views according to specific reporting needs or operational concerns. As a result, the client layer functions not only as a display interface but also as an interactive decision-support environment.

On the server side, the platform handles all backend processing, business logic execution, and communication with the database. The system uses Apache Web Server through XAMPP as its local hosting and development environment. Apache is responsible for receiving requests from users through the browser and returning the appropriate data or interface response. Whenever a user requests information such as revenue reports, filtered analytics, or dashboard updates, the server processes the request before sending the results back to the client interface. Backend scripts execute the necessary logic for report generation, sales aggregation, and analytical calculations. The server also manages user authentication and ensures that only authorized personnel can access restricted system functions. By centralizing business logic on the server, the platform maintains consistent processing and improves overall performance. This setup also enhances security because sensitive processes and calculations remain hidden from the client side. In addition, centralized processing reduces the workload on user devices since computation is handled by the server. Overall, the server component serves as the processing engine of the BREWINSIGHT BI platform.

The backend architecture is also designed to support modular processing, allowing individual system functions such as reporting, analytics, and authentication to operate independently while remaining integrated. This modularity improves maintainability by enabling developers to modify or enhance one module without significantly affecting other components. It also facilitates future expansion should additional analytical features or integrations be required. The centralized backend model therefore contributes to both system efficiency and long-term scalability.

To manage data storage, the system utilizes MySQL as its relational database management system. MySQL functions as the centralized storage repository for all business and transactional data used by the platform. It stores records such as sales transactions, kiosk information, product details, timestamps, user accounts, and generated reports. The database is structured into organized relational tables to support efficient storage and retrieval of information. Through SQL queries, the system can process large volumes of data and generate analytical outputs needed for reporting and dashboard visualization. MySQL also supports data filtering, sorting, aggregation, and comparison functions required by the analytics features of the system. Centralized database management ensures that all sales information from Kape Tubong kiosks is stored in one unified source. This eliminates the inconsistencies and duplication commonly associated with decentralized spreadsheets or manual recordkeeping. Security controls are also applied to protect stored data and restrict unauthorized database access. The use of MySQL therefore provides a reliable and scalable data management solution for the system.

Database normalization techniques are also applied during schema design to reduce data redundancy and maintain data integrity across records. Relationships between tables are carefully structured to support efficient querying and consistent data linkage among transactions, products, branches, and user accounts. Backup procedures may also be implemented to preserve critical business records and ensure recoverability in the event of data loss. These database management practices improve reliability and support long-term operational sustainability.

The communication process within the system follows a structured request-response mechanism between the client, server, and database. When a user performs an action through the dashboard interface, the browser sends a request to the server for the required information or operation. The server then processes the request and retrieves relevant data from the MySQL database. Once the data is collected and processed, the server transforms it into a format suitable for visual presentation. The processed results are then returned to the client interface where they are displayed as charts, graphs, tables, or dashboard summaries. This workflow enables the system to provide dynamic and real-time updates based on user interaction. It also allows multiple users to access the system simultaneously while maintaining consistent and synchronized data. Because all requests are processed centrally, the system ensures data accuracy and uniformity across users. This request-response structure is essential for delivering responsive and interactive analytics features. It enables BREWINSIGHT BI to provide timely and accurate business insights to management.

This communication model also improves processing transparency by clearly separating data input, server-side computation, and presentation output into distinct operational layers. Such separation enhances debugging efficiency and simplifies troubleshooting when issues occur. It further ensures that data transformations and analytical calculations are consistently applied regardless of which user accesses the system. This contributes to analytical reliability and reporting standardization across the platform.

The client-server architecture also offers significant operational advantages for Kape Tubong’s multi-kiosk business setup. Since all sales data is processed and stored centrally, management can monitor performance across multiple kiosk locations from one unified platform. This removes the need for manually consolidating reports from different branches and reduces administrative workload. It also improves consistency in reporting because all users access the same centralized dataset. Remote access allows management to review business performance and analytics even when not physically present at the business location. The architecture further simplifies system maintenance because updates and enhancements can be deployed directly on the server without modifying individual client devices. This ensures that all users automatically access the latest version of the platform. The centralized design also supports future scalability, allowing the system to accommodate additional kiosks or expanded analytics features as the business grows. Security and backup processes are easier to manage in a centralized server environment. Overall, the client-server architecture provides a practical, scalable, and efficient technical foundation for the BREWINSIGHT BI platform.

Additionally, centralized architecture improves organizational control by standardizing data processing rules and reporting procedures across all kiosk locations. This helps maintain consistency in operational monitoring and analytical interpretation regardless of branch size or location. It also supports more accurate branch-to-branch performance comparisons by ensuring that all data is processed using the same analytical criteria. Such consistency is essential for fair and reliable performance evaluation.

In summary, the client-server architecture is an appropriate and effective framework for the implementation of the BREWINSIGHT BI system. It supports centralized processing, secure data management, responsive web access, and efficient multi-user functionality. The architecture aligns well with Kape Tubong’s operational needs by enabling centralized monitoring of distributed kiosk locations. Its web-based design promotes accessibility, convenience, and real-time business oversight. The integration of Apache via XAMPP and MySQL offers a cost-effective and practical environment for deployment and maintenance. Additionally, the architecture allows the platform to remain scalable and adaptable as business requirements evolve. By separating user interaction, system logic, and data management into distinct layers, the platform achieves improved organization and maintainability. This architectural approach also enhances system reliability and analytical consistency. Therefore, the selected architecture provides the necessary technical structure to support the effective operation of the BREWINSIGHT BI platform. It establishes a strong and sustainable foundation for delivering business intelligence and revenue analytics to Kape Tubong.

Furthermore, the chosen architecture positions the system for future technological upgrades and integration opportunities. Should Kape Tubong expand its operations or require more advanced analytical capabilities, the modular and centralized nature of the architecture will support efficient enhancement and scaling. This ensures that the BREWINSIGHT BI platform remains adaptable to future organizational and technological developments.

3.4 Tools and Technologies Used

- **Frontend:** HTML, CSS and JavaScript

- **Backend:** Python Flask
- **Database:** MySQL
- **Development Tools:** Visual Studio Code and XAMPP

3.5 Data Gathering Techniques

- Structured interviews with owner (Bryan Tacayon) and branch managers
- Observation of sales processes at kiosks and main shop
- Document analysis of sales records, receipts, and manual logs
- Surveys with staff on technology comfort and reporting preferences

3.6 Testing Procedures

To evaluate the quality, reliability, functionality, and usability of the BREWINSIGHT BI: Revenue Pattern Analytics System, the researchers implemented a comprehensive and structured testing process composed of Unit Testing, System Testing, and User Acceptance Testing (UAT). These testing procedures were conducted systematically throughout the development lifecycle and prior to the final deployment of the system to ensure that every component functions according to its intended design and operational purpose. The testing process was essential in validating whether the developed platform could effectively support the business intelligence and revenue analytics needs of Kape Tubong while maintaining stability, accuracy, efficiency, and usability in actual business operations.

Each testing level focused on different aspects of the system, beginning with the evaluation of individual modules and progressing toward the assessment of the fully integrated platform. This layered testing strategy enabled the researchers to identify, analyze, and correct errors at an early stage before they could affect other parts of the system. By conducting testing incrementally during development, the researchers established a continuous cycle of evaluation and improvement that strengthened the reliability and overall quality of the platform. Problems identified during each testing phase were carefully documented, investigated, corrected, and retested to verify that the implemented solutions successfully resolved the detected issues. This systematic process reduced the risk of operational failures, minimized software defects, and improved the readiness of the platform for real-world implementation.

The testing procedures also ensured that the system directly addressed the operational concerns identified during the earlier phases of the study, including difficulties in monitoring sales performance, generating reports, analyzing revenue trends, consolidating branch data, and supporting management decision-making processes. Through rigorous testing, the researchers verified that the developed solution was not only technically functional but also practical, dependable, and suitable for the specific operational environment of Kape Tubong. Furthermore, the testing process provided objective evidence regarding the system’s strengths, limitations, and overall performance prior to deployment, allowing necessary refinements and improvements to be completed before the platform was introduced into regular business operations.

In addition to validating technical functionality, the testing process was designed to assess the accuracy, consistency, and reliability of the analytical outputs generated by the BREWINSIGHT BI system. Since the platform is intended to support business intelligence activities and data-driven decision-making, it was necessary to confirm that the reports, dashboards, forecasts, and revenue analyses produced by the system were accurate, timely, and meaningful. The researchers tested the platform using different datasets, transaction records, inventory information, and sales reports to determine whether the system could correctly process, organize, and visualize business information under varying conditions.

The researchers also conducted tests under different workload scenarios to examine the system's performance during peak and non-peak operational periods. This included evaluating how the platform responds when multiple users simultaneously access the system, generate reports, or input transaction data. These tests were important in determining whether the system could maintain stable performance and acceptable response times even during periods of high activity. The researchers monitored processing speed, loading time, system responsiveness, and overall efficiency to ensure that the platform remains reliable and functional regardless of the volume of data being processed or the number of active users.

Moreover, the testing procedures evaluated how the system handles incomplete, inconsistent, or invalid data entries. The researchers intentionally introduced incorrect or missing data into the system to verify whether proper validation mechanisms, warning messages, and error-handling procedures were functioning effectively. This allowed the researchers to assess the platform's resilience, reliability, and ability to prevent data-related errors that could negatively affect business reports or decision-making processes. The system was also tested to ensure that it provides clear notifications and guidance to users whenever invalid inputs, duplicate records, or operational errors are encountered.

The testing process further included security and access-control assessments to verify that sensitive business information remains protected from unauthorized access. User authentication procedures, account permissions, login mechanisms, and data access restrictions were tested to ensure that only authorized personnel can access confidential business records and analytical reports. These evaluations were necessary to protect organizational data integrity and maintain confidentiality within the system. The researchers also verified that the system properly records user activities and transactions to support monitoring, accountability, and audit-related requirements.

By applying multiple levels of testing, the researchers were able to evaluate the BREWINSIGHT BI system from both technical and user-centered perspectives. Technical testing focused on assessing the functionality of program codes, database connectivity, data processing accuracy, system integration, operational stability, and overall performance efficiency. Each module of the system—including data input forms, dashboard interfaces, report generators, analytics modules, and database management functions—was individually tested to verify that it performs according to the specified requirements and design objectives.

The technical evaluation also examined how effectively the system integrates with the existing hardware and software infrastructure used by Kape Tubong. Compatibility tests were conducted to ensure that the platform operates properly across different devices, operating systems, browsers, and network conditions commonly used within the organization. The researchers verified that the system can function consistently without causing software conflicts, compatibility issues, or operational interruptions. These assessments helped ensure that the implementation of the platform would proceed smoothly without requiring major adjustments to the organization's existing technological environment.

User-centered testing, on the other hand, focused on evaluating the usability, accessibility, and practicality of the system from the perspective of actual end-users. This phase involved selected staff members, sales personnel, branch managers, and administrative users who interacted directly with the system and provided feedback regarding their experience while using the platform. The researchers observed how users navigated through the interface, entered information, generated reports, interpreted analytics, and accessed different system functions. Particular attention was given to identifying areas where users experienced confusion, delays, or difficulties while interacting with the system.

To further evaluate usability, interviews and survey questionnaires were administered to participants involved in the testing activities. Users were asked to provide feedback regarding the clarity of the interface, ease of navigation, responsiveness of the system, readability of dashboards and reports, and overall satisfaction with the platform. Additional feedback was gathered regarding the relevance of system features, usefulness of analytical outputs, accessibility of information, and convenience of performing daily operational tasks using the platform.

The researchers also assessed the level of technical knowledge and training required for employees to effectively operate the system. This evaluation helped identify whether additional training materials, user manuals, or technical support mechanisms would be necessary during implementation. By understanding the learning needs and preferences of employees, the researchers were able to recommend strategies for smoother adoption and more effective long-term utilization of the platform within the organization.

One of the most important phases of the testing process was User Acceptance Testing (UAT). During this stage, actual users evaluated the system based on real operational scenarios and business activities commonly performed within Kape Tubong. Users tested the system using actual sales transactions, inventory records, customer order information, and operational reports to determine whether the platform meets their expectations and operational requirements. The primary objective of UAT was to confirm that the developed system is practical, functional, and capable of supporting the organization’s daily activities in a real-world environment.

The feedback gathered during User Acceptance Testing enabled the researchers to identify final adjustments and improvements needed before deployment. Suggestions from users regarding interface enhancements, report formats, workflow adjustments, and feature refinements were carefully reviewed and incorporated whenever necessary. This collaborative process ensured that the final version of the system would align closely with the preferences, work habits, and operational requirements of its intended users.

The comprehensive testing framework significantly improved confidence in the readiness, quality, and effectiveness of the BREWINSIGHT BI: Revenue Pattern Analytics System. Through detailed evaluation and continuous refinement, the researchers ensured that the platform is capable of delivering accurate analytics, stable performance, secure data management, and user-friendly functionality suitable for Kape Tubong’s operational environment. The testing process also established a reliable basis for future maintenance, system updates, and further enhancements as the organization continues to grow and adapt to changing business conditions.

Furthermore, the findings obtained from the testing procedures provided important documentation regarding the performance and behavior of the system under different operational scenarios. These records serve as valuable references for future troubleshooting, maintenance activities, performance monitoring, and system expansion. The testing results also highlighted potential opportunities for future feature enhancements and improvements that may further increase the value and effectiveness of the platform over time.

Ultimately, the rigorous and systematic implementation of Unit Testing, System Testing, and User Acceptance Testing ensured that the BREWINSIGHT BI: Revenue Pattern Analytics System is not only technically sound but also operationally practical, reliable, and highly suitable for real-world business application. The testing procedures validated that the developed system can effectively support revenue analysis, business monitoring, operational efficiency, and data-driven decision-making within Kape Tubong. By thoroughly evaluating every component of the platform, the researchers ensured that the final system is capable of delivering long-term value, improving organizational performance, and supporting the sustainable growth and success of the business in the future.

Unit Testing: was performed to examine individual system components separately and determine whether each module functioned correctly and efficiently on its own before being integrated into the complete system. This testing stage focused on validating the smallest functional parts of the BREWINSIGHT BI: Revenue Pattern Analytics System to ensure that every module performed according to its intended design and specifications. By isolating individual components during testing, the researchers were able to identify errors, inconsistencies, and functional issues more accurately and efficiently, allowing problems to be corrected at the earliest possible stage of development.

During this phase, specific system modules such as the login and authentication module, database connection module, dashboard visualization components, chart display functions, sales filtering tools, report generation logic, user account management, data input forms, and analytics processing features were tested independently. Each module was supplied with different sets of input data to verify whether the expected outputs, calculations, and responses were produced correctly. The researchers carefully evaluated whether each function executed properly under normal operating conditions and whether the module could consistently maintain accurate performance during repeated operations.

The primary objective of Unit Testing was to detect programming errors, logical inconsistencies, calculation mistakes, database communication failures, and unexpected system behavior before the individual components were combined into the fully integrated platform. Conducting tests at the module level made it easier to isolate defects, trace the source of problems, and implement corrective actions without affecting other parts of the system. This approach significantly reduced the complexity of troubleshooting and minimized the risk of unresolved errors propagating into later development stages.

Input validation testing was also performed extensively during this phase to ensure that the system could correctly process both valid and invalid user inputs. The researchers examined whether the system properly accepted authorized and correctly formatted data while rejecting incomplete, duplicate, inconsistent, or invalid entries. Various test cases were applied, including incorrect usernames and passwords, missing required fields, unsupported characters, invalid numerical values, and improperly formatted dates. These tests ensured that the system maintained data accuracy and integrity by preventing erroneous information from being stored or processed within the database.

Error messages, warning prompts, and exception-handling mechanisms were likewise evaluated to determine whether the platform responded appropriately whenever operational issues occurred. The researchers verified that the system displayed clear and understandable messages to users whenever errors were encountered, helping prevent confusion and guiding users toward corrective actions. Exception handling routines were tested to ensure that unexpected failures, connection interruptions, or invalid operations did not cause the system to crash or lose data. This process helped strengthen the reliability and resilience of the platform by ensuring that errors could be managed gracefully without disrupting system functionality.

Additional testing was conducted to evaluate the responsiveness and stability of each module under varying operational conditions. The researchers assessed whether modules continued functioning correctly when subjected to repeated requests, rapid user interactions, or simultaneous operations. This included testing how quickly charts loaded, how efficiently reports were generated, and how accurately filters processed and displayed requested information. Database queries and retrieval functions were also tested to confirm that data could be accessed, processed, and displayed without delays or inconsistencies.

The researchers also performed checks on module compatibility and interaction with supporting technologies such as database servers, web interfaces, and integrated software libraries. These tests ensured that each component could communicate effectively with related system resources and external services required for operation. Particular attention was given to ensuring that data transferred between modules remained accurate, synchronized, and free from corruption during processing.

Furthermore, edge cases and uncommon input conditions were included in the testing process to assess the robustness and dependability of individual modules. Boundary values, empty input fields, invalid credentials, unsupported file uploads, incorrect filter combinations, duplicate records, and unusual transaction conditions were intentionally tested to determine whether the system could continue operating correctly even in non-standard situations. These tests helped identify hidden defects and weaknesses that may not appear during normal use but could potentially cause operational issues in real-world environments.

Performance consistency was another important focus during Unit Testing. The researchers repeatedly executed functions and operations to verify that modules consistently produced the same accurate results over time. Retesting procedures were conducted whenever corrections, modifications, or updates were applied to any module. This process, commonly referred to as regression testing, ensured that newly implemented fixes successfully resolved identified issues without introducing additional errors or negatively affecting previously functioning features.

Documentation of testing results was also maintained throughout the Unit Testing phase. The researchers recorded detected issues, test conditions, corrective actions, and retesting outcomes to create a clear reference for future debugging, maintenance, and system enhancement activities. These records provided valuable insights regarding the behavior and stability of each module and served as supporting evidence that the components had been thoroughly validated prior to integration.

Through Unit Testing, the researchers verified that each individual component of the BREWINSIGHT BI: Revenue Pattern Analytics System was functioning properly, accurately, and consistently before progressing to the integration and system-level testing phases. This testing stage established a strong and reliable foundation for the entire platform by ensuring that all core functions operated correctly on their own. As a result, Unit Testing significantly contributed to improving the overall quality, stability, reliability, and maintainability of the developed system, reducing the likelihood of major technical issues during later stages of development and implementation.

System Testing: Once all individual modules successfully passed the Unit Testing phase, System Testing was conducted to evaluate the performance, functionality, reliability, and stability of the fully integrated BREWINSIGHT BI: Revenue Pattern Analytics System. This stage focused on determining whether all system components worked together correctly as a complete and unified business intelligence platform capable of supporting the operational and analytical requirements of Kape Tubong. Unlike Unit Testing, which examined individual modules independently, System Testing evaluated how the various modules interacted with one another when combined into a single operational environment.

The researchers tested the interaction between the user interface, backend processing logic, database management system, reporting engine, analytics modules, and dashboard visualization components to ensure smooth and accurate communication among all parts of the platform. This phase aimed to verify that information entered into the system flowed properly through each process without loss, corruption, duplication, or inconsistency. By examining the integrated behavior of the platform, the researchers were able to confirm whether the system performed according to the functional and technical specifications defined during the planning and design stages of the project.

During System Testing, common operational tasks and user activities were performed repeatedly under controlled testing conditions. These activities included user login and authentication, account access verification, sales transaction retrieval, report filtering, dashboard visualization, analytics generation, inventory monitoring, revenue trend analysis, and summary report generation. The researchers carefully observed whether the system processed data accurately and displayed correct outputs throughout the platform. Particular attention was given to ensuring that reports, charts, tables, and dashboard indicators reflected updated and synchronized information retrieved from the central database.

The researchers also verified that all modules exchanged data correctly and maintained consistency across the entire platform. For example, updates made within transaction records were checked to determine whether changes were immediately reflected in dashboard analytics, summary reports, and filtering tools. This process ensured that the system maintained real-time or near real-time synchronization between operational data and business intelligence outputs. Any issues involving delayed processing, incomplete updates, mismatched information, duplicate records, or synchronization failures were identified, documented, corrected, and retested to confirm that the implemented solutions resolved the problem successfully.

In addition to functional verification, the researchers evaluated the stability and reliability of the integrated system under regular operational conditions. Repeated testing procedures were conducted to determine whether the platform could maintain consistent behavior during prolonged usage and continuous interaction. The researchers observed whether the system remained stable while performing multiple tasks simultaneously, such as generating reports while processing data queries or updating dashboard visualizations while users navigated through different sections of the platform.

Basic performance testing was also performed during this phase to examine system responsiveness, processing efficiency, and operational speed under normal workload conditions. The researchers monitored loading times, dashboard responsiveness, report generation speed, and database query performance to ensure that the system delivered information within acceptable response times. These evaluations helped determine whether users could interact with the system efficiently without experiencing unnecessary delays or interruptions during daily operations.

To further evaluate performance, the researchers simulated scenarios involving multiple users accessing the platform simultaneously. This allowed them to assess how the system handled concurrent activities such as simultaneous dashboard requests, report filtering, data retrieval, and account access operations. The testing process examined whether increased user activity negatively affected system responsiveness, data accuracy, or operational stability. Through these simulations, the researchers verified that the platform could support regular business operations without significant performance degradation.

System Testing also included compatibility testing to verify that the platform functioned properly across different supported web browsers, operating systems, screen sizes, and hardware devices commonly used within the organization. The researchers tested the system using various browser environments to ensure that interface elements, dashboard components, navigation menus, charts, and reporting tools displayed correctly and behaved consistently regardless of the user’s device or browser preference. Compatibility checks were important in ensuring that all employees and managers could access and operate the system without encountering technical limitations caused by hardware or software differences.

The researchers additionally evaluated the responsiveness and adaptability of the user interface across different display resolutions and device types. This ensured that the platform maintained usability and readability when accessed through desktop computers, laptops, or other supported devices. Interface elements such as charts, buttons, tables, filters, and navigation panels were examined to confirm that they adjusted properly to varying screen dimensions without affecting usability or visual clarity.

Security-related functions were also assessed during System Testing. The researchers examined whether user authentication, access restrictions, account permissions, and session management features operated correctly within the integrated environment. Tests were conducted to verify that unauthorized users could not access restricted information or administrative functions. Logout procedures, session expiration controls, and password validation mechanisms were also evaluated to ensure that the platform maintained appropriate security standards and protected sensitive business data from unauthorized access.

Error handling and recovery mechanisms were another important aspect of the testing process. The researchers intentionally introduced invalid operations, incomplete data entries, interrupted connections, and unexpected user actions to determine how the integrated system responded to unusual conditions. The platform was evaluated to ensure that errors were handled appropriately without causing crashes, data corruption, or operational instability. Clear notifications and guidance messages were also verified to ensure that users received understandable instructions whenever problems occurred.

Data integrity testing was likewise conducted to confirm that information remained accurate and consistent throughout all system processes. The researchers checked whether stored records matched displayed outputs, whether calculations produced correct analytical results, and whether database transactions were completed successfully without missing or altered information. This was particularly important because the BREWINSIGHT BI platform relies heavily on accurate business intelligence outputs to support managerial decision-making and operational analysis.

Documentation of identified issues, corrective actions, testing procedures, and retesting results was maintained throughout the System Testing phase. These records provided a detailed reference regarding the behavior, performance, and stability of the platform under different operational conditions. Maintaining comprehensive documentation also supported future maintenance activities, troubleshooting procedures, and system enhancement efforts after deployment.

Through System Testing, the researchers confirmed that the entire BREWINSIGHT BI: Revenue Pattern Analytics System operated as a fully integrated, stable, and functional business intelligence platform. The testing process verified that all modules worked together efficiently, data was processed accurately, and system features performed according to established functional requirements. Furthermore, System Testing ensured that the integrated platform met the operational, technical, and usability standards necessary for practical deployment within Kape Tubong's business environment.

The successful completion of this phase provided greater confidence in the overall readiness of the platform for real-world use. By thoroughly evaluating functionality, integration, compatibility, security, responsiveness, and data consistency, the researchers ensured that the system was capable of supporting the organization's operational activities, delivering accurate business insights, and assisting management in making informed decisions. Ultimately, System Testing played a critical role in validating the reliability, effectiveness, and deployment readiness of the BREWINSIGHT BI platform before proceeding to the final stages of implementation and User Acceptance Testing.

User Acceptance Testing (UAT):was carried out to determine whether the developed BREWINSIGHT BI: Revenue Pattern Analytics System met the expectations, operational requirements, and practical needs of its intended users. This testing phase focused on evaluating the system in a realistic business environment to confirm that the platform was suitable for actual day-to-day operations within Kape Tubong. Unlike technical testing phases that primarily assessed system functionality and integration, User Acceptance Testing emphasized the experiences, satisfaction, and overall usability perceptions of the people who would directly utilize the system in their daily work activities.

During this phase, the business owner, Bryan Tacayon, together with selected staff members, branch personnel, and administrative users of Kape Tubong, participated in the testing activities. The participants were instructed to use the BREWINSIGHT BI platform in practical and realistic business-related scenarios that closely reflected actual operational conditions. These activities included accessing dashboards, retrieving reports, filtering sales data, monitoring branch performance, reviewing revenue trends, analyzing product sales, generating summaries, and navigating through various analytical features of the platform. By performing real operational tasks, the participants were able to evaluate whether the system effectively supported their responsibilities and business processes.

The primary purpose of User Acceptance Testing was to assess the system from the perspective of actual end-users and determine whether the platform was practical, understandable, efficient, and beneficial in supporting operational and managerial activities. The researchers observed how users interacted with the system and evaluated whether the platform's features, navigation structure, and dashboard presentations aligned with the users' expectations, work habits, and decision-making needs. Particular attention was given to evaluating whether users could easily access the information they needed and whether the analytical outputs generated by the platform were meaningful and useful for monitoring business performance.

Users assessed several important aspects of the platform during testing, including ease of use, clarity of dashboard presentation, responsiveness of the interface, accessibility of reports, speed of data retrieval, accuracy of displayed information, and overall convenience in performing daily operational tasks. The participants also evaluated whether charts, tables, summaries, and analytical visualizations were presented in a clear and understandable manner. Since many of the intended users were not technical specialists, it was important to ensure that the system communicated information in a simple, organized, and user-friendly format that could easily support managerial interpretation and business analysis.

The researchers also examined how quickly users were able to learn and navigate the platform without extensive technical guidance. This helped determine whether the system interface was intuitive enough for employees with varying levels of technological familiarity and experience. Tasks involving dashboard filtering, report generation, navigation between modules, and data interpretation were carefully observed to identify any difficulties or confusion experienced by participants while interacting with the system. The testing process also evaluated whether the terminology, menu labels, and interface controls used within the platform were understandable and appropriate for the organization's operational context.

Feedback was gathered from participants through direct observation, interviews, informal discussions, and structured evaluation forms. Users were encouraged to share their experiences, opinions, suggestions, concerns, and recommendations regarding the system's functionality, appearance, and usability. The researchers documented all feedback related to navigation issues, interface layout, report readability, dashboard organization, processing speed, accessibility of features, and clarity of analytical outputs. Participants also identified areas where additional improvements or refinements could further enhance the usefulness and convenience of the platform.

User Acceptance Testing also provided an important opportunity to evaluate whether the analytical outputs generated by the BREWINSIGHT BI platform were understandable, relevant, and meaningful to non-technical users. This aspect was particularly significant because the primary users of the platform rely on the system for business monitoring, operational analysis, and managerial decision-making rather than advanced technical analysis. The researchers assessed whether users could easily interpret charts, sales trends, revenue summaries, branch comparisons, and product performance reports without requiring extensive technical explanation or assistance.

The feedback obtained during this phase helped refine several aspects of the platform, including dashboard layouts, chart arrangements, report formatting, navigation flow, and interface organization. In response to user suggestions, the researchers implemented necessary revisions and adjustments to improve readability, simplify navigation, and enhance the overall user experience. Modifications were also made to ensure that important information could be accessed more quickly and displayed more clearly to users. These improvements contributed to making the platform more intuitive, accessible, and practical for everyday business use.

The researchers additionally evaluated the responsiveness of the system during actual user interaction. This included observing how quickly dashboards loaded, how efficiently reports were generated, and how smoothly filtering operations were processed while users navigated through the platform. The testing process ensured that users could interact with the system efficiently without experiencing delays or interruptions that could negatively affect productivity or operational workflows.

Another important objective of UAT was to determine whether the system effectively supported the operational workflows already established within Kape Tubong. The researchers examined whether the platform integrated naturally into existing business procedures and whether it simplified tasks related to sales monitoring, report preparation, and performance evaluation. This ensured that the implementation of the BREWINSIGHT BI system would improve operational efficiency rather than introduce unnecessary complexity or disruption to the organization's daily activities.

User Acceptance Testing also allowed the researchers to identify the level of user training and support required for successful implementation. By observing how participants interacted with the platform, the researchers were able to determine whether additional user manuals, tutorials, orientation sessions, or technical support materials would be necessary to facilitate smooth adoption of the system. This information became valuable for planning future training programs and implementation strategies within the organization.

All usability concerns, recommendations, operational difficulties, and enhancement requests identified during UAT were carefully reviewed and analyzed by the researchers. Necessary revisions were implemented, and the affected system components were retested to ensure that the modifications successfully addressed user concerns without negatively affecting other functionalities of the platform. This iterative improvement process ensured that the final version of the system was refined according to actual user experiences and operational requirements.

The successful completion of User Acceptance Testing confirmed that the BREWINSIGHT BI: Revenue Pattern Analytics System was suitable for deployment and aligned with the practical needs of Kape Tubong. The testing process verified that the platform was not only technically functional but also user-friendly, accessible, understandable, and effective in supporting business monitoring and decision-making activities. Furthermore, UAT strengthened user confidence and acceptance of the system by actively involving future users in the evaluation and refinement process.

Ultimately, User Acceptance Testing played a critical role in validating the overall effectiveness, practicality, and usability of the BREWINSIGHT BI platform. Through direct participation from actual users, the researchers ensured that the system was capable of delivering meaningful analytical insights, improving operational efficiency, and supporting informed managerial decisions within the organization. The feedback gathered during this phase contributed significantly to the optimization of the platform, ensuring that the final system was well-suited to the operational environment, business objectives, and long-term needs of Kape Tubong.

3.7 Evaluation Criteria

The developed BREWINSIGHT BI: Revenue Pattern Analytics System will be evaluated using a set of criteria designed to measure its effectiveness, technical quality, and suitability for practical deployment in Kape Tubong's operational environment. These criteria provide a structured basis for determining whether the system meets its intended objectives and performs according to the requirements established during the planning and design phases. Each criterion focuses on a specific aspect of software quality relevant to business intelligence systems, particularly those intended for real-world organizational use, addressing core factors that influence how well the system supports decision-making, integrates with existing workflows, and delivers tangible value to the business. They take into account the specific nature of Kape Tubong’s operations, including daily transaction processes, inventory management practices, sales tracking methods, and reporting requirements that define how the organization functions and makes strategic choices. They also consider the unique characteristics of the coffee shop industry, such as seasonal demand fluctuations, product variety, customer behavior patterns, and market trends that directly impact revenue generation and business performance. By focusing on these specific elements, the evaluation framework ensures that every aspect of the system's functionality is measured against the actual context and requirements of the organization, rather than relying on generic standards that may not apply directly to its operations.

Through evaluation based on these standards, the researchers can determine the strengths and limitations of the platform and identify areas requiring improvement prior to full implementation. This assessment phase is critical to ensuring that the final product is not just a theoretical concept, but a functional tool that can be adopted smoothly and used consistently by the organization's staff. It serves as a bridge between the development stage and actual deployment, allowing adjustments and refinements to be made while the system is still in the testing phase, which reduces risks associated with full-scale implementation and ensures that resources are used efficiently. The evaluation process also ensures that the developed system is not only technically operational but also practical, dependable, secure, and beneficial for its intended users, taking into account factors such as ease of use, reliability under varying workloads, and alignment with the daily operations and long-term goals of the organization. It considers how the system interacts with other tools and processes already in place, whether it can adapt to changes in business needs over time, and how well it addresses specific challenges or gaps that the organization has identified in its current data management practices. It also examines how the system handles different types of data, from basic transaction records to more complex analytical datasets, and whether it can process information efficiently even as the volume of data grows over time. Furthermore, it looks at how the system supports different levels of users within the organization, from frontline staff who input data to management personnel who analyze results and make decisions, ensuring that each group can access the information they need in a format that is appropriate and useful for their specific roles and responsibilities.

By assessing the platform across multiple dimensions, the study can provide a more comprehensive validation of system quality and readiness, moving beyond basic functionality to evaluate how well the system fits within the unique context and operational framework of Kape Tubong. This multi-faceted approach allows for a thorough understanding of not only what the system can do, but also how well it performs those tasks in real-world conditions, how it is perceived by those who will use it regularly, and what impact it is likely to have on overall business performance. It also helps in distinguishing between features that are essential to the system's purpose and those that may be useful but not necessary, ensuring that resources are focused on delivering the most value to the organization. This balanced assessment prevents over-engineering and ensures that the system remains user-friendly and cost-effective, while still providing all the capabilities required to meet the organization’s analytical and operational needs. It also enables a fair comparison between the developed system and existing methods or tools currently used by the organization, highlighting improvements in efficiency, accuracy, and decision-making support that the new platform can offer.

The criteria selected for evaluation are aligned with common software quality standards and reflect the practical needs of Kape Tubong as the intended beneficiary of the platform, balancing technical benchmarks with real-world usability and business relevance. They are derived from established guidelines for software development and system evaluation, as well as direct consultation with the organization’s management and staff to ensure that they capture all elements that matter most to the success of the system. These criteria collectively help determine whether BREWINSIGHT BI successfully fulfills its role as a business intelligence and revenue analytics solution, verifying that it can accurately process data, generate meaningful insights, and support the strategic and operational needs of the business effectively. They cover everything from the accuracy and reliability of data outputs to the clarity and accessibility of information presented to users, ensuring that the system serves as a reliable source of information for planning, analysis, and decision-making at all levels of the organization. They also include considerations related to system performance, such as processing speed, response time, and ability to handle concurrent users, as well as factors related to maintenance, support, and long-term sustainability. Additionally, they address aspects of data integrity, confidentiality, and compliance, ensuring that sensitive business information is protected and that the system operates within legal and ethical standards applicable to business operations and data management.

The evaluation results will serve as evidence of the platform's effectiveness and suitability for supporting business operations, providing clear documentation of performance levels, areas of excellence, and opportunities for further refinement. This information will be valuable not only for the researchers and developers who created the system, but also for Kape Tubong as they consider integrating the platform into their regular operations. It will help the organization understand what to expect from the system, how to maximize its benefits, and what steps may be needed to ensure its long-term success and continued usefulness as business conditions evolve over time. The findings will also provide insights that can guide future updates and enhancements, allowing the system to grow and adapt alongside the organization’s changing needs and market dynamics. Furthermore, the evaluation outcomes can be used to demonstrate the value of the investment made in developing and implementing the system, providing justification for continued support and expansion of its use within the organization. By establishing clear measures of success and performance, the evaluation process creates a foundation for monitoring the system’s impact over time and ensuring that it continues to deliver value long after it has been deployed. The following criteria will be used in assessing the developed system. The system will be evaluated based on the following criteria:

Functionality-Functionality refers to the extent to which the system is able to perform its intended tasks accurately and completely based on the specified requirements. This criterion evaluates whether the BREWINSIGHT BI platform successfully delivers the expected features and services required by Kape Tubong. These include core functions such as dashboard visualization, sales tracking, branch performance comparison, revenue trend analysis, product monitoring, and report generation. The system will be assessed to determine whether these features work properly and consistently produce correct outputs when used. It also examines whether all required modules have been fully implemented and are available to authorized users. Any missing features, inaccurate results, or malfunctioning components may indicate limitations in system functionality. The criterion further checks if the platform responds appropriately to user commands and processes data as expected. Feedback gathered from testing and user evaluation will help determine whether the system meets functional expectations. A system with high functionality demonstrates that it can effectively fulfill its intended purpose as a business intelligence platform. Therefore, this criterion is essential in confirming whether the developed system meets the primary goals of the study.

In addition, functionality assessment will determine whether the analytical outputs generated by the platform accurately reflect the underlying sales and transactional data stored in the database. The correctness of revenue computations, branch comparisons, and product performance summaries will be carefully examined to ensure analytical integrity. The ability of the system to handle filtering, date-range selection, and comparative reporting without producing inaccurate or incomplete results will also be evaluated. Functional completeness will be measured by verifying that all business requirements identified during analysis have been successfully translated into working system features. This ensures that the platform aligns with the operational expectations of Kape Tubong management. Furthermore, the platform’s ability to support simultaneous execution of multiple functions without conflict will be assessed. Strong functionality indicates that the system can serve as a comprehensive and dependable operational support tool. Ultimately, this criterion validates whether BREWINSIGHT BI delivers the intended business intelligence capabilities effectively and accurately.

Moreover, the functionality evaluation will assess the consistency of system operations across different user interactions and operational scenarios. The platform must be capable of maintaining accurate performance regardless of the volume of transactions processed or the number of users accessing the system simultaneously. Features related to data retrieval, report generation, and dashboard updating should operate efficiently without delays, missing records, or processing errors. The system will also be evaluated on its capability to integrate and organize information from multiple branches into a centralized database while preserving data accuracy and consistency. This is important in ensuring that management receives reliable and unified business insights from all operational locations.

Another aspect of functionality involves evaluating the effectiveness of automated processes incorporated within the platform. Automated sales consolidation, real-time data synchronization, and dynamic dashboard updates must function correctly to minimize manual intervention and reduce operational inefficiencies. The platform should also correctly categorize products, summarize transactions, and calculate key performance indicators without requiring extensive user adjustments. Through these automated functions, the system can improve the speed and accuracy of business analysis activities. The evaluation will determine whether these processes contribute positively to operational productivity and decision-making efficiency.

The functionality criterion will also examine the reliability of user access controls and role-based permissions implemented within the system. Authorized administrators, managers, and staff members should only be able to access features and information relevant to their designated responsibilities. This ensures that sensitive sales and financial data remain protected from unauthorized access or unintended modification. The proper functioning of login authentication, account management, and permission settings will therefore form part of the evaluation process. Effective access control mechanisms contribute to both system security and operational organization.

Additionally, the system's capability to generate meaningful and visually understandable reports will be assessed as part of functionality evaluation. The dashboards and analytical summaries presented by the platform should provide clear and interpretable information that supports managerial decision-making. Charts, graphs, and statistical summaries must display accurate data representations that users can easily analyze and compare. The platform should also support customizable reporting options that allow management to examine specific business periods, branches, or product categories according to their needs. This flexibility enhances the usefulness of the business intelligence platform in addressing different operational concerns.

The evaluation will further determine whether the system can effectively support future operational requirements and scalability needs. As Kape Tubong expands its business operations or introduces additional products and branches, the platform should remain capable of accommodating increasing volumes of transactional data and analytical demands. Functional scalability ensures that the system continues to perform efficiently even as organizational requirements evolve over time. This adaptability is important in maintaining the long-term relevance and usefulness of the platform within the business environment.

Furthermore, functionality testing will involve identifying and documenting any system limitations, processing issues, or feature inconsistencies encountered during actual usage. Errors identified during testing will provide valuable insights for system refinement and improvement. User feedback regarding ease of feature execution, report accuracy, and operational reliability will also contribute to the overall assessment of functionality. Continuous evaluation and refinement of system functions help ensure that the final platform operates according to the intended design and business objectives.

Overall, the functionality criterion serves as a comprehensive measure of the system's capability to meet operational, analytical, and managerial requirements. It verifies whether BREWINSIGHT BI can successfully perform all intended tasks while delivering accurate, reliable, and efficient business intelligence services to Kape Tubong. By ensuring that all features operate correctly and support business processes effectively, the study can confirm the practical value and effectiveness of the developed platform.

Usability- Usability pertains to how easy and convenient it is for users to learn, understand, and operate the system. This criterion evaluates whether the BREWINSIGHT BI platform provides an interface that is clear, accessible, and user-friendly for Kape Tubong personnel. The researchers will assess how easily users can navigate the platform, locate features, interpret dashboard data, and perform required tasks. Interface design elements such as menu placement, dashboard layout, labels, charts, and filtering controls will be reviewed for clarity and effectiveness. A highly usable system reduces confusion and allows users to complete tasks with minimal assistance or training. It also improves the overall user experience by making the platform more intuitive and comfortable to use. Feedback from end users during testing will provide valuable insight into how practical and understandable the system is in actual operation. Usability is especially important because even technically advanced systems may be underutilized if users find them difficult to navigate. A platform that is easy to use encourages adoption and consistent use in daily business activities. Therefore, usability helps determine whether the developed system is practical and user-centered.

Additional usability evaluation will focus on how effectively the dashboard communicates analytical information through visual design and layout. Since the platform presents business data through charts, graphs, and visual summaries, it is essential that these elements are understandable even to users without advanced analytical knowledge. The clarity of labels, legends, chart colors, and dashboard grouping will be reviewed to ensure that information can be interpreted quickly and accurately. Navigation efficiency will also be measured by determining how many steps are required for users to complete common tasks such as generating reports or filtering analytics. The responsiveness of the interface across desktop and mobile devices will likewise be considered as part of usability evaluation. A system that remains intuitive across multiple device types improves accessibility and operational flexibility. The researchers will also assess whether the platform minimizes user errors through clear prompts, validations, and guided interactions. Ultimately, usability evaluation confirms whether BREWINSIGHT BI can be effectively adopted and utilized by Kape Tubong personnel in routine operations.

Moreover, usability assessment will examine the learning curve experienced by first-time users when interacting with the platform. The researchers will determine how quickly users can familiarize themselves with the system's features and complete tasks without requiring extensive technical support. Instructions, icons, and navigational cues should be simple and self-explanatory to reduce dependency on manuals or external assistance. A system with a short learning curve enables employees to become productive more quickly and reduces the time needed for training sessions. This is particularly important for small business environments where staff members may have varying levels of technical expertise. By ensuring that the platform is easy to learn, the system can support smoother operational implementation and higher user acceptance.

The usability criterion will also evaluate the consistency of interface behavior throughout the platform. Consistent menu structures, button placements, typography, and color schemes help users predict system behavior and reduce confusion during operation. When interface components behave uniformly across modules, users can navigate the platform more efficiently and confidently. The evaluation will identify whether similar actions produce consistent responses and whether system workflows remain standardized throughout different functions. Consistency contributes significantly to user comfort and improves the overall quality of interaction between the user and the system.

Another important aspect of usability involves assessing the readability and accessibility of displayed information. Text size, spacing, contrast, and overall visual presentation should allow users to read and interpret data comfortably over extended periods of use. Charts and tables should avoid excessive complexity and present information in a concise and organized manner. The platform should also ensure that important information is prominently displayed to support quick decision-making. By improving readability and accessibility, the system can reduce user fatigue and enhance operational efficiency during daily use.

Furthermore, usability testing will evaluate the effectiveness of system feedback and response mechanisms. Users should receive clear confirmations when actions such as data filtering, report generation, or record updates are successfully completed. Likewise, the platform should provide understandable error messages and guidance whenever invalid inputs or operational issues occur. Effective feedback mechanisms help users understand system status and prevent uncertainty during interaction. This contributes to a more reliable and satisfying user experience while minimizing operational mistakes.

The researchers will also examine how the system supports user productivity and task efficiency. The platform should enable users to complete analytical and reporting tasks quickly without unnecessary steps or repetitive procedures. Features such as quick-access menus, automated filtering, searchable records, and streamlined workflows can significantly improve operational convenience. Time efficiency is especially important in business environments where managers and staff must process information promptly to support decision-making. A highly usable system not only improves user satisfaction but also contributes to organizational productivity and operational effectiveness.

In addition, usability evaluation will consider the adaptability of the platform to different user roles and responsibilities within Kape Tubong. Managers, administrators, and staff members may interact with different system features depending on their operational needs. The interface should therefore provide appropriate access and simplified workflows tailored to each type of user. Customized dashboards and role-specific displays can help reduce unnecessary complexity and improve the relevance of information presented to each user group. This adaptability enhances user convenience and ensures that the platform supports diverse operational functions effectively.

The system's visual appeal and overall aesthetic quality will likewise form part of usability assessment. An organized and visually balanced interface can improve user engagement and encourage continued system use. Appropriate use of colors, spacing, icons, and graphical elements can make the platform more attractive while maintaining professionalism and clarity. Although functionality remains the primary concern, visual design significantly influences user perception and comfort during interaction. A visually appealing system can contribute positively to user confidence and satisfaction.

Additionally, the researchers will assess whether the platform supports efficient recovery from user mistakes or operational errors. Features such as undo functions, editable inputs, confirmation prompts, and recovery options can help prevent permanent mistakes and reduce frustration among users. Error tolerance is important in maintaining smooth workflow continuity and minimizing disruptions during system operation. By allowing users to easily correct mistakes, the platform becomes more forgiving and user-friendly in real-world business environments.

Overall, usability serves as a critical measure of how effectively BREWINSIGHT BI supports user interaction, operational convenience, and system accessibility. It determines whether the platform can be comfortably and efficiently used by Kape Tubong personnel regardless of their technical background or experience level. Through comprehensive usability evaluation, the study can confirm whether the developed system promotes ease of use, efficient task completion, and positive user experience. A highly usable platform strengthens user acceptance and ensures that the business intelligence system can be fully integrated into daily operational activities.

Reliability-Reliability refers to the consistency and dependability of the system when operating under normal conditions over time. This criterion evaluates whether the BREWINSIGHT BI platform can function continuously without frequent failures, crashes, or errors. It also examines whether the system consistently produces accurate and stable outputs during repeated use. Features such as dashboard loading, report generation, and data filtering will be tested multiple times to determine consistency in performance. Reliability also includes the system's ability to manage unexpected situations, such as invalid inputs or minor operational errors, without affecting overall performance. The platform should maintain data integrity and remain responsive even when encountering routine issues. A reliable system gives users confidence that the information presented is accurate and dependable for business decision-making. Frequent system interruptions or inconsistent outputs may reduce trust in the platform's effectiveness. Because the system is intended to support important business decisions, reliability is a critical measure of its practical value. Therefore, this criterion ensures that the platform can serve as a dependable analytical tool in day-to-day operations.

Further reliability evaluation will determine the system’s capacity to maintain stable operation during extended periods of continuous use. The researchers will observe whether repeated access, report generation, and simultaneous user activity affect system stability or output consistency. Recovery behavior following minor errors or interruptions will also be assessed to determine how well the platform handles disruptions. Reliable systems should continue functioning predictably without significant degradation in performance over time. The platform’s ability to preserve stored data accurately without corruption or unintended modification will also be examined. Backup and restoration processes may be reviewed to evaluate preparedness against accidental data loss. Reliability is especially important because management decisions based on inaccurate or unstable reports may negatively impact operations. Therefore, the system must demonstrate dependable performance under realistic operating conditions. High reliability indicates that BREWINSIGHT BI can be trusted as a consistent source of business intelligence.

Moreover, reliability assessment will examine the platform’s capability to maintain uninterrupted performance during varying operational workloads. The system should remain stable whether processing a small number of transactions or handling larger volumes of sales data across multiple branches. Sudden increases in user activity or data requests should not cause the platform to become unresponsive or produce incomplete outputs. Consistent responsiveness during peak operational periods is important in ensuring that management can access business information whenever needed. The ability of the system to maintain operational continuity under changing workloads reflects the robustness of its design and implementation.

Another aspect of reliability involves evaluating the consistency of analytical computations and generated reports over repeated executions. Revenue summaries, branch comparisons, and product performance analyses should produce identical and accurate results when the same input conditions are applied. The researchers will verify that calculations remain precise even after multiple report generations or repeated filtering operations. This consistency is essential because inconsistent analytical outputs may lead to confusion and poor managerial decisions. Reliable analytical processing ensures that users can confidently depend on the information presented by the platform.

The reliability criterion will also assess the effectiveness of the system’s error-handling mechanisms. The platform should detect invalid inputs, incomplete records, or operational conflicts without causing system crashes or data corruption. Clear notifications and recovery procedures should guide users whenever problems occur. Effective error handling minimizes disruptions and helps maintain stable operation even during unexpected situations. The ability to recover gracefully from minor faults contributes significantly to the dependability of the system in real-world operational environments.

In addition, the researchers will evaluate the platform's capability to maintain database integrity during transactions and data updates. All recorded sales, inventory information, and analytical records must remain complete, accurate, and synchronized throughout system operations. The platform should prevent duplication, accidental deletion, or inconsistent modification of stored data. Reliability in data management is critical because inaccurate or incomplete information may compromise the quality of business analysis and reporting. By preserving database consistency, the system can provide trustworthy analytical outputs for management use.

System availability will likewise form an important part of reliability evaluation. The researchers will assess whether the platform can remain operational and accessible during regular business hours with minimal downtime. Unexpected outages or prolonged inaccessibility can interrupt operations and reduce user confidence in the platform. A dependable system should provide continuous access to dashboards, reports, and analytical tools whenever required by management and staff. High system availability ensures that the platform effectively supports ongoing operational activities and decision-making processes.

Furthermore, reliability testing will include evaluating the platform's capability to support concurrent users without operational conflicts or performance failures. Multiple users accessing dashboards, generating reports, or filtering data simultaneously should not negatively affect system stability. The platform should maintain consistent speed, responsiveness, and accuracy regardless of concurrent activity levels. This capability is especially important for organizations with multiple branches and managerial personnel accessing the system at the same time. Reliable multi-user support enhances operational coordination and overall system dependability.

The researchers will also assess the reliability of data synchronization and real-time updating mechanisms within the platform. Information entered from different branches should be accurately reflected within the centralized dashboard without delays or inconsistencies. Real-time synchronization is important in ensuring that managers are viewing current and reliable operational data when making business decisions. Delayed or inconsistent updates may reduce the usefulness of the platform as a business intelligence tool. Therefore, the consistency and timeliness of data synchronization processes will be carefully evaluated.

Additionally, the platform's backup and recovery capabilities will be examined as part of reliability assessment. The system should be capable of preserving critical business information through regular backup procedures and secure storage mechanisms. In the event of accidental deletion, software failure, or technical malfunction, the platform should allow efficient restoration of lost or damaged data. Reliable recovery procedures help minimize operational disruptions and protect important organizational records. This contributes to the long-term stability and sustainability of the business intelligence platform.

Another important consideration is the platform's ability to maintain reliability after software updates, feature enhancements, or system modifications. New updates should not negatively affect existing functions or introduce operational inconsistencies. The researchers will determine whether the platform continues to operate correctly after maintenance activities or configuration changes. Stable post-update performance reflects effective system architecture and proper software management practices. Maintaining reliability during system evolution is important in ensuring continuous operational support for Kape Tubong.

Overall, reliability serves as a critical indicator of the platform's stability, consistency, and trustworthiness in supporting business operations. It confirms whether BREWINSIGHT BI can continuously deliver accurate analytical outputs, preserve data integrity, and maintain operational performance under normal and demanding conditions. Through comprehensive reliability evaluation, the study can determine whether the system is dependable enough to support managerial analysis and decision-making processes effectively. A highly reliable platform strengthens organizational confidence and ensures that the system can function as a stable and long-term business intelligence solution for Kape Tubong.

Efficiency - Efficiency refers to the speed and resource effectiveness of the system when performing its functions. This criterion measures how quickly the BREWINSIGHT BI platform can process requests, retrieve data, and display results to users. The researchers will evaluate the response time of the system when loading dashboards, generating reports, and applying filters to analytical data. Fast and smooth performance is essential in ensuring that users can access business insights without unnecessary delay. Efficiency also involves assessing how well the system utilizes available computing and server resources during operation. A system that performs tasks efficiently contributes to improved productivity and a better user experience. Slow processing times may hinder usability and reduce the practical value of the platform in operational settings. Since business decisions often require timely access to updated information, efficiency is especially important in analytical systems. A highly efficient platform enables users to retrieve and analyze data quickly, thereby supporting more responsive management decisions. Therefore, efficiency is a key factor in determining the practicality of the developed system.

In addition, efficiency assessment will include evaluating the optimization of database queries and backend processing logic to ensure that large transaction datasets can be handled without excessive delay. As Kape Tubong expands and accumulates more sales records over time, the system must remain capable of processing increasing volumes of data efficiently. The platform's ability to maintain acceptable performance despite data growth will therefore be considered during evaluation. Efficient data handling improves scalability and supports long-term usability of the platform. The researchers will also assess loading times for visual analytics and determine whether dashboard components render smoothly across different devices and internet conditions. Efficient performance reduces waiting time and enhances user satisfaction during operational use. Resource optimization additionally minimizes server strain and improves hosting sustainability. Overall, this criterion ensures that BREWINSIGHT BI can deliver business intelligence outputs promptly and efficiently.

Moreover, efficiency evaluation will examine the speed at which the platform processes and updates transactional information within the centralized database. Sales records entered from multiple branches should be reflected in the dashboard within an acceptable time frame to ensure that management receives updated business insights promptly. Delays in data synchronization may reduce the effectiveness of real-time monitoring and affect the accuracy of operational analysis. The system should therefore maintain fast processing performance even when handling simultaneous data submissions from different locations. Efficient real-time updating contributes to improved responsiveness and supports timely managerial decision-making.

Another important aspect of efficiency involves assessing the system’s capability to execute complex analytical operations without excessive consumption of computing resources. Revenue analysis, comparative reporting, trend visualization, and filtering operations should be completed quickly while maintaining stable performance. The researchers will evaluate whether analytical computations can be performed efficiently without causing lag, freezing, or delayed dashboard responses. Efficient analytical processing ensures that users can interact with the platform continuously without interruptions. This is especially important for managers who rely on immediate access to analytical reports during daily operations.

The efficiency criterion will also assess the responsiveness of the user interface during navigation and interaction. Menus, charts, filters, and reporting tools should react promptly to user commands without noticeable delay. Smooth transitions between dashboard pages and fast loading of graphical components improve user satisfaction and operational convenience. The platform should minimize unnecessary waiting time so that users can complete tasks more efficiently. A responsive interface contributes significantly to the practicality and usability of the system in real-world business environments.

Furthermore, the researchers will evaluate how efficiently the system handles concurrent user activity. Multiple users accessing the platform simultaneously should not significantly reduce system speed or affect operational performance. The system must maintain acceptable response times even when managers, administrators, and staff are using different modules at the same time. Efficient handling of concurrent operations supports organizational coordination and prevents workflow interruptions. This capability is particularly important for businesses with multiple branches that depend on centralized access to shared analytical information.

Efficiency testing will also involve measuring the platform’s startup and loading performance during initial access. The time required for user authentication, dashboard initialization, and retrieval of analytical data will be carefully observed. A system that initializes quickly improves accessibility and reduces delays during operational use. Long startup times may discourage frequent use and negatively affect user productivity. Efficient initialization ensures that users can access business intelligence tools whenever needed without unnecessary inconvenience.

In addition, the researchers will assess the optimization of data storage and retrieval mechanisms implemented within the platform. Efficient indexing, query processing, and database organization can significantly improve the speed of information retrieval and report generation. The platform should be capable of locating and processing specific records rapidly even as the database grows larger over time. Proper database optimization contributes not only to performance efficiency but also to overall system scalability and sustainability. Efficient storage management also reduces the likelihood of performance degradation as business operations expand.

The efficiency criterion will further examine the platform’s capability to perform automated functions within minimal processing time. Automated report generation, sales consolidation, and dashboard refreshing should occur smoothly without disrupting ongoing operations. Efficient automation reduces manual workload and enhances operational productivity by allowing repetitive processes to be completed quickly and accurately. The researchers will determine whether these automated features contribute positively to workflow efficiency and time management within Kape Tubong operations.

Another consideration in efficiency evaluation is the adaptability of the platform under varying internet connectivity conditions. Since users may access the system from different locations and devices, the platform should maintain reasonable responsiveness even with moderate network limitations. Efficient data transmission and optimized dashboard rendering help ensure smoother performance across different operational environments. This flexibility improves accessibility and supports uninterrupted use of the platform regardless of connection quality.

Additionally, energy and resource efficiency may also be considered during system evaluation. Efficient systems consume fewer server and hardware resources while still maintaining high performance levels. Reduced resource consumption can lower operational costs related to hosting, maintenance, and system management. By optimizing resource utilization, the platform becomes more sustainable and cost-effective for long-term deployment. Efficient resource management also contributes to improved system stability and reliability.

The researchers will likewise evaluate whether the platform maintains efficiency after prolonged periods of continuous use. Systems that gradually slow down during extended operation may negatively affect productivity and user experience. The platform should therefore sustain stable processing speed and responsiveness even after long operational hours. Consistent efficiency over time reflects effective system architecture and proper software optimization practices. This ensures that the platform remains dependable during routine daily business activities.

Overall, efficiency serves as a critical measure of the platform's speed, responsiveness, and effective resource utilization in supporting business operations. It determines whether BREWINSIGHT BI can process, analyze, and present business information promptly without compromising performance quality. Through comprehensive efficiency evaluation, the study can verify whether the system is capable of delivering timely analytical outputs that support fast and informed managerial decision-making. A highly efficient platform enhances productivity, improves user satisfaction, and strengthens the overall operational value of the business intelligence system for Kape Tubong.

Security - Security refers to the system's ability to protect data, user access, and internal processes from unauthorized use, manipulation, or breaches. This criterion evaluates whether the BREWINSIGHT BI platform includes adequate safeguards to secure Kape Tubong's confidential sales and operational information. Security measures such as user authentication, password validation, role-based access control, and restricted system permissions will be examined during evaluation. These mechanisms help ensure that only authorized personnel can access sensitive reports and administrative functions. Security also involves protecting stored data against unauthorized modification, accidental deletion, or corruption. The platform's session management and secure handling of database interactions will likewise be assessed. Since the system contains financial and sales-related information, strong data protection is essential for maintaining confidentiality and trust. Weak security measures may expose the business to operational risks and unauthorized data access. Therefore, evaluating the security of the platform is necessary before full deployment. This criterion helps ensure that the system is safe and trustworthy for actual business use.

Additional security evaluation will examine whether the system properly enforces access restrictions based on user roles and responsibilities. The researchers will verify that lower-level users cannot access administrative or confidential functions beyond their authorization. Session timeout mechanisms and logout controls will also be reviewed to reduce the risk of unauthorized access from unattended devices. Input sanitization and validation measures will be assessed to determine whether the system is protected against common vulnerabilities such as SQL injection and malicious form submissions.

Data backup and recovery safeguards may also be considered as part of overall data protection measures. Strong security controls increase confidence in the system's integrity and reliability for operational deployment. Protecting sensitive business information is especially critical in analytical systems that centralize financial and operational records. Therefore, security evaluation ensures that BREWINSIGHT BI meets acceptable standards for safeguarding Kape Tubong's digital business assets.

Moreover, security assessment will evaluate the effectiveness of the platform's authentication mechanisms in verifying user identity before granting access to the system. Login procedures should prevent unauthorized individuals from accessing confidential information through the use of secure credentials and properly validated authentication processes. Password policies such as minimum length requirements, password complexity, and secure password storage practices will also be reviewed. These measures help reduce the risk of unauthorized account access and strengthen overall system protection. Reliable authentication mechanisms are essential in maintaining accountability and protecting organizational data from misuse.

Another important aspect of security involves assessing the confidentiality of data transmitted between the user interface and the database server. Sensitive information such as sales records, user credentials, and financial reports should be transmitted securely to minimize exposure to interception or unauthorized access. The researchers will examine whether secure communication practices are implemented during system transactions and data exchanges. Protecting data during transmission is especially important in web-based systems where information may travel across networks and internet connections. Secure data communication contributes significantly to maintaining the integrity and confidentiality of business operations.

The security criterion will also examine the platform's ability to maintain accurate activity records and audit trails. Logging mechanisms should record important user actions such as login attempts, report access, data modifications, and administrative activities. These records help administrators monitor system usage and identify suspicious or unauthorized activities when necessary. Audit trails improve accountability by allowing management to trace operational actions within the system. Effective monitoring capabilities strengthen organizational control and support security incident investigation if problems occur.

Furthermore, the researchers will assess the platform's resistance to common cybersecurity threats and operational vulnerabilities. The system should be capable of preventing unauthorized database manipulation, malicious data entry, and unauthorized privilege escalation. Proper input validation, secure coding practices, and controlled database access help minimize exposure to security risks. The evaluation will determine whether the platform can maintain stable and secure operation even when exposed to potentially harmful or invalid inputs. Strengthening system defenses against vulnerabilities improves the overall resilience and reliability of the platform.

In addition, security evaluation will consider the protection of stored data within the centralized database. All business records, sales information, and analytical reports should remain safeguarded against accidental loss, unauthorized alteration, or corruption. The system should implement appropriate database management controls to preserve data accuracy and integrity. Backup procedures and recovery mechanisms should also ensure that important records can be restored in the event of technical failure or accidental deletion. Secure data storage practices are critical in ensuring the long-term protection of organizational information assets.

The researchers will also evaluate how effectively the platform separates user privileges and operational responsibilities. Administrators, managers, and staff members should only have access to functions necessary for their assigned roles. Limiting access privileges reduces the risk of accidental misuse and prevents unauthorized exposure of confidential information. Proper role segregation additionally supports organizational discipline and operational accountability. By ensuring that access rights are carefully managed, the platform can provide stronger protection for sensitive business data.

Another important consideration in security assessment involves evaluating the system’s capability to maintain operational security during prolonged use. The platform should preserve secure session handling, user authentication, and access restrictions consistently over time without introducing vulnerabilities. Security controls should remain functional even after system updates, maintenance activities, or prolonged operational periods. Stable and continuous enforcement of security measures reflects the reliability and maturity of the platform’s design. This ensures that the system remains dependable and protected throughout its operational lifecycle.

The security criterion will further examine whether the platform includes safeguards against accidental operational mistakes that may compromise data integrity. Confirmation prompts, restricted deletion privileges, and controlled editing functions can help minimize unintended modifications or data loss. Such safeguards are important because human errors may negatively affect business reports and operational records. By reducing the likelihood of accidental mistakes, the platform can better preserve the accuracy and reliability of stored information.

Additionally, the researchers will evaluate user awareness and system guidance related to security practices within the platform. Clear notifications regarding invalid login attempts, password requirements, and session expiration can help users understand and follow secure operational procedures. Security-related prompts and reminders improve user awareness and encourage responsible system usage. Educating users indirectly through interface guidance contributes to a stronger overall security environment within the organization.

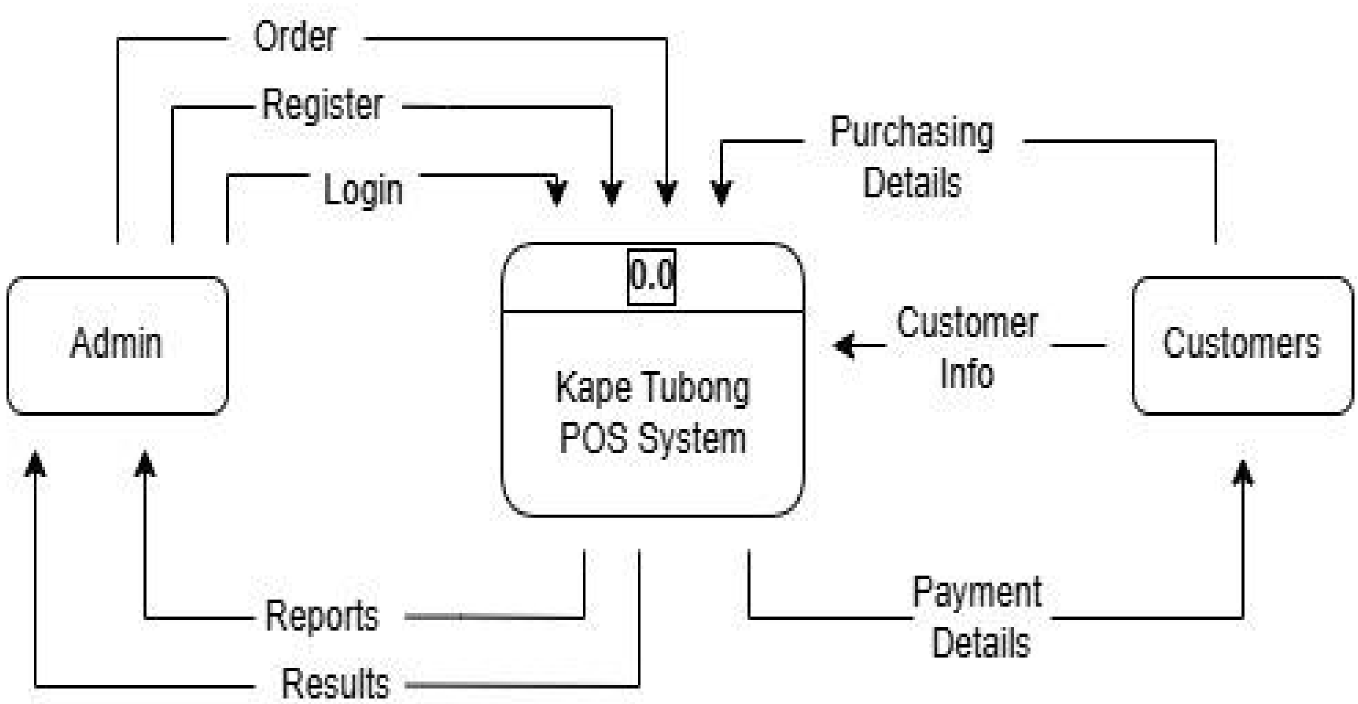
The platform’s ability to maintain system integrity during simultaneous user access will likewise be considered during security evaluation. Multiple users interacting with reports, records, and dashboard features should not create conflicts that compromise data consistency or system protection. Secure transaction handling and controlled database access are essential in preventing data overlap, unauthorized modifications, or operational inconsistencies. This capability is particularly important in centralized business intelligence systems accessed by multiple branches or departments.

Overall, security serves as a critical measure of the platform’s capability to protect business information, maintain confidentiality, and prevent unauthorized access or misuse. It determines whether BREWINSIGHT BI can provide a safe and controlled environment for managing Kape Tubong’s financial and operational data. Through comprehensive security evaluation, the study can verify whether the platform meets acceptable standards for safeguarding digital assets, maintaining data integrity, and supporting secure operational processes. A highly secure platform strengthens organizational trust, minimizes operational risks, and ensures that the system can be confidently utilized as a reliable business intelligence solution.

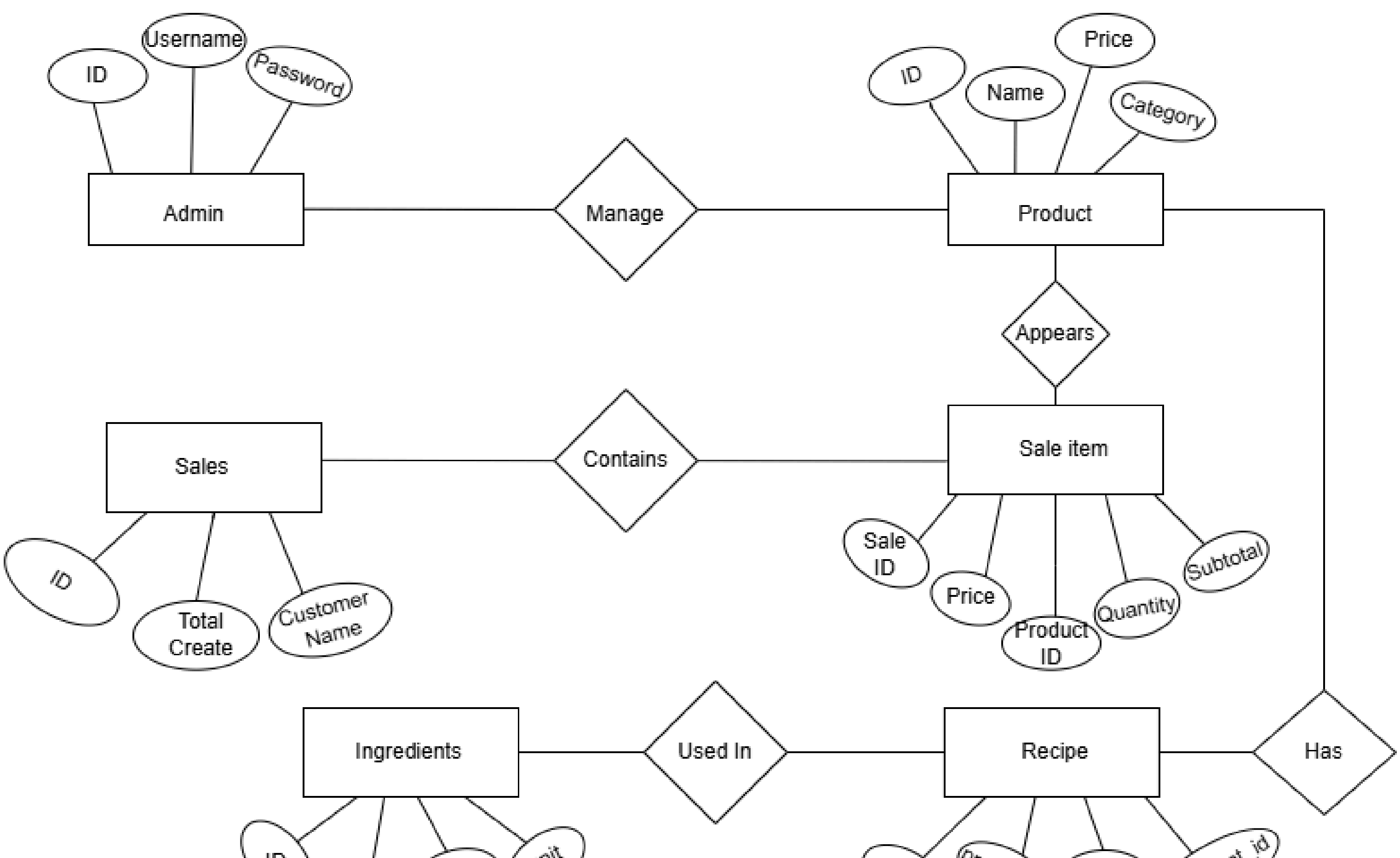
SUPPORTING DIAGRAMS (Included within Chapter 3)

Data Flow Diagram (DFD – Level 0)

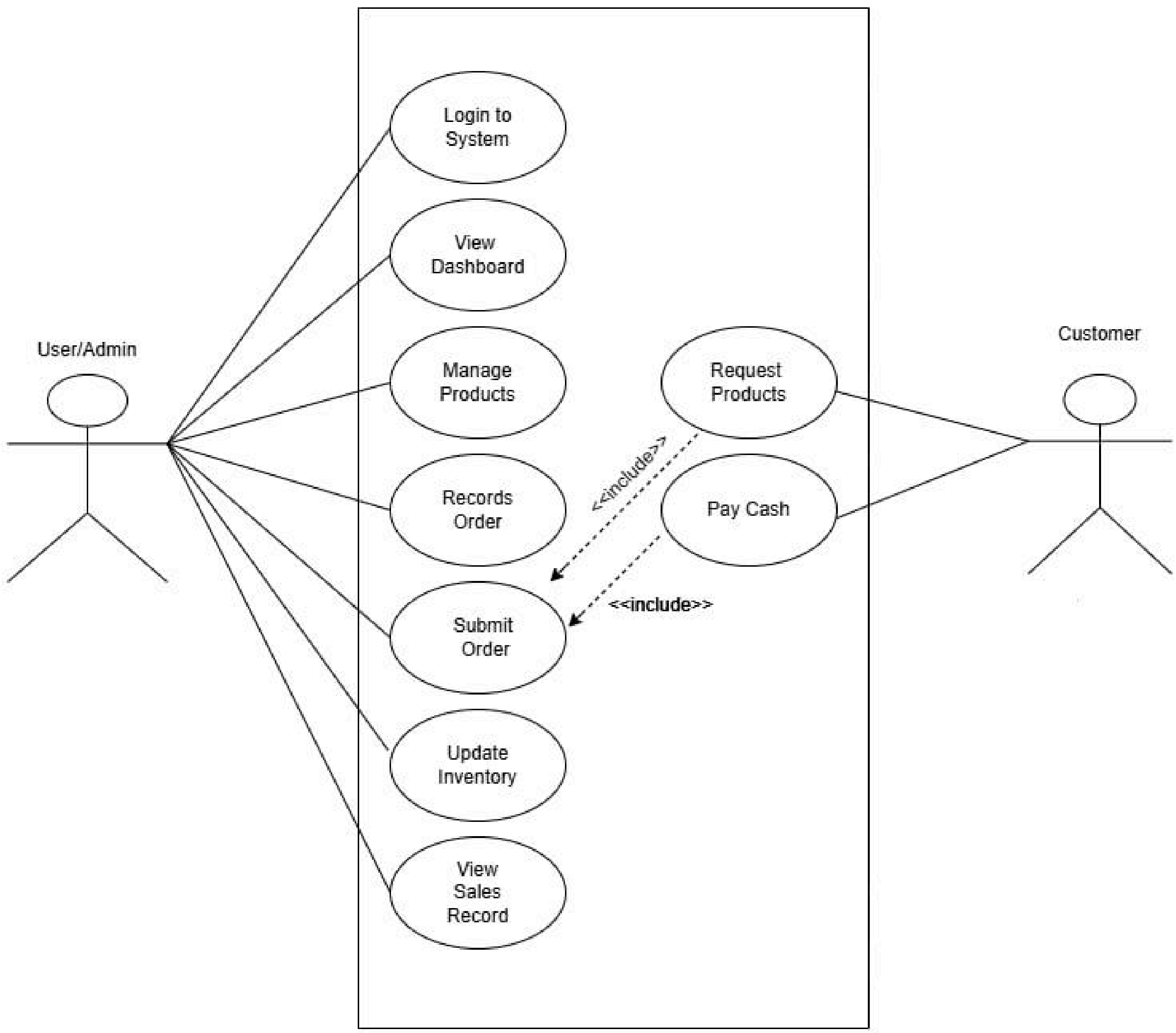
Admin → Pos System→ Customers



Entity Relationship Diagram (ERD)



Use Case Diagram



End of Proposal

Note to the instructors handling this subject

CHED BSIT ALIGNMENT STATEMENT

This capstone project complies with CHED BSIT standards by demonstrating competencies in application development, database management, systems analysis and design, and emerging technologies such as machine learning. The project follows outcomes-based education principles and integrates ethical considerations, user safety, and data privacy.